



GENETIC IMPROVEMENT OF FARMED TILAPIAS (GIFT)

FINAL REPORT

(March 1988 to December 1997)



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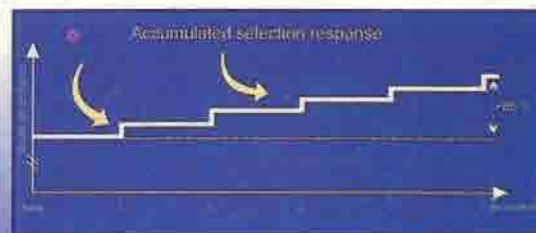
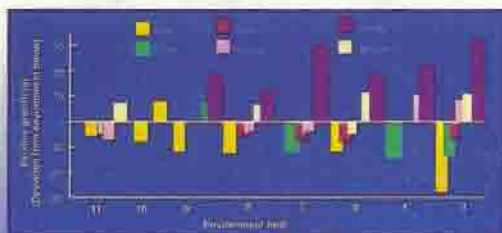
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GIFT



**SUPER
TILAPIA**

INTERNATIONAL, INC.



United Nations
Development Programme
Project No. G12/95/016



International Center for Living Aquatic
Resources Management



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**United Nations Development Programme
Sustainable Energy and Environment Division
Project No. GLO/90/016**

GENETIC IMPROVEMENT OF FARMED TILAPIAS (GIFT) PROJECT



FINAL REPORT

**(Phase I – UNDP Project # INT/88/019 and INT 91/005;
Phase II – UNDP Project # GLO/90/016)**

March 1988 to December 1997

Part 1

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(March 1, 1988-December 31, 1997)

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August 1998

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GENETIC IMPROVEMENT OF FARMED TILAPIA PROJECT FINAL REPORT (1988-1997)

Prepared By

DR. AMBEKAR E. EKNATH
Ms. BELEN O. ACOSTA

August 1998

A Collaborative Research Project between the International Center for Living Aquatic Resources Management (ICLARM), Philippines; National Freshwater Fisheries Technology Research Center/Bureau of Fisheries and Aquatic Resources (NFFTRC/BFAR), Philippines; Freshwater Aquaculture Center, Central Luzon State University (FAC-CLSU), Philippines; University of the Philippines-Marine Science Institute (UP-MSI); and the Institute of Aquaculture Research (AKVAFORSK), Norway (through the Norwegian Centre for Agricultural Development)

Cover: The GIFT Project logo and the logos of four major projects and initiatives established by ICLARM and its partners as a result of the emerging lessons and findings of the GIFT Project: the DEGITA, INGA, GICA and the GIFT Foundation. Descriptions of these initiatives are in pages xii & xiii. The two diagrams represent some of the significant results from the GIFT Project's experiments. The rigorous genotype x environment interaction experiment (left figure) carried out during generation 1 reveals consistency in growth performance of individual strains across test environments and the potential for growth and magnitude of possible differential growth performance among strains across a wide range of environments (details in section 3.1.3.). The observed response to selection across five generations of selection is shown in the right figure. The accumulated genetic gain in relation to the base population was 85%. *Artist: Alan Esquillon*

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ABOUT THIS REPORT

The GIFT Project is one of the longest multidisciplinary research and development program implemented by ICLARM and its partners. It began on March 1, 1988 and was successfully concluded on December 31, 1997. Condensing the ten years of cumulative experience gained in implementing the GIFT Project into a report of few pages is a task that we have set out to accomplish in this report.

The focus of this report, as suggested by the principal donor of this initiative, the United Nations Development Programme (UNDP), has been on describing the major findings, recommendations and lessons learnt. This Part 1 of the final GIFT Project report is organized into three sections: (a) extended executive summary. The objective of this section is to provide a "stand-alone" account of the entire GIFT Project for readers wishing to gain a quick comprehensive insight into the wide variety of activities implemented by ICLARM and its partners; (b) assessment and brief description of project activities in relation to the stated immediate objectives of the Project. This section also includes a brief description of activities carried out through the DEGITA project. In anticipation of a wider range of target audience, we have, to the extent possible, kept the technical jargon to a minimum; and (c) recommendations and lessons learnt. Detailed technical descriptions and other relevant materials are included in a separate Part 2 of the report. Part 2 also contains references to all relevant literature on the subject.

The GIFT project as a whole, like any pioneering projects, is a "story" of research successes and also setbacks. It has evolved continuously through rigorous peer review and by taking stock of experiences gained at the end of each year of implementation. All these experiences have been captured in annual progress reports, the tripartite reviews convened by the UNDP, and the External Review Panel reports submitted to the UNDP.

To facilitate proper implementation of the complex range of activities nested within the stated objectives of the GIFT Project and in response to the challenges and opportunities presented by the donor community, ICLARM developed three major parallel projects/initiatives: (a) the Dissemination and Evaluation of Genetically Improved Tilapia Species in Asia (DEGITA) Project with principal focus on the dissemination of the improved Nile tilapia germplasm and *ex ante* evaluation of impact on equity, environment and biodiversity; (b) the International Network on Genetics in Aquaculture (INGA) as a forum for the participating national institutions to exchange research methods, experiences and germplasm; and (c) the GIFT Foundation International Inc. as a mechanism for continuing the selective breeding research initiated by the GIFT Project. Consequently, a comprehensive insight into the complete story of the GIFT project can only be gained by reading this report in conjunction with the following reports available on request from ICLARM:

- The DEGITA Final Report (an extended summary of the report is included as an attachment to Part 1 of the GIFT Final Report)
- The INGA Reports including the breeding plans prepared for Indonesia (Nile tilapia, milkfish and carps) and Vietnam (Nile tilapia, carps and silver barb); and the new research initiative on "Genetic Improvement of Carps in Asia".



- The GIFT Project Database: Data Dictionary and Documentation.
- The GIFT Manual of Procedures - developed according to the UNDP Train-X methodology for training purposes.
- The annual Tripartite Review Reports and the annual External Review Panel Reports
- Agreements between ICLARM and its partners for successful termination of the GIFT Project and establishment of the GIFT Foundation International Inc.
- Recommendations of the Meeting on "International Concerns in the Use of Aquatic Germplasm" (June 1992) — this document laid the foundation for establishing the GIFT Phase II, DEGITA and INGA projects.

At the time of compilation of this report, a number of scientific papers for primary literature were in various stages of preparation.

Ambekar E. Eknath
Belen O. Acosta



FOREWORD

Meryl J. Williams
Director General
ICLARM

This report describes the conduct of a unique 10-year research and capacity building partnership. The context within which the GIFT project was conducted is also a unique period in the history of development, food production and aquaculture.

Feeding the world with affordable, nutritious food, produced in farm systems which respect the environment and biodiversity is one of the central aims of today's development. Many are familiar with the achievements of the 'Green Revolution' in cereals production and the large gains in production of livestock. These developments have helped to lower the prices of grains and animal meat and other products almost continuously over the last several decades. Billions of lives have been saved as a result and the nutrition of many more improved. World populations continue to grow, especially in developing countries, and thus improved production and production efficiency in crops and livestock is still being sought through genetic improvements, even as controversial genetic engineering and other biotechnologies are applied to help. The diversity within the germplasm of crop and animal species is being painstakingly documented, characterised and tested for its future use. Some efforts, but not sufficient, are being made to conserve the remaining natural diversity on earth. With a few exceptions among temperate species, genetic improvements in fish and other aquatic resources have lagged far behind those of terrestrial food species. So too have the efforts to characterize, document and use aquatic biodiversity.

In 1988, when ICLARM and its main partners (the Philippine Bureau of Fisheries and Aquatic Resources, the Central Luzon State University, and the Norwegian Aquaculture Research Institute AKVAFORSK) embarked on a project, supported by the United Nations Development Program and the Asian Development Bank, to test whether genetic improvement was also possible for a tropical aquaculture species, the Nile tilapia, the parties little imagined that the pressure to increase fish production would escalate so fast during the nearly 10 year course of the project. In this period, a crisis in world fisheries production was recognized as many natural fish stocks became fully and over-exploited. This led to increasing scarcity of supply and prices rose for many aquatic commodities. Indeed, the subject of the GIFT project, the Nile tilapia, *Oreochromis niloticus* changed from being a low value commodity of little more than local domestic interest in many parts of the world to being an internationally traded commodity of great commercial interest. Due to the biology and culture characteristics of Nile tilapia, however, it still remains a fish which small farmers can grow successfully and which many poor consumers can still obtain.

Against this world backdrop, the GIFT project partnerships developed. From the start and over the decade, scientific opinions differed over the approaches being taken. Each of the debates helped focus and improve the science and the technical procedures, from the genetic studies to the questions of fish introductions. Real environmental concerns over the development of improved breeds and their impact on the environment led the project partners to examine this question and led ICLARM to develop a policy position on fish introductions. To this day, we have not permitted the repatriation of the GIFT fish to Africa where interactions with natural populations of Nile tilapia could ultimately impact existing biodiversity. Concerns over possible disease introductions through fish transfers led to the development of stringent quarantine arrangements which are still constantly under review.



It is our hope that the results of the GIFT project will live on in its legacies, namely:

- the demonstration that tropical aquaculture species respond well to selective breeding;
- the existence of a new strain of Nile tilapia suitable for many different farming environments;
- proof that biological diversity does matter in fish farming and fish selection;
- a small research collection of tilapia germplasm;
- many trained project partners and staff;
- an International Network for Genetics in Aquaculture which begins to bring fish genetics research and development more into the mainstream of aquaculture research and development;
- a pilot fish multiplication and dissemination institution - the GIFT Foundation International Inc.;
- contributions to science;
- contributions to understanding the different impacts of new fish strains on different socioeconomic groups in five Asian countries; and
- follow-on studies on the genetic improvement of carps.

None of this would have been possible without the unique GIFT partnership among researchers, government, development assistance agencies, farmers and hatchery owners.



MESSAGE

Trygve Gjedrem
Professor
AKVAFORSK, Norway

It is widely accepted that capture fisheries have reached their maximum potential of around 100 million tons per year. The increased requirement for aquatic protein in the future must therefore be supplied by aquaculture. It is estimated that to keep pace with the demands of the rapidly growing human population, aquaculture production will have to expand at a faster rate in the future as it has done during the successful period 1984-1996. By 2025 fish production from aquaculture will have to increase by 5% per year to meet the projected demand.

Aquaculture is similar in most respects to animal production on terrestrial farms. However, so far aquaculture has not adequately utilized selection techniques in breeding to increase productivity and improve product quality. It is well documented that systematic selection over a long period has greatly increased the production of farm animals and cultivated plants and also improved the utilization of nutrients and other resources. Today, it is impossible to imagine terrestrial farm animal production based on wild, unimproved stocks. Yet, the aquaculture industry continues to do exactly that. Even if the productivity of aquaculture operations can be increased substantially by better management and improved feeding practices, the future demand for fish cannot be met with these improvements alone.

The reasons why aquaculture has not focused on improving the breeds of farmed fish are diverse and, to some extent, due to less knowledge of reproduction of aquatic organisms compared to terrestrial animals. Twenty to thirty years ago it was believed that the prospects of genetic improvement in fish and shellfish were not as good as for farm animals and plants, since selection experiments with common carp in Israel and with tilapia in Israel, USA and Taiwan had not shown successful results.

Selection experiments with Atlantic salmon and rainbow trout were started in early 1970s in Norway to study the prospects of genetic improvement in fish. These extensive experiments showed a large genetic variation for all economic traits studied, with the response to selection for growth rate at 10 to 15 % per generation. Similar results were later shown in selection experiments with coho salmon and channel catfish in USA. However, the less promising results from the experiments with common carp and tilapia seemed to support the view that selection did not work in farmed tropical fish.

Given the imperative to develop a more efficient and productive aquaculture, it was necessary to investigate and document what could be accomplished through selection in tropical aquaculture species. In 1985, ICLARM contacted AKVAFORSK, the research institute in charge of the Norwegian salmonid breeding programs, to support a selection experiment with Nile tilapia. The two institutions worked out a research plan. UNDP recognized the strategic importance of the proposal and gave financial support, together with ADB, for three years. The GIFT project (Genetic Improvement of Farmed Tilapias) was started in 1988 at Muñoz, Philippines in co-operation with the Bureau of Fisheries and Aquatic Resources (BFAR) and the Freshwater Aquaculture Center of the Central Luzon State University (FAC/CLSU). The project was extended several times and was terminated in 1997 after 10 years of research. It is the most extensive fish breeding project ever carried out.

The GIFT project has provided valuable information which could lead to a revolution in tropical aquaculture production if applied to commercial operations. Substantial genetic variation was found for all economic traits studied and the response to selection for growth rate was, on average, 12-17 % per generation. Growth rate was doubled during the first 6 generations of



selection. These results are supported by more recent studies in which substantial response to selection has been reported for other tropical species, such as in rohu carp and shrimp, Penaeus vannamei.

From the early stages of the GIFT project, collaboration arrangements were established with tilapia farmers in the Philippines to carry out comparison tests between the tilapia strains traditionally used by the farmers and the improved GIFT tilapia. The superiority of the improved fish was confirmed under commercial conditions, and generated a demand in the industry for GIFT fingerlings.

The GIFT project has also led to the acceptance by scientists working with tropical aquaculture that systematic selection and domestication has great potential for increase productivity and improving efficiency. In 1993, UNDP supported an initiative from the GIFT project to form the International Network on Genetics in Aquaculture (INGA) to strengthen partnerships to advance the science of fish genetics and development of national fish breeding programs. At present, INGA has 13 member countries representing the bulk of the present tropical aquaculture production and a wide range of important farmed species and farming systems. In order to realize the opportunities offered by systematic selection and domestication, breeding programs should be started in many countries for every important farmed species.

With the assistance of INGA, GIFT tilapia has been distributed to and thoroughly tested in several Asian countries through the ADB funded project "Dissemination and Evaluation of Genetically Improved Farmed Tilapias (DEGITA)." Breeding plans have been designed both for tilapia and for other species. However, many INGA member countries, lack the funding to develop their own breeding program. Since the majority of aquaculture production takes place in developing countries, foreign aid could be used to help establish such breeding programs.

Alternatively, breeding programs could be established by farmer's co-operatives or private investors assisted by experts in quantitative genetics, as has been done for many farm animal species. However, the termination of the GIFT project has once again proved that the commercialization of breeding programs in aquaculture is a rather difficult exercise. Time and patience are required to establish and to run an economically viable breeding program. Necessary testing facilities need to be developed, programs for processing field data and estimation of breeding values must be generated and a marketing organization has to be built. The Norwegian salmonid breeding programs were operated through the first 14 years by a research institute (AKVAFORSK), partly based on research funding, and then for another 8 years in collaboration with national fish farmer's organizations, before it was completely commercialized in 1992. The GIFT project legacy is facing the challenge to establish a sustainable commercial base, after 10 years of existence during which it enjoyed the solid support from UNDP and the project partners.

To conclude, the GIFT project has proved the great potential of selection and domestication of farmed tropical fish in the efforts to face the future requirements of rapid increases in aquaculture production. The project has also encouraged scientists in several major tropical aquaculture countries to focus on the development of selection programs for farmed fish. The challenge will be to secure financial support for this and to create mechanisms for successful commercialization of research based selection projects.



MESSAGE

Graham A. E. Gall
Chairperson, Expert Advisory Panel to the GIFT Project
University of California, Davis

The GIFT project was the first successful effort to demonstrate for Asia, the appropriateness of traditional animal breeding technology for the breed improvement of fish populations. The project provided leadership in extending this knowledge to many countries now committed to genetic improvement of tilapia. The potential result will be a profound impact on the volume and efficiency in the production of a primary source of protein. I had the pleasure of serving as the chair of an expert advisory panel charged with advising the United Nations Development Programme, the sponsor of the project. The vast volume of research results of the project are beginning to appear in the scientific literature. These papers, along with the reports prepared by the project, will allow anyone with an interest in the breed improvement of tropical fish full access to the protocols, successes, and failures experienced by the research team. However, for this old scientist, the greatest reward was observing, and being able to make a small contribution to, the enormous energy and commitment exhibited by a group of young scientists pushing the project to completion. It was the effort of this group of individuals, who worked endlessly day after day, that made the GIFT project more than a scientific and technological success. It was a demonstration to all agencies and groups funding development projects that success comes about, not by the imposition of ideas from without, but by encouraging and supporting young minds at the local level. The young scientists associated with the GIFT project were committed to making a difference, regardless of how small, and to sharing their experiences with anyone interested in listening.

One highly innovative and productive approach used by the research team was to actively involve tilapia farmers in the Philippines to carry out comparison tests between the tilapia strains traditionally used by the farmers and the improved GIFT tilapia. Not only did the effort establish the superiority of the genetically modified fish, it, more importantly, demonstrated to the local farmers that genetic improvement would enhance their productivity and success. This experience then was easily transmitted to officials and farmers in other countries. The outcome was an overwhelming respect for breed improvement programs, an appreciation of the effort of the research team, and a realization that genetic improvement program results in direct benefits to the fish farmers.

The project also showed clearly that just the mere presence of a group of scientific experts to provide assistance to local "workers" does not guarantee success. Rather, a lot is required from both parties. There must be a high level of respect on the part of both the experts and the local scientists. There must be a commitment on the part of the experts to let the local scientist grow and mature. And finally, there must be continuous, if often intense, discussions every step of the way. The Norwegian group of experts from AKVAFORSK exemplified this caring and sharing attitude, even if there were difficulties at times. I like to feel also that the Expert Advisory Panel system chosen by UNDP made a significant contribution. The panel appeared to function as an effective intermediary, in addition to its expected scientific role, so communications among the many parties involved in the project was interpreted, translated, and circumscribed for all involved. UNDP must be complemented for this outward looking attitude.

The GIFT project was a scientific success. It has demonstrated the great potential of selection and domestication of farmed tropical fish. It provided stimuli for young scientist to follow their beliefs and ambitions. It provided leadership that will be invaluable to other young scientists facing the challenges and demands for increased aquacultural production. The development of selection programs for farmed fish by a wider array of young scientists in many developing countries is the challenge for the future.



A TRIBUTE TO KEY CONTRIBUTORS

Ambekar E. Eknath
ICLARM Senior Scientist and GIFT Project Leader
(March 1, 1988 to December, 31, 1997)

Over 500 individuals from different walks of life - international donor representatives, politicians, planners, NGO representatives, scientists, farmers and consumers --- have contributed to the development and implementation of the GIFT Project and the associated initiatives. It has been a great decade-long learning experience.

At the very outset, I wish to express my sincere appreciation and gratitude to my colleagues in the GIFT Project who have been largely responsible for the evolution of the GIFT Project concepts and also for my personal development. As a team, we have braved many challenges, including some natural calamities (earthquakes, typhoons, and Mount Pinatubo eruption). I also, in behalf of the entire GIFT Project core staff, wish to record our appreciation to the Executive Management of the collaborating institutions and the management of UNDP/DGIP and UNOPS for their support throughout the execution of this project.

The commitment of our partners in the Philippine national institutions (BFAR and CLSU) to the objectives of the GIFT project has been extraordinary, a fact that finds a pride of place in all annual External Advisory Panel review reports. I can confidently say that the complex range of project activities could not have been implemented anywhere else in the world! The value of in-kind contribution (staff time, facilities, feeds and fertilizers, and other major operating expenses) of BFAR and CLSU probably matches the total cash contribution of UNDP. They have remained goodwill ambassadors of the project in all national and regional level meetings. They have enthusiastically shared the products of GIFT research (germplasm and results) with all Asian countries.

My intention here is to single out for special mention a few individuals, **outside of the core GIFT Team**, who, in my view have played a significant role in the evolution of the GIFT concept and for their unfailing support in implementing this complex project.

Preparation of research project proposals and putting together a team of scientists is a task that do not just happen overnight. All the necessary background work, spanning nearly a decade, from 1978 to 1988, that eventually led to the development of the first successful proposal to the Asian Development Bank, was done almost single-handedly by Dr. Roger S. V. Pullin of ICLARM. The initial research plans including the guiding principles of collaboration among the partner institutions were laid down as early as 1987 by Roger Pullin and Prof. Trygve Gjedrem of AKVAFORSK. Trygve Gjedrem has been a source of inspiration for the entire team throughout the GIFT project period.

As the GIFT Project started to metamorphose from a relatively small research project during the Phase I into a major research, training and national capacity strengthening initiative during 1993-1994, the Project Staff were confronted with substantial changes in the management structure at ICLARM, and also in other collaborating institutions. The arrival of Meryl J. Williams as the new Director General of ICLARM paved the way for strategic changes in the operations of the GIFT Project and in redefinition of collaborative linkages among partners on a professional footing. The structure of annual reporting of progress and preparation of workplan and budget for UNDP and EAP reviews were substantially improved through her personal intervention. Dr. Williams also paved the way for rapid development of new initiatives for ICLARM, namely, the DEGITA, the INGA, the GICA, and the GIFT Foundation International Inc.



Dr. Williams will always be remembered by the GIFT core staff for her personal touch in all her communications and her sincere concern for their welfare.

At UNDP, Dr. Alva App has been involved with the GIFT Project ever since its formative days and was largely responsible for securing UNDP funds for the first phase. The collective wisdom of Dr. Alva App, Dr. Nyle Brady, Mr. Philip Reynolds, and Dr. Timothy Rothermel, was instrumental in establishing the INGA. These gentlemen will be remembered for their defense during annual UNDP budget hearings against any budget cuts for the GIFT Project. Much of the debate against the research strategies followed in the GIFT Project, especially, during the initial GIFT Phase I, were cleared by UNDP through organization of independent external reviews. All GIFT staff owe a sense of deep gratitude to Phil Reynolds for organizing these reviews, for instituting a permanent "External Advisory Panel" to review the annual GIFT Project workplans, and for his moral support throughout.

The most difficult role of conducting annual review of project activities and providing clear directions to the project was shouldered by the External Advisory Panel consisting of Prof. Graham A. E. Gall (University of California, Davis), Dr. Brian Davy (SIFR / IDRC, Canada) and Dr. Bernard Chevassus (INRA, France) and Dr. Vo Tong Xuan (University of Cantho, Vietnam). They have, in addition, played the critical role of facilitating relationships among collaborating partners and individuals; and also in advising on research strategies, directions, and priorities. The wide spectrum of activities of the GIFT Project would not have been implemented without the technical support and directions provided by the Panel.

The various independent initiatives developed under the auspices of the GIFT Project needed experienced pro-active people to lead them. I record here my sincere appreciation and gratitude to: Dr. Madan Dey for his leadership and direction to the highly successful DEGITA Project; to Dr. D. V. Seshu for initiating the INGA Network, based on his 20 years of experience in leading the highly acclaimed INGER Network at IIRI; to Dr. M. V. Gupta for developing sound strategies and research proposals to continue the INGA; to Mr. Basilio M. Rodriguez Jr. for assuming the role of Executive Director of the GIFT Foundation International Inc. and for developing business strategies and research priorities; and to Dr. Rex Dunham for taking the responsibility of leading the GICA Project and for his able leadership to the ICLARM GEB Program.

Finally, with some notable exceptions to the rule mentioned above, I must express my gratitude to Dr. Hans Bernard Bentsen of AKVAFORSK, for being there at all times. Hansi has been a great sparring partner, a friend, and a guide throughout my tenure as Project Leader of GIFT. I am also grateful to Dr. Morten Rye of AKVAFORSK for his dedication and assistance in organizing the entire GIFT database and for initiating the data analysis in preparation for the major scientific publications to follow.

I sincerely appreciate the efforts of Dr. Rex Dunham for securing and maintaining the data sets and for carrying out data analysis from several strategic experiments. Dr. Dunham and Dr. Gupta have edited the initial drafts of this report.

Of course, like any leader should, I assume the responsibility for opportunities missed and for any strategic errors that may have occurred.

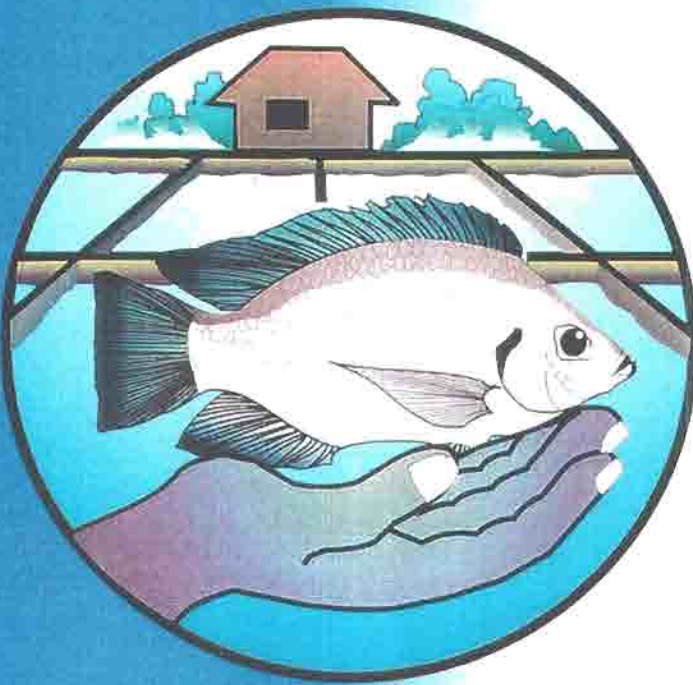


LIST OF ABBREVIATIONS

ACIAR	- Australian Centre for International Agricultural Research
ADB	- Asian Development Bank
AKVAFORSK	- Institute of Aquaculture Research of Norway
BFAR/NFFTRC	- Bureau of Fisheries and Aquatic Resources/National Freshwater Fisheries Technology Research Center
CIFA	- Central Institute of Freshwater Aquaculture (India)
CGIAR	- Consultative Group of International Agriculture Research
DEGITA	- Dissemination and Evaluation of Genetically Improved Tilapia in Asia
EAP	- External Advisory Panel
FAC/CLSU	- Freshwater Aquaculture Center of the Central Luzon State University
GEBP	- Germplasm Enhancement and Breeding Program of ICLARM
GFII	- GIFT Foundation International Incorporated
GICA	- Genetic Improvement of Carp in Asia
GIFT	- Genetic Improvement of Farmed Tilapia
IAB	- Institute of Aquatic Biology (Water Research Institute) of Ghana
ICLARM	- International Center for Living Aquatic Resources Management
IDRC	- International Development Research Centre of Canada
INGA	- International Network on Genetics in Aquaculture
INGER	- International Network for Genetic Evaluation of Rice
INRA	- Institut National de la Recherche Agronomique (France)
IRRI	- International Rice Research Institute
NORAGRIC	- Norwegian Centre for International Agricultural Development
PMC	- GIFT Project Management Committee
PNTBP	- Philippine National Tilapia Breeding Program
RIA	- Research Institute of Aquaculture No. 1; No. 2 of Vietnam
RICF	- Research Institute for Coastal Fisheries
RIFF	- Research Institute for Freshwater Fisheries
ROS/DA	- Research Outreach Station of the Department of Agriculture (Philippines)
RSCF	- Research Station for Coastal Fisheries (Indonesia)
SIFR	- Strategy for International Fisheries Research
SINGER	- Systemwide Information Network on Genetic Resources
TAC	- Technical Advisory Committee
TPR	- Tripartite Review of GIFT
UNDP/DGIP	- United Nations Development Programme/Division for Global and Interregional Programmes
UNDP/SEED	- UNDP/Sustainable Energy and Environment Division
UNDP/UNOPS	- UNDP/United Nations Office of Project Services
UPMSI	- Marine Science Institute of the University of the Philippines



**The GIFT Project and the four major independent projects/initiatives
established by ICLARM and its partners as a result of the emerging lessons
and findings of the GIFT Project.**



The GIFT Project and the four major independent projects/initiatives established by ICLARM and its partners under the auspices of the GIFT Project.

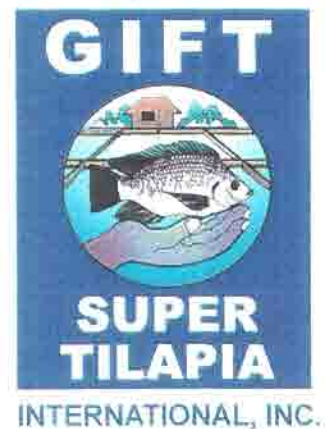


DEGITA

Dissemination and Impacts



GIFT FOUNDATION





The Genetic Improvement of Farmed Tilapias (GIFT) Project:

is a collaborative research project implemented by ICLARM in cooperation with NFFTRC/BFAR, FAC/CLSU, UP/MSI and AKVAFORSK. It was executed in two phases: Phase I from March 1988 to July 1992; and Phase II from January 1993 to December 1997. It was based in the Philippines with outreach to 8 Asian countries.

GIFT Phase I was co-financed by the Asian Development Bank through a Regional Technical Assistance Grant to ICLARM (ADB RETA No. 5279) and the Division for Global and Interregional Programmes of the United Nations Development Programme (UNDP Project Nos. INT/99/019 and INT/91/005/1/01/42).

GIFT Phase II represents the evolution of the GIFT project from a relatively small research project during Phase I into a major strategic research, training, and national capacity strengthening initiative in germplasm enhancement and Breeding. It was financed by the UNDP/DGIP (Project No. GLO/90/016), ICLARM core budget, and participating national institutions.

Although executed in two phases, the GIFT Project as a whole represents one major ICLARM program with three major themes: (a) evaluation of Nile tilapia genetic resources and their utilization in national breeding programs; (b) dissemination and monitoring of the adoption of improved Nile tilapia breeds with regard to their potential impacts on equity, the environment, and biodiversity; and (c) human resources development and establishment of a forum for national institutions to exchange methods and genetic materials.

Dissemination and impact assessments was carried out through a separate parallel initiative known as the DEGITA Project. The International Network on Genetics in Aquaculture (INGA, see below), established in 1993, focuses on providing assistance to National institutions in developing breeding plans and in exchange of genetic materials. The GIFT Foundation International Inc. was established almost towards the end of the Project to continue the selective breeding research initiated by the GIFT Project in partnership with institutional partners of the GIFT Project and the private sector in the Philippines. All 18 GIFT Project core staff members are employed by the GIFT Foundation.

The Dissemination and Evaluation of Genetically Improved Tilapias in Asia (DEGITA) Project: was developed by ICLARM to: (a) disseminate the improved germplasm, the "GIFT Strain", developed by the GIFT Project to countries in the region requesting for it. Dissemination has followed rigorous quarantine procedures in consonance with the International Codes of Practices in Fish Transfers; (b) assess ex ante the impacts of improved genetic material on social equity, environment and the biodiversity.

The DEGITA project was implemented by ICLARM from June 1994 to June 1997 in cooperation with national institutions in Bangladesh, People's Republic of China, the Philippines, Thailand and Vietnam. It was funded by the Asian Development Bank (RETA grant No. 5588), ICLARM Core funds, and the participating national institutions.

The International Network on Genetics in Aquaculture (INGA): was established in July 1993 as a forum for exchange of knowledge, research methods, and where applicable, genetic materials among participating countries. The present focal species are tilapias and carps. Thirteen countries in Asia and Africa are collaborating in research and exchange of genetic materials: Bangladesh, Peoples Republic of China, Cote d'Ivoire, Egypt, Fiji, Ghana, India, Indonesia, Malawi, Malaysia, the Philippines, Thailand, and Vietnam. ICLARM is the member-coordinator. INGA was established with funds from UNDP. It is now operating mainly through funds from ICLARM Core and supplementary grants from the Canadian IDRC and the Norwegian Ministry of Foreign Affairs.



The GIFT Foundation International Inc.: was established in 1997 by the institutional partners of the GIFT project as an independent non-stock non-profit organization to undertake the commercialization of the GIFT strain developed by the GIFT project through partnerships with the private sector. Commercialization of the GIFT strain was chosen as a strategy to generate the funding required to continue the selective breeding and research on the genetic materials assembled by the GIFT project and to sustain the "GIFT legacy" comprising the highly trained and dedicated GIFT project core staff members (18) in the Philippines, the improved GIFT strain, the gene banks, and the data base meticulously gathered over the past 10 years.

The GIFT Foundation at present distributes the GIFT strain using a network of private hatcheries that become members of the Foundation through a formal licensing agreement with the Foundation. This formal agreement includes provisions obligating the participating hatcheries to make regular research and development contributions to the Foundation in exchange for the privilege of using the Foundation breeders and the Foundation trade name and trademark. The Foundation became operational in January 1998 and operates its breeding and production facilities in the former GIFT Project site now called the "Center for Applied Fish Breeding and Genetics", built with funds from the UNDP, located in the campus of the CLSU. The GIFT Project partners have formally agreed to allow the Foundation rent-free use of these facilities for the first five years of its operations.

The Genetic Improvement of Carp Species in Asia (GICA) Project: was developed by ICLARM following the successful implementation of the GIFT and the DEGITA Projects and in response to the request from the carp producing countries in Asia to initiate a GIFT-type strategic research and training program on Asian Carps. The project became operational in June 1997 and is being implemented by ICLARM in cooperation with national institutions in Bangladesh, People's Republic of China, India, Indonesia, Thailand and Vietnam. During its present Phase I (June 1997 to May 2000), the focus is on: (a) assessing the current status of carp genetic resources in Asia, including their systematic documentation and evaluation; (b) establishing strategic research partnerships and networking arrangements; (c) development of criteria for prioritizing research areas, species, farming systems and breeding strategies. The GICA Project is funded by the ADB RETA No. 5711, ICLARM Core funds, and participating national institutions.



EXECUTIVE SUMMARY

Challenges and Development Context

The GIFT Project was established to address three challenges. First: Could the application of basic principles of genetics and applied breeding technology, as practiced for centuries in plant and animal production, to the emerging aquaculture industry result in enhanced production efficiency? This challenge was highly relevant within the context of: (i) stagnating fish production from capture fisheries (direct harvests from marine and inland waters) and aquaculture (farming of aquatic animals and plants) is increasingly seen as a panacea for the looming gap between the demand for and supply of food fish; (ii) glaring lack of conscious application of genetics to tropical aquaculture, even though the tradition of fish farming dates back hundreds of years. Consequently, the fish stocks used in aquaculture are generally similar to wild undomesticated stocks and in some situations there is strong evidence for genetic deterioration. In fact, before the GIFT Project, it had been widely assumed that genetic intervention will require extensive resources and, therefore, should be considered as an option for the future.

The second challenge pertains to the dissemination of improved fish breeds, once they are developed. It was understood right from the beginning of the program that genetic improvement and development of improved breeds was only half of the story. The other crucial aspect is their dissemination to reach the targeted beneficiaries effectively, including monitoring of the impact and adoption of improved breeds. Management of and access to germplasm are also becoming increasingly important issues, under the new Convention on Biological Diversity, which went into force in 1993. It was critically important for a nascent scientific discipline of applied fish breeding that it consciously and responsibly derive guiding principles from the complexities unique to aquaculture in a developing country context and also from the social lessons learned from the green revolution in crop production.

Ways and means of catalyzing the development of applied fish breeding programs in countries that needed them most was the third challenge addressed by the GIFT Project. Relevant issues in this context included, objective assessment of needs and opportunities for developing applied fish breeding programs in each participating country; prioritization of species and farming systems; and development of the most viable mechanisms for international cooperation and provision of fora for exchange of research methods and where necessary, genetic materials. ICLARM had already declared its intention to adopt and apply internationally accepted codes of practices for responsible transfer of germplasm in relation to the GIFT Project.

The GIFT Project was implemented by ICLARM in cooperation with the National Freshwater Fisheries Technology Research Center of the Bureau of Fisheries and Aquatic Resources (NFFTRC-BFAR), the Freshwater Aquaculture Center of the Central Luzon State University (FAC-CLSU), the Marine Science Institute of the University of the Philippines (UP-MSI) and the Norwegian Institute of Aquaculture Research (AKVAFORSK).

Objectives

The three challenges described above formed the three immediate objectives of the GIFT Project, namely: (a) to develop improved breeds of Nile tilapia and provide those fish breeds to national testing programs and thence to the fish farmers; (b) to strengthen national institutions in aquaculture genetics research; and (c) to establish a mechanism for international exchange and evaluation of improved breeds and research methods. The overall development objective was to increase the quantity and quality of protein consumed in low income rural and urban populations in tropical developing countries in all regions of the world and increase the income of low-income



producers. As with future agricultural and aquacultural developments, the objective was to aim for sustainable systems, in harmony with social equity, the environment and biodiversity.

Nile tilapia (*Oreochromis niloticus*) was chosen as a prominent test species because of their growing importance in aquaculture, their omnivorous diet which suits low cost aquaculture, and their short generation time of about 8 months, making them an excellent model species for investigating the application of genetics in aquaculture, from conservation of genetic resources to breeding programs. The research thrust was to develop methodologies and policy guidelines that can be applied or adapted to a wide range of finfish species in national fish genetic improvement programs.

Sponsors and Implementation

The Project was implemented by ICLARM and its partners in two consecutive phases from March 1, 1988 to December 31, 1997. The Phase I (March 1988 to June 1992) was co-financed by ADB and the UNDP/DGIP and the Phase II (January 1993 to December 1997) was financed by UNDP/SEED and ICLARM Core funds.

The stated objectives and plan of activities of the GIFT project has evolved over the years. The Project, as a whole, is a concerted effort on the part of ICLARM and its partners to build on the successes and experiences of various other initiatives in the region. The Project research plan was developed following the successful experiences of AKVAFORSK with salmon and trout breeding programs in Norway. The expertise required to implement the project was secured by bringing together institutions with experience in: (a) tilapia farming (BFAR, FAC, ICLARM); (b) documentation and characterization of tilapia genetics resources (UPMSI and ICLARM); and (c) application of genetic improvement strategies in fish (AKVAFORSK and ICLARM). The research was carried out in strengthened research facilities in FAC-CLSU and NFFTRC-BFAR. The Project plans were circulated for peer-review before implementation.

To facilitate proper implementation of the complex range of activities nested within the stated objectives of the GIFT Project and in response to the challenges and opportunities presented by the donor community, ICLARM, under the auspices of the GIFT Project, and very early during the GIFT Phase II, developed: (a) the Dissemination and Evaluation of Genetically Improved Tilapia Species in Asia (DEGITA) Project with principal focus on the dissemination of the improved Nile tilapia germplasm and *ex ante* evaluation of impact on equity, environment and biodiversity; and (b) the International Network on Genetics in Aquaculture (INGA) as a forum for the participating national institutions to exchange research methods, experiences and germplasm. Thirteen countries in Asia and Africa are collaborating in research and exchange of genetic materials: Bangladesh, Peoples Republic of China, Cote d'Ivoire, Egypt, Fiji, Ghana, India, Indonesia, Malawi, Malaysia, the Philippines, Thailand, and Vietnam. ICLARM is the member-coordinator. INGA was inaugurated in 1993 by seed funding from the GIFT Project and has been supported since by ICLARM core funds and project funds from the Government of Norway, ACIAR and IDRC, as well as operational funds from the member countries.

Main Results and Outputs

The main focus of this report is on the results and outputs from the GIFT Project implemented by ICLARM and its partners since 1988. The major findings from the ADB funded DEGITA project is also included to demonstrate the two major themes addressed by ICLARM under the auspices of its rapidly evolving Germplasm Enhancement and Breeding Program: (a) evaluation of Nile tilapia genetic resources and their utilization in national breeding programs; and (b) monitoring of the adoption of improved Nile tilapia breeds with regard to their impact on equity, the environment and biodiversity.



Development of improved breeds of Nile tilapia has involved: (a) assembling of tilapia germplasm from their wild habitats in Africa and from "domesticated" Asian aquaculture stocks; (b) estimation of genotype x environment interaction in diverse Asian farming systems; (c) a complete diallel crossing experiment to determine the magnitude of heterosis and thereby the appropriate breeding strategy – pure-breeding or crossbreeding; (d) establishment of a base population and initiation of selection to improve growth performance.

The program began with consultations with fish farmers and experts from various disciplines. The relative economic importance was considered for each of the diverse tilapia farming systems including the backyard fishpond, rice-fish culture, cage culture and some more intensive systems. A grow-out period of about 90 days (from the fingerling stage when the fish are about 3-7g to their harvest weight of about 120 g) was chosen as a representative production cycle. Standardized research methods were then developed for mating, mass marking of individual fish, sampling, anesthesia, recording of traits and pedigree, management of databases, statistical analyses, and quarantine and safe dispersal of fish.

Assembling Nile tilapia germplasm. While the global wealth of tilapia genetic resources are restricted to Africa and the Levant, the main aquaculture industries are at present in Asia. The established farm stocks in Asia derive from very small founder stocks, and possibly suffer from inbreeding and widespread introgression of genes from other less desirable feral tilapia species. ICLARM and its partners collected wild Nile tilapia germplasm from Egypt, Ghana, Sénégal and Kenya. Philippine commercial Nile tilapia strains were also gathered, giving a total of eight strains for study. These have been described using biochemical markers and truss morphometrics. Spermatozoa from founder stocks and from subsequent generations are being cryopreserved.

Genotype x Environment interaction (GE): The most important criterion determining the number of strains to be developed is their relative performance in different target environments. In genetic parlance, this is called genotype x environment interaction. If there is little or no GE, then the best strains in one environment will be the best in most or all environments. On the other hand, if there is high GE then special strains must be developed for specific environments.

The GIFT project's first major experiment was to determine the growth performance of strains and the magnitude of GE over two consecutive generations. The 11 test environments chosen covered a wide range of tilapia farming systems: earthen ponds fertilized with organic manure, ponds fertilized with on-farm agricultural residues, rice-fish systems, cages, and hatcheries located in different agro-climatic regions. A total of 11,000 individuals (7650 and 3420 individuals during first and second generation) were communally reared in all test environments for 90 days. Results indicated highly significant differences in growth performance among the strains. Except for Ghana strain, the wild African strains performed as well as or better than the 'domesticated' strains used by the Philippine farmers. The widely cultured Israel strain in the Philippines performed poorly. The GE was low. Overall, the growth performance of the strains were relatively consistent across test environments. It was concluded that development of specialized strains for each of the farming systems in the Philippines may not be necessary. However, the GIFT plan incorporates testing of breeding materials across all target environments to ensure that the response to selection will be valid in almost all grow-out environments.

Breeding strategy. The major alternative breeding systems are selection and crossbreeding. Selection as a breeding strategy focuses on additive genetic variance – choosing the best performing individuals as breeders in every generation. Selection programs are in general resource efficient, simple in design and easier to implement than crossbreeding programs, and the genetic gain is sustainable during generations of reproduction outside the breeding system. By nature, the selection approach is 'non-protective'. Crossbreeding focuses on non-additive genetic variance – choosing the right combination of strains/ individuals. The hybrid vigor (or heterosis) should be substantial to adopt crossbreeding as a breeding strategy.



The GE experiment performed during the first generation experiments was followed by a complete diallel crossing experiment producing all 64 hybrid and pure strain combinations among the eight founding strains to estimate the magnitude of heterosis. Growth performance was recorded in a total of 23,739 individually tagged progeny after a grow-out period of about 90 days. Least square means (LSM) of body weights at harvest were computed for each strain combination within and across all test environments, and additive genetic effects, maternal (reciprocal) effects, and non-additive genetic effects (heterosis) were estimated. The LSM heterosis of all strain crosses across all test environments was 4.3 %, and the cross that expressed the largest non-additive genetic effect showed a least square mean heterosis of 14 %. However, only seven out of the 22 crosses with significant heterosis performed better than the best pure strain, and the largest gain was then about 11%.

Although some strains performed better as sire strains and others as dam strains, it was concluded that specific line or strain crossing schemes will result in only marginal improvements in growth performance and may show an increased sensitivity to GE interactions. Therefore, a regular selection program based on additive genetic performance was adopted for rapid genetic improvement of Nile tilapia in the Philippines.

Building the base population and initiating selection: Based on their growth performance of 64 different strain combinations, individuals from the 25 best performing purebred and crossbred groups were selected to build a base population with a broad genetic base and initiate a selection program, with a combined family and within family selection strategy. Another major experiment was set up to estimate additive genetic parameters. 50 sires were mated with 150 dams in a hierarchical mating scheme. About 25,000 individually tagged fingerlings were grown for 90 days in seven test environments.

The heritability estimates for body weight at harvest were consistent across test environments. The estimates based on sire and dam components of variance, respectively, were 0.23 and 0.53; heritability for survival based on the sire component of variance was 0.08; and the half-sib correlation between body weight at harvest and survival was 0.2. The observed response after the first generation of selection in the synthetic GIFT strain, estimated as a difference between progeny of individuals with average breeding values and those from selected individuals, was 26%. In on-station trials, the synthetic GIFT strain was about 60% heavier at harvest than the 'Israel' strain.

In every successive generation, about 20,000 individually tagged fingerlings from 120-183 selected full-sib families (within 100 half-sib families) have been evaluated in a variety of test environments. The genetic gain per generation across five generations of selection for growth performance has been about 12-17%. The accumulated genetic gain in relation to the base population was 85%.

Several strategic experiments initiated during the GIFT Project are now being carried out by the GIFT Foundation (see below). These include: evaluation of GE in specific target environments such as cages, rice-fish and earthen ponds; genotype x season interaction; procedures for developing late-maturing Nile tilapia; and procedures for a multi-trait selection plan involving growth, survival and late maturation.

Regional On-Farm Trials And Early Impacts

Under the **DEGITA Project**, the production potential of GIFT and existing Nile tilapia strains (referred as 'non-GIFT' strains because of their diversity of origins) were assessed through on-station and on-farm experiments in a range of aquaculture systems in Bangladesh, Peoples Republic of China, the Philippines, Thailand and Vietnam. GIFT fish were distributed after one generation of selection.



Production per unit area or yield depends on stocking density, average weight at harvest and survival. After accounting for the wide heterogeneity of input use levels, production environments and other factors, the percent change in average weight at harvest and survival due solely to the mean superiority of the GIFT strain over the existing non-GIFT strains were highly significant across different countries. The magnitude of the effect of the GIFT strain on growth rate, however, varied among countries, ranging from 18% in China to 66% in Bangladesh. In terms of survival, the GIFT Strain showed higher survival compared to the existing strains only in China (3%) and the Philippines (13%). The magnitude of impact, however, varied according to the history of tilapia introductions and farmers' experience. Overall, the results indicated that the cost of production per unit of fish produced is lower for the GIFT strain than for the non-GIFT strains: more than 30% lower in Bangladesh and the Philippines; about 20% lower in Peoples Republic of China, Thailand and Vietnam.

The estimated yield potential of the GIFT strain was more than 50% higher than that of the best existing strains, ranging from about 54% percent higher in Vietnam to about 97% higher in Bangladesh. The estimated Yield Gap II, the difference between potential and actual yields in farmers' fields, ranges from around 3000 Kg ha in China to around 850 Kg ha in the Philippines (which has a shorter growing period and poorer production environment). These estimated gaps show the extent to which yield might be increased through the invention of appropriate technology to overcome production constraints such as disease, water quality, etc.

In the Philippines and Thailand, where about 80-90% of the tilapia production is marketed locally and the demand for tilapia is price elastic, both producers and consumers will benefit from technological interventions (such as the GIFT technology). As low-priced tilapia is consumed mainly by poor people whose demand is inversely related to price, the major portion of the consumers' benefit will go to poor consumers. In countries like Bangladesh, where tilapia fish farmers consume about 70% of their produce, the major economic gain from GIFT technology will accrue to the producers. In subsistence farming, the reduction in market price due to higher production has relatively little influence on producers' income.

The participating countries in the DEGITA project are now in the process of developing their respective national tilapia breeding programs with the GIFT strain as a base starting genetic material.

The GIFT Foundation

ICLARM and its partners established the GIFT Foundation as an independent non-stock non-profit organization primarily to continue the selective breeding research on Nile tilapia initiated by the GIFT Project and to undertake commercialization of the GIFT strain through partnerships with the private sector. Commercialization of the GIFT strain was chosen as a strategy to generate the funding required to continue the selective breeding and research on the genetic materials assembled by the GIFT project and to sustain the "GIFT legacy" comprising the highly trained and dedicated GIFT project core staff members (18) in the Philippines, the improved GIFT strain, the gene banks, and the data base meticulously gathered over the past 10 years. The Foundation continues to maintain strategic research partnerships with the GIFT Project institutions.

The GIFT Foundation at present distributes the GIFT strain using a network of private hatcheries that have become members of the Foundation through a formal licensing agreement with the Foundation. This formal agreement includes provisions obligating the participating hatcheries to make regular research and development contributions to the Foundation in exchange for the privilege of using the Foundation breeders and the Foundation trade name and trademark. The Foundation became operational in January 1998 and operates its breeding and production facilities in the former GIFT Project site now called the "Center for Applied Fish Breeding and Genetics", built with funds from the UNDP, located on the campus of the CLSU. The GIFT Project



partners have formally agreed to allow the Foundation rent-free use of these facilities for the first five years of its operations.

Conclusions and Recommendations

The selective breeding program for Nile tilapia developed and implemented by ICLARM and its partners is the first systematic attempt to apply the principles of animal breeding theory to tropical aquaculture in a developing country setting with the primary objective of demonstrating: (a) the importance of genetics in improving production efficiency in aquaculture, (b) methods and mechanisms for dissemination of results and products to reach targeted beneficiaries effectively, and (c) methods to estimate *ex ante* the potential benefits to a wide spectrum of aquaculture enterprises. Successful demonstration could mean not only better tilapia breeds for fish farmers but also new breeding procedures for use with other species. ICLARM and its partners have also proceeded a step further in developing the non-stock non-profit GIFT Foundation International as a mechanism for continuation of the GIFT Program for the benefit of the tilapia industry in the Philippines and elsewhere. Depending on the structure of the tilapia farming industry in other countries, other mechanisms may be more appropriate in those countries.

The program has demonstrated the enormous gains in economic performance of farmed tilapia that are possible from a systematic selection program. The GIFT program has stimulated planning of and implementation of national fish breeding programs in Bangladesh, India, Indonesia and Vietnam. An International Network on Genetics in Aquaculture (INGA) was established in 1993. Bilateral and multilateral research initiatives under the INGA have begun to include other fish species, especially the carps which account for most of the finfish production in Asia. A collaborative research and training project on the genetic improvement of carps in Asia (GICA) is already in progress with funds from ADB.

ICLARM is now well poised to establish itself as one of world leaders in Fish Germplasm Enhancement and Breeding. The decade long experience and the significant lessons learned in implementing the GIFT and the DEGITA projects, the research forum offered by the GIFT Foundation, the INGA and the GICA initiatives, mark the beginnings of exciting research and development opportunities for ICLARM. ICLARM's unique position in the Consultative Group on International Agricultural Research as the only center devoted to fisheries research also brings with it new challenges. In executing its new responsibilities, ICLARM should build on and must forge new and innovative working relationships with national fisheries centers.

The selective breeding technology generated through the GIFT Project is not an end in itself. It is in many ways the beginning of new research challenges and directions for ICLARM and its partners. Some of the research challenges emanating from the GIFT Project are as follows:

The need for systematic documentation and conservation of tilapia genetics resources was felt very strongly throughout the course of GIFT project implementation. It was also recognized that there exists a vast but scattered body of local knowledge about fish genetic resources and habitats which also are in as urgent need of conservation as the germplasm itself. The ICLARM's FISHBASE program is a step in the right direction. This activity, however, should be strengthened and could benefit through appropriate partnerships with projects and agencies which are currently engaged in such activities.

The development of GIFT Manual of Procedures Volume 1, a training manual of GIFT Project techniques based on TRAIN-X methodology, has been completed and will be distributed at the next INGA Steering Committee meeting in March 1999. It is expected to directly benefit the targeted users (research partners from INGA member countries). However, the reproduction and distribution of this document would still require financial assistance. The preparation of GIFT Manual of Procedures Volume 2, more specifically focused on development of tilapia breeding



programs, is also important but compilation of this manual must follow after the publication of the core text material (scientific papers on project's major experiments) and identification of future financial resources. Given these constraints, it is essential that either ICLARM or INGA seek donor's support to continue the activity.

The main bottlenecks experienced in the GIFT breeding program are: (a) the amount of time required in producing the required numbers of families; (b) the time required to rear these families to a size when they can be tagged; and (c) the number of families that can be effectively tagged; and (d) the tag-losses experienced during rearing. Although corrections for the effects of various factors (initial size, initial age, etc.) affecting the early life history through novel statistical procedures and manipulations were possible, it is important that procedures for synchronous maturation of tilapia breeders and more efficient tagging techniques be developed. DNA fingerprinting is one possible technique of enhancing the efficiency of the selection program.

The focus of the GIFT program has been on a relatively simple but extremely important breeding objective: growth rate, with recording of other traits such as survival, occurrence of diseases, and maturation rate. The purpose of this exercise was to demonstrate the effectiveness of a properly planned selection program. In future experiments, it is important that the breeding objectives (in relation to the targeted species) are carefully examined. This should necessarily include a formal study on the benefits to the fish farmers arising from improvements in each trait.

The rigorous genotype x environment experiments conducted under the GIFT Program did not suggest a need for the development of several parallel strains for different production systems. However, it should be recognized that the rice-fish farming system has received very little attention in the GIFT project and that all environments tested were in the Philippines where temperature changes were limited. It is possible that the importance of genotype x environment may emerge later, particularly when the GIFT strain is used in other countries.

The GIFT breeding nucleus should remain open in the medium term. There should be continuous comparative evaluations to screen potential genes for possible inclusion in the on-going selection program, and if the potential benefits from such genetic mixing will outweigh setbacks. Sex manipulation procedures (hormonal sex reversal, ploidy induction, or YY-male technologies) are important in the distribution process.

As economies are transformed and agriculture in general becomes more commercialized, the private sector is also expected to play an increasingly important role in technology generation, in food production, and in ensuring food security. For ICLARM to advance its strategic research agenda of improving productivity in aquaculture to close the looming gap in demand for and supply of food fish estimated at about 100 million mt by the year 2025 and also to establish a strong research base for the evolving tropical aquaculture industry, it is important that ICLARM reaches out to building strategic alliances with the private sector. ICLARM has recently developed a pro-active policy document describing the specific guidelines for mutually beneficial relationships with the private sector as a first step in this direction.

Future ICLARM Tilapia Breeding

Based on the success of the GIFT Project, ICLARM will continue and expand its Program on Tilapia Germplasm Enhancement and Breeding. One of the primary purposes of the GIFT Foundation is to continue the GIFT selection program as selection plateaus have not been reached and further genetic enhancement is likely. ICLARM will collaborate with the GIFT Foundation, BFAR and FAC/CLSU to continue the selection program. Additionally, ICLARM has initiated selective breeding of tilapia in Africa. Great interest has also been expressed by NARS partners in Southeast Asia to have ICLARM assist in tilapia breeding programs. Maximum genetic gain and maximum impact will be obtained by combining the GIFT selection program with other traditional and new biotechnological breeding programs. ICLARM intends to continue selective



breeding, to combine genetic enhancement approaches, and to document and preserve tilapia genetics resources including thorough environmental risk assessment. Dissemination and impact assessment research are critical to affect global food security. ICLARM has ongoing research in this area with plans to expand such research in the future. The GIFT project demonstrated the potential of germplasm enhancement in tilapia. The continuation and/or expansion of this research will require investment of new resources and funds, but the benefits are clear.

Global Impact of GIFT

The GIFT project had and is expected to continue to have global impact in a number of ways. During the GIFT project over 650,000 GIFT brood stock were disseminated throughout the Philippines. An additional 45,000 GIFT brood stock were disseminated in Bangladesh, China, Fiji, Indonesia, Thailand and Vietnam. Together, these countries are among the major tilapia aquaculture countries in the world and several are major exporters of tilapia. These fish also form the base for further genetic enhancement in each country. Inevitably, in several countries which have received GIFT fish, the GIFT line is being crossed with local unimproved tilapia and thus the improved genome is entering other locally available tilapia. This impact, although difficult to measure, is that of top crossing and serves to upgrade the local fish.

Another mechanism bringing widespread impact is the transfer of the GIFT technology to other countries, as well as training in the technology. The trained staff from the project form the nucleus of staff for the GIFT Foundation International in the Philippines, thus continuing to use their skills in applied fish breeding research, development and management. Two hundred twenty-four researchers have been trained in 11 countries in quantitative genetics research and analysis methods. ICLARM is seeking further funds through INGA and other sources such as UNDP's TCDC program to further extension of the technology particularly to Asian countries.

The International Network for Genetics in Aquaculture has been formed and now comprises 13 developing country members, 10 associate members which are advanced scientific institutes. Funding is being provided and from ICLARM's core resources and alternated from donors such as Norway and IDRC. ICLARM in all its international fora, continues to publicize the work of INGA as a flagship research network, thus continuing to globalize the outcomes of the GIFT project. The GIFT research methods are available, and are being applied in the GIFT Foundation and in national breeding programs under INGA. The methods are simple, rigorous and effective in breeding better fish. Better still, the techniques result in immediate improvement soon after selection studies are started. This contrasts to many other genetic techniques for which the promised results are still 5-10 years off. The GIFT Foundation International model for public/private sector cooperation in delivering new breeds of fish to the farmers is receiving international attention. Several INGA member countries such as Indonesia and Vietnam are developing national tilapia breeding programs based on GIFT methods.

The method of germplasm collection, the selective breeding experiments and the chosen dissemination route are leading to pioneering studies on the definitions of intellectual property rights for fish. ICLARM has used this and other experience and knowledge to develop its Policy of Intellectual Property Rights for Germplasm, a first for aquatic resources. The policy is drawing international interest as a possible model for use or adoption by other agencies.

The GIFT strain at this stage appears to be better than most competitors and is developing a reputation in the Philippines for being more profitable to grow than other strains. Early indications from talking to the hatchery and feeds sector in the Philippines and Europe are that the existence of a faster growing strain of tilapia is spurring the need to examine other efficiency factors in tilapia production. These factors include feed improvements, better breeding and hatchery controls, better farm management practices and close attention to marketing of the fish. Ex-ante assessment of likely impact, borne out by early farm results, are that the GIFT tilapia and other selective breeds will lower the price of tilapia for the consumer.



Continuing impact from the GIFT project is also expected. The GIFT Foundation has been formed and continues to distribute GIFT fish and further enhance Nile tilapia. Additional GIFT fish are distributed, but without the GIFT trademark by BFAR in the Philippines. ICLARM retains a breeding population at the GIFT Foundation for further enhancement and future dissemination. The GIFT Foundation cooperates with INGA for further international distribution. Recent generations of GIFT have been shipped to India, Thailand and Vietnam. Further international distribution may occur through private enterprise.

Funding and economic change may make either government or private enterprise more effective at different locations for disseminating germplasm to best benefit poor consumers. However, it is essential that government mechanisms are established to ensure that the poor fish farmer benefits as well.

Recommendations for Future Research

The successful GIFT program leads to the need to address the following research issues.

1. Selection to improve growth rate should be continued for GIFT Nile tilapia as selection plateaus have not been reached.
2. Crossbreeding and selection should be combined and evaluated to maximize genetic gain for growth in tilapia.
3. YY male technology and sex reversal should be combined with GIFT selection procedures and evaluated.
4. Correlated responses to selection for other important economic traits should be evaluated.
5. Additional heritabilities and genetic correlations should be calculated for the design of multiple trait selection programs.
6. Enough time has elapsed since the original distribution that post impact assessment of GIFT Nile tilapia should be conducted. In fact, since GIFT fish have been distributed by both government and private enterprise, an opportunity to assess the relative merits of different dissemination exists.
7. Research techniques should be evaluated to allow transfer of the successful GIFT technology to Africa. The UNDP/TCDC program is a possible mechanism to accomplish this.

ICLARM is seeking to address these research needs with the appropriate partner institution and follow-up funding is needed to maximize the benefit from the groundwork laid by the GIFT project.



Photographs:

Page 1:

Plate 1: Part of the GIFT Core team during the First Training Workshop on Quantitative Genetics of Tilapias.

Plate 2: The First landmark meeting laying the foundation for the establishment of the International Network on Genetics in Aquaculture"

Page 2: Research Activities of the GIFT Project

Plate 1: The scale of experimentation in selective breeding of fish can be seen from this view of the research ponds at the BFAR/ CLSU, containing some 500 specially designed net cages to hold breeding groups of tilapia and their progeny.

Plate 2: Inspecting Founder Broodstock in the "Live Gene Bank" at the BFAR/NFFTRC Complex.

Page 3:

Plate 3: Cryogenic preservation of fish sperm at the "Frozen Nile tilapia Gene Bank".

Plate 4: Part of the GIFT Team admiring the Founder stocks. The Project Leader, Dr. Ambekar Eknath (second from left).



Part of the GIFT core team during the 6th Training Workshop on Quantitative Genetics of Tilapias.



National program leaders during the First Consultative Meeting laying the foundation for the establishment of INGA.



The scale of experimentation in selective breeding of Nile tilapia can be seen from this picture at the BFAR/CLSU complex, containing some 500 net cages to hold breeding groups of tilapia and their progeny.



Inspecting Founder Broodstock in the "Live Gene Bank" at BFAR/NFFTRC complex.



Cryogenic preservation of fish sperm.



Part of the GIFT team with the Project Leader, Ambekar Eknath (second from left).

1.0. INTRODUCTION

ICLARM's long quest to develop a strategic research program for application of genetics in aquaculture began in the late 1970s, with the realization that more productive and profitable aquaculture in developing countries would depend upon interactive research for the development of better breeds of farmed aquatic organisms and better farm environments. In the early 1980s, ICLARM, in collaboration with Philippine national institutions, undertook a series of studies on Asian farmed tilapia stocks and found a poorly documented and worrying situation. Farmers spoke of poor growth performance. Population genetics studies revealed extensive introgressive hybridization with feral populations of a much less desirable species for tilapia farming - the Mozambique tilapia (*Oreochromis mossambicus*). Clearly, the lack of any systematic attempt to breed tilapias for aquaculture was a major bottleneck to increasing the productivity and expansion of the industry.

The landmark ICLARM workshop on "Tilapia Genetic Resources for Aquaculture" in March 1987, in Bangkok, Thailand, marked the beginnings of the GIFT Project. A long-term research program to investigate the potential for genetic improvement of farmed tilapias was conceived during this workshop. It was anticipated that successful results could mean not only better tilapia breeds for fish farmers, but also new research methodologies for use with other species.

The GIFT Project was established in March 1988. It was implemented by ICLARM in cooperation with NFFTRC-BFAR, FAC-CLSU, UP-MSI and AKVAFORSK in two consecutive phases from March 1, 1988 to December 31, 1997.

The stated objectives and plan of activities of the GIFT project has evolved over the years. The Project, as a whole, is a concerted effort on the part of ICLARM and its partners to build on the successes and experiences of various other initiatives in the region. The Project research plan was developed following the successful experiences of AKVAFORSK with salmon and trout breeding programs in Norway. The expertise required to implement the project was secured by bringing together institutions with experience in: (a) tilapia farming (BFAR, FAC, ICLARM); (b) documentation and characterization of tilapia genetics resources (UPMSI and ICLARM); and (c) application of genetic improvement strategies in fish (AKVAFORSK and ICLARM). The research was carried out in strengthened research facilities in FAC-CLSU and NFFTRC-BFAR. The Project plans were circulated for peer-review before implementation.

1.1. Challenges addressed

The GIFT Project was established by ICLARM and its partners to address three challenges.

First: Could the application of basic principles of genetics and applied breeding technology, as practiced for centuries in plant and animal production, to the emerging aquaculture industry result in enhanced production efficiency? This challenge was highly relevant within the context of: (i) stagnating fish production from capture fisheries (direct harvests from marine and inland waters) and aquaculture

(farming of aquatic animals and plants) is increasingly seen as a panacea for the looming gap between the demand for and supply of food fish; (ii) glaring lack of conscious application of genetics to tropical aquaculture, even though the tradition of fish farming dates back hundreds of years. Consequently, the fish stocks used in aquaculture are generally similar to wild undomesticated stocks and in some situations there is strong evidence for genetic deterioration. In fact, before the GIFT Project, it had been widely assumed that genetic intervention will require extensive resources and, therefore, should be considered as an option for the future.

The second challenge pertains to the dissemination of improved fish breeds, once they are developed. It was understood right from the beginning of the program that genetic improvement and development of improved breeds was only half of the story. The other crucial aspect is their dissemination to reach the targeted beneficiaries effectively, including monitoring of the impact and adoption of improved breeds. Management of and access to germplasm are also becoming increasingly important issues, under the new Convention on Biological Diversity, which went into force in 1993. It was critically important for a nascent scientific discipline of applied fish breeding that it consciously and responsibly derive guiding principles from the complexities unique to aquaculture in a developing country context and also from the social lessons learnt from the green revolution in crop production.

Ways and means of catalyzing the development of applied fish breeding programs in countries that needed them most was the third challenge addressed by the GIFT Project. Relevant issues in this context included, objective assessment of needs and opportunities for developing applied fish breeding programs in each participating country; prioritization of species and farming systems; and development of the most viable mechanisms for international cooperation and provision of fora for exchange of research methods and where necessary, genetic materials. ICLARM had already declared its intention to adopt and apply internationally accepted codes of practices for responsible transfer of germplasm in relation to the GIFT Project. Companion and follow-on projects covering the dissemination and evaluation of the GIFT tilapia (the DEGITA Project) and the genetic improvement of carps in Asia (GICA), are beginning to address this challenge.

1.2. The Project Focus

The Project focus is on Nile tilapia (*Oreochromis niloticus*). It was chosen as a prominent test species because of their growing importance in aquaculture, their omnivorous diet which suits them to low cost aquaculture, and their short generation time of about 8 months, making them an excellent model species for investigating the application of genetics in aquaculture, from conservation of genetic resources to breeding programs. The research thrust was to develop methodologies and policy guidelines that can be applied or adapted to a wide range of finfish species in national fish genetic improvement programs.

1.3. Objectives and Scope

The overall **development objective** was to increase the quantity and quality of protein consumed in low income rural and urban populations in tropical developing countries in all regions of the world and increase the income of low-income producers. As with future agricultural and aquacultural developments, the objective was to aim for sustainable systems, in harmony with the natural environment, to benefit producers and consumers.

The project's **immediate objectives** were to:

- Develop improved breeds of tilapia and provide those fish breeds to national testing programs and thence to fish farmers;
- Strengthen national institutions in aquaculture genetics research; and
- Establish a mechanism for international exchange and evaluation of improved breeds and research methods.

The immediate objectives were accomplished by pursuit of the following activities:

Core Research Activities:

- a) Documentation, collection and genetic characterization of tilapia genetic resources in Asia and Africa.
- b) Evaluation of new African germplasm along with existing cultured stocks in the Philippines in a wide range of farming systems and under various agro-climatic conditions.
- c) Estimation of the magnitude of non-additive genetic effects (heterosis) and thereby determine the breeding strategy: selection and crossbreeding
- d) Building up of base population using promising strains and crosses.
- e) Selective breeding of Nile tilapia
- f) Investigation of genetics of other important traits
- g) Preservation of tilapia germplasm as live collections and as cryopreserved spermatozoa.
- h) Development of database management systems
- i) Scientific Publications
- j) Regional dissemination of improved germplasm and assessment of impacts (carried out under the auspices of the **DEGITA Project**).

Training and International Cooperation: (partially carried out under the INGA established in 1993)

- a) Dissemination of information and training of personnel from various national and regional institutions through a series of training workshops.
- b) Distribution of genetically improved fish through national broodstock distribution channels and testing programs and finally to fish farmers.
- c) Formulation of plans and policy guidelines for effective management of cultured stocks and establishment of national fish breeding programs.
- d) Defining from this work with tilapia, methods and approaches to fish genetic improvement for application to other species.

1.4. Implementation Arrangements

1.4.1. Donor/institutional arrangements

The **GIFT Phase I** was co-financed by the Asian Development Bank (ADB, Regional Technical Assistance Project, RETA # 5279) and the United Nations Development Programme/Division for Global and Interregional Programmes (UNDP/DGIP Project # INT/88/019 and INT/91/005/1/01/42). It was executed by ICLARM in cooperation with NFFTRC/BFAR, FAC/CLSU, UPMSI and AKVAFORSK (through NORAGRIC). The Project's Phase I began in 1988 with initial funding for three years. Following the review of the overall progress of the project and request from ICLARM, the Bank agreed to an extension of the Project. The UNDP/DGIP granted a one-year supplementary funding to the project (INT/91/005/1/01/42). The extension phase of the UNDP/DGIP grant was executed by the World Bank. The Phase I of the Project was successfully completed in July 1992.

The **GIFT Phase II** officially started on 1 January 1993. It represents a major evolution of the GIFT Project from a relatively small research project during the GIFT Phase I into a major complex strategic research, training and national capacity strengthening initiative in germplasm enhancement and breeding. The basic principles for research and cooperation were laid down during a consultative meeting convened by ICLARM and the UNDP on "International Concerns in the Use of Aquatic Germplasm" (June 1-5, 1992). The Phase II of the Project was financed mainly by the UNDP (UNDP/DGIP Project # GLO/90/016). The Project document for GIFT Phase II was officially signed between ICLARM and the UNDP/OPS in January/February 1993.

Separate Memoranda of Understanding (MOU) were forged between ICLARM and its Philippine research partners (FAC/CLSU and NFFTRC/BFAR) and AKVAFORSK (Institute of Aquaculture Research in Norway). The MOU encompassed the institutional responsibilities over the five years and the protocols for development, control, and the use of improved germplasm, data and technology developed through the GIFT Project. The Philippine government's contributions to the Project were in terms of manpower, research facilities and farm inputs. The GIFT Phase II was officially terminated on 31 December 1997.

To facilitate efficient implementation of the complex range of activities nested within the stated objectives of the GIFT Project and in response to the challenges and opportunities presented by the donor community, ICLARM, under the auspices of the GIFT Project, and very early during the GIFT Phase II, developed: (a) **the Dissemination and Evaluation of Genetically Improved Tilapia Species in Asia (DEGITA) Project** with principal focus on the dissemination of the improved Nile tilapia germplasm and *ex ante* evaluation of impact on equity, the environment and biodiversity. The DEGITA project was implemented by ICLARM from June 1994 to June 1997 in cooperation with national institutions in Bangladesh, the People's Republic of China, the Philippines, Thailand and Vietnam. It was funded by the Asian Development Bank (RETA grant No. 5588), ICLARM Core funds, and the participating national institutions; and (b) **the International Network on Genetics in Aquaculture (INGA)** as a forum for the participating national institutions to exchange research methods, experiences and germplasm. Thirteen countries in Asia and Africa are collaborating in research and exchange of genetic materials: Bangladesh, Peoples Republic of China, Cote d'Ivoire, Egypt, Fiji, Ghana, India, Indonesia, Malawi, Malaysia, the Philippines, Thailand, and Vietnam. ICLARM is the member-coordinator. INGA was inaugurated in 1993 by seed funding from the GIFT Project and since then has been supported by ICLARM core funds and project funds from the Government of Norway, ACIAR and IDRC, as well as operational funds from the member countries.

1.4.2. Resource Allocation

Four categories of Project staff were formed: (a) seconded project staff - permanent staff members of collaborating institutions (BFAR and FAC/CLSU) seconded to the GIFT project and who spent between 20 and 100% of their time in the project; (b) Project-hired staff - staff hired on an *ad hoc* basis on prevailing national salary scales and benefits; (c) ICLARM GIFT project staff - staff hired under ICLARM's terms and conditions; and (d) Project staff based in AKVAFORSK, Norway. Staff in categories (b) and (c) worked full time in the project. In AKVAFORSK, the time spent in various project activities was logged up to effective man hours and budgeted accordingly. Details on allocation of resources (personnel) in relation to project tasks and activities are indicated in the annual Project Workplans previously submitted to UNDP.

The names, institutional affiliations of personnel and duration of their direct involvement in the project are presented in Annex 1.

1.4.3. Project Management

The management of the GIFT Project including preparation of annual research plans and budgets was through the Project Management Committee (PMC) set up according to the principles outlined in the Memorandum of Agreements between ICLARM and its research collaborators. The Project Leader was the Chairperson of the PMC.

A total of 29 PMC meetings were held/organized throughout the duration of the project.

1.4.4. Dedicated Research Facilities and Equipment

With funds from the UNDP/SEED, the following research facilities were built during the project period:

- The Philippine Tilapia Germplasm Reference Collection Center. The facility was built at the NFFTRC/BFAR in April 1989 to hold the GIFT founder stocks. This is a self-contained facility comprising 7 units of twin holding tanks (14 tanks), 3 units of quarantine tanks, a sump pond, a deep well and a water tank.
- Dedicated Research Facilities (8 ha) known as the "Center for Applied Fish Breeding and Genetics Research" was built at the CLSU as a common research facility for use of ICLARM and its Philippine national research partners (FAC/CLSU and NFFTRC/BFAR). This was formally inaugurated on 9 October 1996.
- For long-term storage of tilapia genes, a modest cryopreservation facility was built at the NFFTRC/BFAR. The Institute of Aquaculture, University of Stirling, Scotland provided the technical expertise in setting up this facility.
- The list of fixed assets accumulated during the entire project duration and their present location is in Annex 2.

1.4.5. Project Reviews

The progress and direction of the GIFT Project were reviewed periodically through: (i) the annual review by the External Advisory Panel (EAP), instituted by UNDP since 1993; (ii) the annual Tripartite Review (TPR) convened by UNDP/SEED from 1993 to 1996; and (iii) through regular ICLARM Program Reviews and the TAC/CGIAR External Program Reviews. The full account of the TPR conducted yearly are given in the TPR reports submitted previously to UNDP Manila/New York.

1.4.6. Archiving

ICLARM will function as the long-term permanent depository of all project data, reports and manuscripts. This process has already been implemented.

2.0. SUMMARY OF ASSESMENT OF PROJECT ACTIVITIES IN RELATION TO THE STATED PROJECT OBJECTIVES

The status of implementation of all project activities is presented in the Logframe (Table 1 – Please note that the codes in column 1 of Table 1 refers to the corresponding sub-sections of Section 3.0 on Results and Outputs and Section 4.0 on Publications).

Overall, the majority of the activities in pursuit of the various project activities have been accomplished. During the entire duration of the Project, the one major activity that suffered the most was the preparation of scientific publications. Although a number of publications have been prepared (see section 4.0), about 10 scientific papers are in

various stages of preparation. ICLARM and its partners are committed to complete these papers as they are crucial to understanding the whole range of research activities conducted under the auspices of the GIFT Project and their reviews by scientific peers is also critical to finalizing the project outcomes.

3.0. RESULTS AND OUTPUTS

3.1. CORE RESEARCH ACTIVITIES

Overview:

The program began with consultations with fish farmers and experts from various disciplines. The relative economic importance was considered of each of the diverse tilapia farming systems including the backyard fishpond, rice-fish culture, cage culture and some more intensive systems. A grow-out period of about 90 days (from the fingerling stage when the fish are about 3-7g to their harvest weight of about 120 g) was chosen as a representative production cycle. Standardized research methods were then developed for mating, mass marking of individual fish, sampling, anesthesia, recording of traits and pedigree, management of databases, statistical analyses, and quarantine and safe dispersal of fish. Published manuscripts from the early standardization methods are in Final Report Part 2 Attachment 1.

The standardized research methods have been thoroughly documented in the GIFT Manual of Procedures Vol. 1, prepared according to the UNDP Train-X format. This manual has been prepared for use during regular training of national scientists participating in INGA. The Manual of Procedures Vol. 2 was not initiated as earlier planned as not being a high priority in the 1997 workplan. It will, however, be an important additional instructional manual for developing country partners wishing to establish breeding programs for the first time. This manual will describe the broad techniques of developing the tilapia breeding program and will comprise the following procedures: (i) determination of breeding goals; (ii) systematic documentation and evaluation of available genetic resources; (iii) identification of genetic base; (iv) determination of the number of strains to be developed; (v) knowing the breeding strategy; (vi) establishment of the base population; (vii) implementation of selection program for identified traits; (viii) production and dissemination of improved strains and (ix) assessment of social, economic and environmental impacts. Completion of this volume will require identification of additional donor support.

The core research activity comprised 9 **major research experiments**, spanning nine tilapia generations. The first three experiments (generation 1 - 3) were devoted to a search for the most effective breeding strategy. This involved: (a) collection of a planned set of wild and locally available Nile tilapia genotypes; (b) comparison of these different genotypes in different farming systems to quantify any interactions between the genotypes and the range of targeted environments (farming systems); (c) comparison of strains and their two-way crosses in the form of a complete diallel cross experiment to quantify any specific or general benefits from heterosis effects; and (d) combining the best strain crosses and pure strains to build a base population. Selective breeding using the base population began from generation 4 onwards. The breeding objective was to improve growth performance within the pre-determined rearing period in grow-out

environments. Records of other traits, e.g. survival, occurrence of diseases and deformities, and maturation rate (defined as the frequency of spawning females within a family) were also recorded. The potential for developing late maturing tilapias was also investigated. The genetic material and data gathered in each generation of experiments were utilized to conduct parallel studies (known here as **complimentary studies**) to answer specific research questions on the biology and genetics of tilapia.

In the following sections, only a brief account of the major experiments conducted are presented. The various complimentary studies conducted during each major experiment are listed. Detailed technical description of both major experiments including the experimental environments used and the complimentary studies are in appropriate Attachments in Final Report Part 2.

3.1.1. Documentation, Collection, Quarantine and Genetic Characterization of Nile Tilapia Germplasm

3.1.1.1. Documentation, collection and quarantine

One of the major recommendations of the 1987 Bangkok Workshop on "Tilapia Genetic Resources for Aquaculture", organized by ICLARM in preparation for the GIFT Project, was that the established farm stocks in Asia do not form the best genetic base to begin a genetic improvement program because of small founder populations originally imported, possible inbreeding and widespread introgression of genes from other less desirable feral tilapia species.

Based on the identification of sources of pure Nile tilapia genetic resources from the above workshop, the GIFT team made in 1988 and 1989 the first ever collections and direct transfer of Nile tilapia germplasm from Africa to tropical Southeast Asia. Breeders and/or fingerlings were collected in Egypt, Ghana, Kenya and Sénégal, in collaboration with the University of Hamburg; the Musée Royale de l'Afrique Centrale, Tervuren, Belgium; the Institute of Aquatic Biology (IAB), Ghana; the Suez Canal University, Egypt; and Baobab Farms, Mombassa, Kenya. The details of tilapia germplasm collections in Africa are in Table 2. The fish from Africa were held in isolation at the NFFTRC/BFAR. The team worked with the BFAR Fish Health Unit in developing quarantine procedures for three to seven months: a model for use elsewhere. The experience of the project team on quarantine methods followed are fully described in the GIFT Manual of Procedures Volume 1.

Domesticated stocks of four Philippine commercial Nile tilapia strains (known in the Philippines as 'Israel', 'Singapore', 'Taiwan' and 'Thailand' strains) were also gathered, giving a total of eight strains for study. The origin of the four farmed Nile tilapia strains in the Philippines are presented in Table 3. The eight founder stocks were held in newly constructed "Tilapia Germplasm Reference Collection Center" at the NFFTRC/BFAR. Spermatozoa from tilapia founder stocks, as well as from the breeders from each generation of selection are being cryopreserved, as part of a small tilapia genebank, for further research. About 761 samples have been preserved, as described below.

3.1.1.2. Biochemical and Morphometric Characterization of Nile tilapia Germplasm held by the Project

Genetic characterization using morphometric, electrophoretic and mitochondrial DNA techniques were performed on the eight founderstocks. Isozyme analysis and the resulting dendrogram of genetic distance values revealed the close identity of the Philippine strains with those from Egypt and Ghana - confirming that the origin of Philippine strains is Egypt and Ghana. The wider separation of Kenya strain supports its recognition by Dr. E. Trewavas as a different subspecies (*O. niloticus vulcani*) from all others (*O. niloticus niloticus*). Morphometric (truss) analysis, however, indicated very little morphological difference among the eight strains.

The technical details in the form of six published manuscripts are in Final Report Part 2 Attachment 2.

Outputs:

- Collection of diverse Nile tilapia germplasm useful for further strategic research to support national breeding programs.
- Development of effective quarantine procedures for tilapia stocks - a model experience in the region. This experience is fully documented in the GIFT Manual of procedures Vol. 1.
- Thorough biochemical and morphometric characterization of diverse tilapia strains.
- Six scientific papers published (Part 2 Attachment 2).

3.1.2. Maintenance of Live and Cryopreserved Nile Tilapia Genebank

3.1.2.1. Maintenance of live genebank

In addition to the eight founder stocks (four directly introduced from Africa and four domesticated strains of Nile tilapia) and their replacements, the live collections include top ranking crosses (from the diallel crossing ---the base population) and selected lines produced from every generation of selection. Representative samples from these collections were used regularly as controls for measuring the magnitude of genetic gains achieved in the selection program. The selected lines were kept separately in ponds by generation of selection and sex in the new breeding facilities (Center for Applied Fish Breeding and Genetics Research). The germplasm collections used as founder stocks in the GIFT Project are presently being kept at the NFFTRC/BFAR. A complete list of all Nile tilapia germplasm held by the GIFT project, based on an actual inventory of stocks conducted in December 1997 is presented in Table 4.

3.1.2.2. Maintenance of cryopreserved tilapia genebank

As of November 1997, spermatozoa of about 761 fish samples representing the founder stocks and selected lines have been cryopreserved (Table 5). The protocol followed in the cryopreservation of tilapia spermatozoa was patterned after the technology developed at the Institute of Aquaculture, University of Stirling, Scotland.

Following the recommendations of the External Review team of ICLARM Genebank Operations in November 1995, all the cryopreserved tilapia germplasm collections (founder stocks and selected stocks) were replicated for safety. Negotiations with the Philippine Carabao Center at CLSU to hold a duplicate collection is on-going.

On termination of the GIFT Project in December 1997, the genebank is being maintained by BFAR/NFFTRC and the GIFT Foundation.

3.1.2.3. Database on the GIFT Nile Tilapia Germplasm Reference Collections

Under Microsoft Access Version 2, a comprehensive relational database covering importation, quarantine, and live and cryopreserved germplasm collections has been developed. This database is a sub-system within the larger GIFT database system, and thus can be further linked to other databases such as breeding and grow-out databases. The database consists of a master list of PIT tag numbers which facilitates the systematic recording of cryopreservation data linked with the breeding data was also established. A data dictionary was prepared to describe each variable used.

The cryopreservation datasets were made available to the System-wide Information Network on Genetic Resources (SINGER) an information network system geared at handling the genetic resources documented by the CGIAR centers.

Outputs:

- Establishment of live and cryopreserved tilapia germplasm collections for further strategic research.
- Protocol for maintenance of live germplasm collections (Part 2 Attachment 3).
- Development of comprehensive relational database for live and cryopreserved tilapia germplasm collections.

3.1.3. Evaluation of Culture Performance of the Eight Strains in a Wide Range of Farming Systems and Under Various Agro-climatic Conditions (Generation 1)

The most important criterion determining the number of strain to be developed is their relative performance in different target environments. In genetic parlance, this is called genotype X environment interaction. If there is little or no GE, then the best strains in one environment will be the best in all. On the other hand, if there is high GE then special strains must be developed for specific environments. In determining the target

environments, emphasis should be on the relative importance and proportion of commercial production represented by each of the environments and whether or not existing environments will prevail in the foreseeable future.

The GIFT project's first major experiment was to determine the growth performance of strains and the magnitude of GE over two consecutive generations (Schematic diagram of the experiment in Fig. 1). The 11 test environments chosen covered a wide range of tilapia farming systems: earthen ponds fertilized with organic manure, ponds fertilized with on-farm agricultural residues, rice-fish systems, cages, and hatcheries located in different agro-climatic regions. A total of 11,000 individuals (7650 and 3420 individuals during first and second generation) were communally reared in all test environments for 90 days. Results indicated highly significant differences in growth performance among the strains. Excepting Ghana strain, the wild African strains performed as well as or better than the 'domesticated' strains used by the Philippine farmers. The widely cultured Israel strain in the Philippines performed poorly. The GE was low. Overall, the growth performance of the strains were relatively consistent across test environments. It was concluded that development of specialized strains for each of the farming systems in the Philippines may not be necessary. However, the GIFT plan incorporates testing of breeding materials across all target environments to ensure that the response to selection will be valid in almost all grow-out environments.

Technical details on this strategic experiment and on the other complimentary studies conducted during generation 1 are provided in Final Report Part 2 Attachment 4.

List of complimentary studies conducted during generation 1:

- Predictability of individual growth rates of Nile tilapia
- Growth and reproduction of individually tagged Nile tilapia of different strains
- A practical quantitative method to estimate relative reproductive activity in Nile tilapia.

Outputs:

- Strategic decision 1. Development of specialized strains and diverse selection programs for each of the investigated diverse farming system is not necessary. The best performing strains could be mixed together to build one base population.
- Four scientific papers published (Part 2 Attachment 4).

3.1.4. Estimation of the Magnitude of Additive- and Non-additive Genetic Effects (Heterosis) and Determination of the Breeding Strategy: Selection or Crossbreeding (Generation 2)

The major alternative breeding systems are selection and crossbreeding. Selection as a breeding strategy focuses on additive genetic variance – choosing the best performing individuals as breeders in every generation. Selection programs are in general resource efficient, simple in design and easier to implement than crossbreeding programs, and the genetic gain is sustainable during generations of reproduction outside the breeding system. By nature, the selection approach is 'non-protective'. Crossbreeding focuses on non-additive genetic variance – choosing the right combination of strains/ individuals. The Hybrid vigor (or heterosis) should be substantial to adopt crossbreeding as a breeding strategy.

The GE experiment performed during the first generation experiments (see above) was followed by a complete diallel crossing experiment producing all 64 hybrid and pure strain combinations among the eight founding strains to estimate the magnitude of heterosis (see schematic diagram in Fig. 2). Growth performance was recorded in a total of 23,739 individually tagged progeny after a grow-out period of about 90 days. Least square means (LSM) of body weights at harvest were computed for each strain combination within and across all test environments, and additive genetic effects, maternal (reciprocal) effects, and non-additive genetic effects (heterosis) were estimated. The LSM heterosis of all strain crosses across all test environments was 4.3%, and the cross that expressed the largest non-additive genetic effect showed a least square mean heterosis of 14%. However, only seven out of the 22 crosses with significant heterosis performed better than the best pure strain, and the largest gain was then about 11%.

Although some strains performed better as sire strains and others as dam strains, it was concluded that specific line (or strain) crossing schemes will only result in marginal improvements in growth performance and may show an increased sensitivity to GE interactions. Therefore, a regular selection program based on additive genetic performance was adopted as the breeding strategy for rapid genetic improvement of Nile tilapia in the Philippines.

Technical details on this strategic experiment and on the other complimentary studies conducted during generation 2 are provided in Final Report Part 2 Attachment 5.

List of complimentary studies conducted during generation 2:

- Early growth and survival of eight strains of Nile tilapia and their crosses
- Fry Production of eight strains of Nile tilapia and their crosses
- Progeny sex ratio in a complete diallel cross with eight strains of Nile tilapia.

Outputs:

- Strategic decision 2: A combined family selection strategy commencing with a mixed base population with a broad genetic variability is the best approach to genetic improvement in this case.
- Strategic decision 3: Establishing the base population. Based on their additive genetic performance, breeders from the 25 best performing purebred and crossbred groups (from the 64 evaluated) were chosen to build the mixed based population.
- Three scientific papers published (Part 2 Attachment 5).

3.1.5. Building the Base Population and Starting the Selection Program (Generation 3)

The objective of this experiment was to estimate the genetic parameters (heritability, genetic variation and genetic correlations) of survival, growth and reproduction traits.

One hundred fifty breeding units in three sets (or 450 units) were established in a nested mating design (one male mated to three different females) to produce 150 full-sibs groups. Of the 150 full-sibs groups planned and actually produced, 27 groups were not tagged due to low numbers of fish at tagging. A total of 25,000 individually tagged fingerlings were communally reared for 90 days in 7 different test environments. Tagged fingerlings of the commercial strain most commonly used by fish farmers ('Israel' strain) were also stocked in three different locations in Central Luzon to compare the growth performance.

Significant results from this experiment were as follows:

- (i) the heritability estimates for body weight at harvest were consistent across all test environments. The average heritability value based on the sire component of variance was 0.23. This indicates a substantial additive genetic variance for growth rate in the mixed base population. The prospects for rapid genetic improvement for this trait through selective breeding are therefore very promising.
- (ii) the heritability values for growth rate based on the dam component of variance were high, averaging 0.53 across test environments. This may indicate a significant maternal genetic, non-additive genetic or environmental effect common to full-sibs for body weight at harvest.
- (iii) the heritability for survival based on the sire component of variance was 0.08 - also indicating a significant genetic variance for this trait also. The percentage survival at harvest of the base populations was higher than the commercial strain.
- (iv) correlation between body weight and survival based on the sire component was positive (0.2) indicating a favorable genetic correlation between these two economic traits.

- (v) the performance of the commercial strain was very poor. It was equal to the second lowest ranking full-sib group (out of the 123 groups tested). The GIFT project's base population was about 60% faster growing than the most widely cultured 'Israel' strain across the range of environments tested.

Outputs:

- Strategic decision 4: There is substantial genetic variance for growth and survival in the base population. Rapid genetic improvement is possible.
- An improved tilapia breed which performs better than the Philippine commercial 'Israel strain'. Preliminary economic analysis showed that: (i) under experimental conditions when the project's base population and the commercial 'Israel' strain were communally reared, the average gross revenue from using GIFT base population was 1.6 times (or 60%) greater than the commercial strain; (ii) a survey of optimal production conditions in three provinces of Central Luzon indicated that tilapia reach a harvest size of about 80 g after a grow-out period of 5-6 months - about 2 crops per year. The GIFT fish reaches the same harvest size in only 3 months at a slightly higher stocking density - about 3 crops per year. This translates into doubling of annual production and gross revenue.
- two major scientific papers in preparation

3.1.6. Selection and estimation of genetic gains (Generations 3 to 9)

The primary objectives of this study were: (i) to estimate the response to selection (magnitude of genetic gain); (ii) to initiate procedures for recording of other economically important traits such as maturation, carcass quality, survival etc. and (iii) test release genetically improved fish in farmer participatory research in a wide range of farming systems.

3.1.6.1. Selection for growth (Generations 4 to 9) and for two traits (growth and frequency of spawning females (Generation 7 to 9).

The generation 3 experiment described above (3.1.5) represents the base reference material (the base population) for comparisons and documentation of observed genetic gains in growth performance in subsequent selection experiments (Generation 4 to 9).

Generations 4 to 9 focused on selection for growth of tilapia. A two-trait selection experiment (Generation 7) was initiated in 1996 but this was not repeated in the following year (Generation 8) because records on female maturity were not gathered. Records to perform a two-trait selection were again collected in Generation 9. The Generation 9 experiment was completed in March 1998 within the resources of the GIFT Foundation. Data analysis in preparation for a full two trait selection experiment is in progress.

The regular procedures followed during each selection cycle involved production of about 200 families by mating 100 selected males with 200 selected females in a nested mating design (1 male: 2 different females; schematic diagram in Fig. 3). Representative samples of progeny from each family were individually tagged and distributed to different test environments. After a grow-out period of about 120 days, the fish from all the test environments were harvested and individuals ranked based on their growth (single trait selection) or growth and reproductive performance for performing a two-trait selection. Breeders for the next round of selection were selected based on their estimated breeding values.

Reproduction (or maturation) was recorded as the frequency of spawning females per individual family, about halfway through the rearing period, when substantial fry production was observed in rearing ponds. Maturation records were obtained by total draining of the ponds and sampling of the entire population.

Magnitudes of genetic gains were estimated by comparisons with standard controls (progeny of breeders with average breeding values; progeny from repeat mating of breeders used in previous generations) and also with non-standard controls (progeny from conventional commercial strains in the Philippines, namely, 'Israel' and 'Thailand'). The latter comparisons were carried out for demonstration purposes. Representative family materials from each selection cycle were also used for carrying out specific research experiments, as described in following sub-sections.

Estimation of genetic parameters and response to selection for growth:

Estimation of heritabilities (with standard errors) for growth performance in Generation 3 through 8 were all estimated with a single-trait animal model, using a derivative-free Restricted Maximum Likelihood algorithm (dfREML). Within each generation and within each test environment, the dfREML model included the random additive genetic effect and a random environmental effect common to members of a full-sib group. The fixed effect part of the model included the effect of sex and the initial age variable was included as a covariate. For analysis across test environments (within a given generation), the same model was applied, except that the fixed effect part now contained the sub-class effect of environment by sex. For all analyses, pedigree information was traced to the base population (Generation 3). The total number of individual records used in the univariate dfREML analyses and information on the number of test environments, sires and dams in each generation are presented in Table 6.

For the estimation of selection response within each generation, the dfREML model was expanded to include the fixed effect of genetic group ('selected' individuals and 'controls'), and the response was estimated as the contrast between these. The latter was performed across test environments in each generation.

The estimated heritabilities within each environment, and across all environments, during each generation is in the Final Report Part 2 Attachment 6. The summary of heritability estimates for growth performance and the observed response to selection for growth in each generation is in Table 7.

Survival

Percent survival in each test environment in generation 3 through 8 is given in Table 8. The survival across environments and across generations (generation 3 - 7) has been almost consistent with few significant deviations. The low survival recorded during generation 8 were mainly due poorer water quality and water circulation in the test environments during this particular generation. Furthermore, the tag-losses were also very high during this experiment.

Analyses of survival records using the Gianola and Foulley Categorical analysis procedure (GFCAT) has been initiated. The GFCAT procedure performs a categorical data analysis and provides estimates for additive variance component in terms of the underlying, non-observable normal distribution.

Results so far indicate no or low additive genetic variation for survival. This is probably due to extensive tag-losses, which unfortunately is difficult to separate from the actual survivorship. Consequently, the planned estimation of genetic correlations between survival and growth performance had to be dropped.

Genetic correlations: Growth performance in different environments

Growth performance of representative full-sib groups in different environments were considered as different traits, and their genetic correlations were estimated. The analyses were done by applying a multi-trait animal model including the random effect of additive genetic effect and common environment, as described above. Sex was included as a fixed effect and the target variable for analysis, observed body weights at harvest, were all pre-corrected for variation in initial age at stocking before analysis. Analyses were done on records obtained from generation 6, 7 and 8. The estimates of genetic correlations across environments were highly variable. In general, the genetic correlations were significantly high (0.7 to 0.8) for similar environments, the earthen pond environments of BFAR and FAC. Between earthen pond and the cage environments, the genetic correlations were low (0.0 to 0.4). Interestingly, the genetic correlation between earthen pond environments and the rice-fish environments were very high (0.9 to 1.0).

Inbreeding

The level of inbreeding in each generation was calculated by tracing the pedigree to the base population (generation 3) using the conventional tabular method.

Outputs (Generations 4 to 8):

- A genetically improved tilapia breed and estimates of its genetic gains.
- The survival across test rearing environments and across generations (generation 3 to 7) has been almost consistent (73%) with few significant deviations. There was no or low levels of additive genetic variation for survival.
- Genetic correlations between growth performance of sib groups in different environments were highly variable across test environments. The relatively lower

values of genetic correlations between growth performance in earthen ponds and in the cages should be investigated further.

- Five generations of selection completed. The genetic gain per generation has been about 12-17% across five generations of selection. The accumulated genetic gain in relation to the base population was 85% (see Fig.4).
- The estimated rates of inbreeding per generation from Generation 4, when selection was initiated, to Generation 8 were, respectively: 0.138, 3.82, 5.34, 7.09, and 7.09.
- Initiation of research work for developing late-maturing Nile tilapia; and procedures for a multi-trait selection involving growth and late maturation (as described below).
- Standardized techniques for controlled mating, anesthesia, mass individual tagging, recording of ancestry, hatching data, growth and survival.
- Improved research methodologies and associated software for estimation of genetic parameters and response to selection, widely applicable in developing country aquaculture research and development.
- The models and results obtained needs to be further reviewed before preparation of scientific papers can proceed smoothly (see section 4.0).

Specific core-research experiments conducted from Generation 5 - 9

3.1.6.2. Estimation of the magnitude of genotype (families) x season interaction (using the same sires and dams used in the previous generation)

Two seasonal lines were produced for testing in two different growing seasons. The over-all objective was to determine if there was a need for establishing two separate lines depending on their relative performance during the two major growing seasons.

Repeat mating of the same sires and dams used in Project's seventh generation experiment were bred again in 1995 to produce progenies for testing in three different environments (BFAR, FAC and Pantabangan cage). After 120 days of culture all the fish from the three environments were harvested. Data gathered were analyzed to correlate the growth performance of progenies reared during cold and summer months.

Examining correlations between growth of winter and summer lines in each of the three environments, showed no significant correlation on the growth of 28 full sib families in two environments (BFAR pond and Pantabangan cage) and significant correlation in only one environment (FAC, 0.44 at $p < 0.05$). Implications for setting up of two separate selection lines (winter/cold and summer) should be considered in the strategic planning of the GIFT Foundation.

3.1.6.3. Estimation of genetic parameters for maturation (frequency of spawning females per family)

Separate fishponds were prepared and stocked with family materials from the Project's seventh generation experiment for recording of maturation (age at first spawning). Progenies from 155 families from the Generation 5 were communally stocked in two replicate earthen ponds in FAC. Total draining of ponds and sampling of all fish were done about two weeks after the appearance of fry. Female's spawning conditions were recorded at this intermediate sampling. Females were scored for their spawning activity (either as "0" or "1", depending on whether they had spawned or not). Body weights were recorded at this intermediate sampling. The fish were further grown and harvested at the end of the experiment, when spawning and body weights were again recorded.

The estimated heritability (based on sire component of variance) for growth, heritability of frequency of spawning females and genetic correlation between frequency of spawning females and growth at intermediate sampling were, respectively, 0.34, 0.37 and 0.70. The corresponding values based on the dam component of variance were 0.55, 0.80, and 0.43. These results indicate significant positive correlations between growth and maturation, i.e., faster growing families had significantly high frequency of spawning female individuals. A strategic decision was therefore made to include maturation as the second important trait in the selection program.

The estimated individual breeding values for growth and family values for frequency of spawning females (assuming a heritability of 0.25) were combined in a selection index (assuming equal economic weights).

3.1.6.4. Response to selection for late and early spawning females

Progenies from families with high and low values for maturity (late and early spawning families) were used in the experiment. A total of 3,566 individually tagged fingerlings from 9 late maturing families and from 19 early maturing families were stocked communally in three 600 m² earthen ponds. Total sampling was done two weeks after each appearance of swim-up fry in the ponds. Two intermediate samplings were done before the fish were harvested. Data such as body weight, standard length, body depth and papilla condition of the female (for determination of maturation) were recorded during samplings. The experiment was terminated after a grow-out period of 120 days.

A highly significant response to selection was observed for frequency of early maturing females (33.5% mature females in the progeny of late maturing broodstock as compared to 55% in progeny of early maturing broodstock at the time of recording). Furthermore, a significant correlated response was observed for body weight at harvest in male progeny. Progeny of early maturing broodstock was heavier than progeny of late maturing broodstock (205 and 182 g. respectively). A similar, but non-significant trend was observed in female progeny.

The results of this experiment further confirmed the need to apply a simultaneous selection for both traits in an applied breeding program. One scientific paper describing the details of the experiment has been accepted for publication (Final Report Part 2 Attachment 7).

3.1.6.5. Determination of Indirect markers to improve rate of Survival

This study was carried out as part of a Ph.D Thesis dissertation by a Thai student in AKVAFORSK. The study was carried out in two parts: (a) Determination of genetic variation in lysozyme and spontaneous haemolytic activities of blood serum; and (b) Determination of genetic associations of serum lysozyme and serum spontaneous haemolytic activities with survival and body weights. Details of these studies are in Final Report Part 2 Attachment 8.

3.1.6.6. Specific Genetic Studies to Design Efficient Selection Programs

The following experiments were completed: (a) Replicate effects of rearing of families; (b) Effects of communal and separate rearing of test groups; and (c) sex reversal.

Manuscripts from (a) and (b) are being reviewed, and for (c), a manuscript on "Combined effects of genetic improvement through selection and sex reversal on survival and growth of Nile tilapia" (Final Report Part 2 Attachment 9) was prepared and presented at the Second International Workshop on Genetics in Aquaculture and Fisheries Management in Asia, held in Bangkok, Thailand in November 1994. Reanalysis of the data was completed following the review of paper. Revision of the manuscript is being done based on the revised analyses.

3.1.6.7. Growth performance in different farming systems

Studies to evaluate the performance of the GIFT Strain in various practical farming systems were conducted under the FAC/CLSU research component. Seven complementary studies were completed as follows: (i) Response of the GIFT strain to various feed formulations; (ii) Growth evaluation in earthen ponds fertilized with on-farm crop residues; (iii) Growth performance under intensive fertilization and supplemental feeding regimes; (iv) Growth performance in rice-fish environments; (v) Use of Bagasse (sugarcane waste) in freshwater aquaculture; (vi) The influence of stocking density on the growth, size-at-harvest, and survival rate of GIFT fish; and (vii) the response of the GIFT strain in polyculture systems.

Two manuscripts on the above studies have been prepared for submission to scientific journals for publication (Final Report Part 2 Attachment 10). These papers are now being circulated for review.

3.1.7. Commissioned Reviews to Investigate Genetics of Resistance to Biotic and Abiotic Stresses

The purpose of the commissioned reviews was to obtain an authoritative account on a chosen subject and provide advice to ICLARM on possible approaches and the investment needed to implement new research initiatives. Two topics were chosen for review: (a) Genetics of Disease Resistance in Fish; and (b) genetics of tolerance to saline waters in tilapias. Prof. Trygve Gjedrem of AKVAFORSK prepared the review on

genetics of disease resistance. The paper has already been reviewed, finalized and submitted to ICLARM for publication. A copy of the review is in Final Report Part 2 Attachment 11. The second review on tolerance to saline waters could not be implemented due to lack of resources. However, a special session on "Development of Salt Tolerant Tilapia" was organized and conducted in conjunction with the 3rd INGA Steering Committee meeting in Cairo, Egypt, 8-12 July 1996 to discuss among INGA member countries the possibility of developing a salt-tolerant Nile tilapia. The output of the session is in Final Report Part 2 Attachment 12.

3.1.8. Database Management

3.1.8.1. Standardization of datasets and preparations of data dictionary

Datasets from all core-research experiments conducted from Generations 1 to 9 were standardized to fit into a user-friendly database system in SAS-format. There are four sets of files for each generation experiment corresponding to its different stages: breeding, fry collection, early rearing and growth and grow-out. The accompanying Data dictionary and documentation for Generations 1 to 9 describes all the variables used in the GIFT database.

3.1.8.2. User-friendly software (within SAS) for routine analysis of genetics data – the GIFT Stat.

Applications development was one of the major tasks carried out by the Database Management Unit of the Project. The objective was to produce stand-alone software or a utility program which is user-friendly. The task was started in 1994 and was completed in April 1996. The GIFT Stat acts as repository of all standard statistical, genetic and data management programs developed and used by the GIFT Project since 1988.

3.1.9. Regional On-Farm Trials and Assessment of Impacts

This activity was inter-phased with the **DEGITA Project**. An extended summary on the DEGITA Project is included as Annex 3 to this report. The complete DEGITA Project Final Report has already been submitted to the UNDP and is also available on request from ICLARM. The main findings are as follows:

The production potential of GIFT and existing Nile tilapia strains (referred as 'non-GIFT' strains because of their diversity of origins) were assessed through on-station and on-farm experiments in a range of aquaculture systems in Bangladesh, Peoples Republic of China, the Philippines, Thailand and Vietnam.

Production per unit area or yield depends on stocking density, average weight at harvest and survival. After accounting for the wide heterogeneity of input use levels, production environments and other factors, the percent change in average weight at harvest and survival due solely to the mean superiority of the GIFT strain over the existing non-GIFT strains were highly significant across different countries. The magnitude of the effect of the GIFT strain on growth rate, however, varied among countries, ranging from 18% in

China to 66% in Bangladesh. In terms of survival, the GIFT Strain showed higher survival compared to the existing strains only in China (3%) and the Philippines (13%). The magnitude of impact, however, varied according to the history of tilapia introductions and farmers' experience. Overall, the results indicated that the cost of production per unit of fish produced is lower for the GIFT strain than for the non-GIFT strains: more than 30% lower in Bangladesh and the Philippines; about 20% lower in Peoples Republic of China, Thailand and Vietnam.

The estimated yield potential of the GIFT Strain was more than 50% higher than that of the best existing strains, ranging from about 54% percent higher in Vietnam to about 97% higher in Bangladesh. The estimated Yield Gap II, the difference between potential and actual yields in farmers' field, ranges from around 3000 Kg ha in China to around 850 Kg ha in the Philippines (which has a shorter growing period and poorer production environment). These estimated gaps show the extent to which yield might be increased through the invention of appropriate technology to overcome production constraints such as disease, water quality, etc.

In the Philippines and Thailand, where about 80-90% of the tilapia production is marketed locally and the demand for tilapia is price elastic, both producers and consumers will benefit from technological interventions (such as the GIFT technology). As low-priced tilapia is consumed mainly by poor people whose demand is inversely related to price, the major portion of the consumers' benefit will go to poor consumers. In countries like Bangladesh, where tilapia fish farmers consume about 70% of their produce, the major economic gain from GIFT technology will accrue to the producers. In subsistence farming, the reduction in market price due to higher production has relatively little influence on producers' income.

The participating countries in the DEGITA project are now in the process of developing their respective national tilapia breeding programs with the GIFT strain as a base starting genetic material.

3.2. TRAINING AND INFORMATION DISSEMINATION

3.2.1. Training of GIFT Project Staff and Scientists from National Institutions participating in INGA

A summary of the trainings conducted for technical personnel in national institutions and GIFT Project is given in Table 9. Twenty weeks were spent by the project for conducting the trainings organized for GIFT core staff and researchers/scientists from INGA member countries. A total of 224 researchers/ scientists from GIFT Project and national institutions from 11 INGA member countries had been trained during the project period.

Prior to implementation of the main project activities, a two-week training program in quantitative genetics was conducted for the leaders of the GIFT collaborating institutions on May 30 to June 10 1988 at the Institute of Aquaculture Research, Agricultural University of Norway. The training was followed by a workshop held at the AKVAFORSK (Sunndalsora) on June 26-28, 1988 to finalize the project research plan.

In addition to hands on-training, about six training workshops for the 18 project personnel from national collaborating institutions directly involved in the implementation

of the project were subsequently conducted from 1989 to 1993. Scientists from other national and regional institutions likewise participated in some of these training workshops. These training workshops focused on designing of genetic experiments, management and analysis of large databases, computer software applications and scientific writing.

An intensive training program on the "Application of Quantitative Genetics in Aquaculture" was successfully conducted for the GIFT Project core research staff and INGA researchers from November 6 to December 2, 1995 in Cavite, Philippines. The objective was to strengthen the capacity of GIFT core research staff and scientists from national institutions in the field of quantitative genetics and its application to increase sustainable aquaculture production. A full account of this activity is described in a separate training report submitted to UNDP in January 1996.

3.2.2. Training and supervision of graduate students

Five students directly involved in the Project successfully completed their Master of Science degrees. The Project Leader and senior research partners from FAC and AKVAFORSK provided guidance for completion of the theses. Of the five thesis studies, two received "Outstanding" rating from the Central Luzon State University and another one from the University of the Philippines. In addition, six students sponsored by the Philippine Bureau of Agricultural Research have completed their Master of Science degrees under the auspices of the GIFT Project. These students were supervised directly by the senior FAC research partners.

A compilation of all theses abstracts is in Final Report Part 2 Attachment 13.

3.2.3. Training and Dissemination of GIFT germplasm in support of National Tilapia Breeding Program in the Philippines

Organization of trainings/workshops

One of the objectives of the Project was to provide assistance to national institutions in planning national fish breeding programs. To accomplish this, nineteen training/courses were conducted by the Project from March 1993 to September 1997, either as a main organizer or in support to its national research partner (NFFTRC/BFAR). Shown in Table 10 is the summary of these training/courses.

These series of training courses are classified into four, depending on their main objective:

- (i) training courses to formulate distribution strategies of GIFT fish

About eighty-two (82) personnel from various levels of organizations of the Philippine Department of Agriculture were trained during the four different training workshops conducted between March 1993 to April 1994. The purpose of the meetings was to develop strategies for distribution of improved tilapia germplasm.

- (ii) training courses to gather feedback on GIFT fish performance from farmers and research outreach station;

A total of ninety two (92) Philippine farmers and technicians were trained during the five training workshops conducted to gather feedback on performance of GIFT fish at farmers' level, during the period December 1994 to May 1996

- (iii) training to disseminate information on GIFT fish

A total of four hundred fifty (450) participants representing Philippine tilapia farmers, teachers, and aquaculturists were trained during the seven training workshops aimed at disseminating information on GIFT technology, conducted during the period February 1994 to September 1997.

- (iv) training courses/workshops which led to formation of the GIFT Foundation International Inc.

Eight training workshops and discussion meetings were conducted from December 1994 to June 1997, aimed at formation of the self-sustaining national tilapia breeding program.

Dissemination of the GIFT Strain in the Philippines

The distribution of GIFT fish to Research Outreach Station (ROS) was initiated in 1993. BFAR took the leadership role in this activity. Recipients of the fish have gone into mass production and dispersal of fingerlings to grow-out farmers in their respective regions. Details of dissemination of GIFT fish to government stations (ROS) and private tilapia hatcheries are given in Table 11, 12 and 13. A total of about 688,250 of GIFT brood stock were disseminated during the duration of the project.

3.3. INTERNATIONAL COOPERATION

3.3.1. Establishment of INGA and continued support to its activities

Under the auspices of the GIFT Project, ICLARM with assistance from UNDP organized a series of consultative meetings with national program leaders from Asia and Africa to explore possible mechanisms for establishing a formal global forum for national scientists to exchange ideas, research methods, and genetic materials (see Table 14). Details of these workshops/meetings were given in the reports earlier submitted to UNDP. These meetings eventually led to the establishment of the International Network on Genetics in Aquaculture (INGA), patterned after a similar network on rice breeding (the International Network for Genetic Evaluation of Rice) coordinated by IRRI.

Thirteen countries in Asia and Africa are collaborating in research and exchange of genetic materials: Bangladesh, Peoples Republic of China, Cote d'Ivoire, Egypt, Fiji, Ghana, India, Indonesia, Malawi, Malaysia, the Philippines, Thailand, and Vietnam. ICLARM is the member-coordinator. INGA was inaugurated in 1993 by seed funding from the GIFT Project and has been supported since by ICLARM core funds and project

funds from the Government of Norway, ACIAR and IDRC, as well as operational funds from the member countries.

The GIFT Project team continued to provide significant technical and logistical support in the organization of the INGA Steering Committee meetings convened by the INGA Secretariat at ICLARM and the host country institutions. Specifically, significant contributions of GIFT were given during the Committee meetings held in 1994 (Thailand), 1995 (India) and 1996 (Egypt) and also during the SIFR/UNDP convened "International Fisheries Development Donors Consultation meeting held in Paris in October 1996.

3.3.2. Production and distribution of germplasm/improved fish breeds for evaluation in national programs

Fingerlings of improved tilapia breed (GIFT strain) were produced and disseminated to six countries in Asia and Pacific (Vietnam, China, Indonesia, Thailand, Fiji, Bangladesh) according to standard quarantine procedures developed in the GIFT Project. A total of 45,163 fingerlings were sent to these countries through INGA and DEGITA Projects from May 1994 to August 1997. A record of tilapia fingerlings transported to the INGA/DEGITA member countries is shown in Table 15. Protocols for testing of genetically improved breeds in national and regional institutes and protocols for dissemination of these fish breeds were developed by the GIFT Project in consultation with international experts and national program scientists.

3.3.3. Preparation of Standard Research Methodologies and Reporting Procedures for Evaluation of Fish Genetic Materials

The GIFT Project contributed to the preparation of the Draft Procedures for Evaluation of Fish Genetic Materials. The document was presented at the first INGA Steering Committee Meeting held in Bangkok, Thailand in 1994 and subsequently finalized in consultation with INGA Research Coordinator for use by national institutions of participating INGA member countries.

3.3.4. Preparation of the GIFT Manual of Procedures

The GIFT Manual of Procedures contains user-friendly training material prepared by the GIFT team. It has been developed using the United Nations TRAIN-X methodology. This manual documents the technical procedures developed by the GIFT Project and was based mainly on the Project experiences. The Manual's targeted audience/beneficiaries are the technical staff of the INGA member countries.

Two training workshops to finalize the Manual were organized in 1997: (a) Instruction for Trainers' was conducted in March 1997; and (b) Training to validate the Manual of Procedures (July/August 1997). Revisions from the Validation course conducted in July/August 1997 are presently being incorporated into the entire standardized training package.

Reports of the training courses conducted in relation to the preparation and validation of the Manual have been submitted to UNDP.

3.3.5. Training of Scientists from National Institutions Participating in INGA.

The training of scientists from national institutions was a significant activity in the GIFT Project, with emphasis on quantitative genetics and planning of national fish breeding programs.

The intensive training program on the "Application of Quantitative Genetics to Aquaculture" was held on 6 November - 2 December 1995 in Cavite, Philippines and attended by researchers from INGA national institutions. It was a major initiative undertaken by the Project. Apart from this, short-term training programs/study visit were also organized (see Table 16). A training program/study visit to the Philippines (GIFT Project site) in April 1994 was organized for three Indian scientists from Central Institute of Freshwater Aquaculture (CIFA) participating in the India Rohu Breeding Program.

The Project also provided significant technical assistance to national programs in developing their national fish breeding programs. For example, the Project Leader participated as a resource person during a national workshop on "Broodstock Management and Procedures for Avoiding Inbreeding in Fish Hatcheries" held in Bangladesh in March 1994. Apart from this, and under INGA through financial support from the Norwegian Ministry of Foreign Affairs, AKVAFORSK assisted national program scientists in Vietnam and Indonesia to develop breeding plans for several fish species. In Vietnam, breeding plans were developed in cooperation with scientists from Vietnam's Research Institute for Aquaculture 1 and RIA 2 for utilization of the GIFT strain Nile tilapia, mrigal and silver barb (see Final Report Part 2 Attachment 14). In Indonesia, breeding plans were developed in cooperation with scientists from the Research Institute for Freshwater Fisheries (RIFF), Research Station for Coastal Fisheries (RSCF) and the Research Institute for Coastal Fisheries (RICF) for tilapia, milk fish and common carp.

3.4. DEFINITION OF METHODS AND APPROACHES FOR APPLICATION TO OTHER SPECIES

3.4.1. Genetic Improvement of Carps in Asia

A proposal for collaborative research and training on "Genetic Improvement of Carps in Asia" was developed for implementation during the years 1997-2000. The proposal eventually led to a three-year regional technical assistance grant of ADB to implement the study in six countries of Asia (Bangladesh, China, India, Indonesia, Thailand, and Vietnam). The GIFT Project provided technical inputs during the Planning Workshop for the ADB Genetic Improvement of Carps Project held in India on July 24-28, 1997. The objective of this workshop was to discuss in detail the objectives, scope of work, workplans and expected outputs.

3.4.2. Future Directions for Research in Fish Germplasm Enhancement and Breeding

A Consultative workshop to develop "Strategies for Collaborative Research and Training in Application of Genetics to Increase Sustainable Aquaculture Production" was organized in conjunction with the INGA Steering Committee meeting held in June 1995 in Hyderabad, India

In 1996, the Project Leader submitted and presented to the ICLARM Board of Trustees a concept paper on ICLARM's future directions for research in germplasm enhancement and breeding. The primary focus of the paper was on developing a near term (5 years, 1996-2000) research agenda for work on carps and tilapias.

3.5. THE GIFT FOUNDATION INTERNATIONAL INC.

ICLARM and its partners established the GIFT Foundation as an independent non-stock non-profit organization primarily to continue the selective breeding research on Nile tilapia initiated by the GIFT Project and to undertake commercialization of the GIFT Strain through partnerships with the private sector. Commercialization of the GIFT strain was chosen as a strategy to generate the funding required to continue the selective breeding and research on the genetic materials assembled by the GIFT project and to sustain the "GIFT legacy" comprising the highly trained and dedicated GIFT project core staff members (18) in the Philippines, the improved GIFT strain, the gene banks, and the data base meticulously gathered over the past 10 years. The Foundation continues to maintain strategic research partnerships with the GIFT Project institutions.

The present structure of the Foundation has evolved through the years. For nearly four years, from 1993-1996, ICLARM in cooperation with the Philippine national institutions vigorously pursued a public sector approach (involving BFAR and CLSU) to continue the GIFT legacy. Thereafter, the partners realized that the complex breeding activities of the project and the maintenance of the core of highly trained staff would require a more commercially-oriented approach. A non-stock, non-profit Foundation was therefore established by the partners.

The GIFT Foundation at present distributes the GIFT strain using a network of private hatcheries that become members of the Foundation through a formal licensing agreement with the Foundation. This formal agreement includes provisions obligating the participating hatcheries to make regular research and development contributions to the Foundation in exchange for the privilege of using the Foundation breeders and the Foundation trade name and trademark. The Foundation became operational in January 1998 and operates its breeding and production facilities in the former GIFT Project site now called the "Center for Applied Fish Breeding and Genetics", built with funds from the UNDP, located in the campus of the CLSU. The GIFT Project partners have formally agreed to allow the Foundation rent-free use of these facilities for the first five years of its operations.

Additionally, many steps have been taken to ensure the widespread use of GIFT fish and GIFT technology in and outside the Philippines. The GIFT project had and is expected to continue to have global impact in a number of ways. During the GIFT project over 650,000 GIFT brood stock were disseminated throughout the Philippines. An additional 45,000 GIFT brood stock were disseminated in Bangladesh, China, Fiji, Indonesia,

Thailand and Vietnam. Together, these countries are among the major tilapia aquaculture countries in the world and several are major exporters of tilapia. These fish also form the base for further genetic enhancement in each country. Inevitably, in several countries which have received GIFT fish, the GIFT line is being crossed with local unimproved tilapia and thus the improved genome is entering other locally available tilapia. This impact, although difficult to measure, is that of top crossing and serves to upgrade the local fish.

Another mechanism bringing widespread impact is the transfer of the GIFT technology to other countries, as well as training in the technology. The trained staff from the project form the nucleus of staff for the GIFT Foundation International in the Philippines, thus continuing to use their skills in applied fish breeding research, development and management. Two hundred twenty-four researchers have been trained in 11 countries in quantitative genetics research and analysis methods. ICLARM is seeking further funds through INGA and other sources such as UNDP's TCDC program to further extension of the technology particularly to Asian countries.

The International Network for Genetics in Aquaculture has been formed and now comprises 13 developing country members, 10 associate members which are advanced scientific institutes. Funding is being provided and from ICLARM's core resources and alternated from donors such as Norway and IDRC. ICLARM in all its international fora, continues to publicize the work of INGA as a flagship research network, thus continuing to globalize the outcomes of the GIFT project. The GIFT research methods are available, and are being applied in the GIFT Foundation and in national breeding programs under INGA. The methods are simple, rigorous and effective in breeding better fish. Better still, the techniques result in immediate improvement soon after selection studies are started. This contrasts to many other genetic techniques for which the promised results are still 5-10 years off. The GIFT Foundation International model for public/private sector cooperation in delivering new breeds of fish to the farmers is receiving international attention. Several INGA member countries such as Indonesia and Vietnam are developing national tilapia breeding programs based on GIFT methods.

The method of germplasm collection, the selective breeding experiments and the chosen dissemination route are leading to pioneering studies on the definitions of intellectual property rights for fish. ICLARM has used this and other experience and knowledge to develop its Policy of Intellectual Policy Rights for Germplasm, a first for aquatic resources. The policy is drawing international interest as a possible model for use or adoption by other agencies.

The GIFT strain at this stage appears to be better than most competitors and is developing a reputation in the Philippines for being more profitable to grow than other strains. Early indications from talking to the hatchery and feeds sector in the Philippines and Europe are that the existence of a faster growing strain of tilapia is spurring the need to examine other efficiency factors in tilapia production. These factors include feed improvements, better breeding and hatchery controls, better farm management practices and close attention to marketing of the fish. Ex-ante assessment of likely impact, borne out by early farm results, are that the GIFT tilapia and other selective breeds will lower the price of tilapia for the consumer.

Continuing impact from the GIFT project is also expected. The GIFT Foundation has been formed and continues to distribute GIFT fish and further enhance Nile tilapia. Additional GIFT fish are distributed, but without the GIFT trademark by BFAR in the Philippines. ICLARM retains a breeding population at the GIFT Foundation for further enhancement and future dissemination. The GIFT Foundation cooperates with INGA for further international distribution. Recent generations of GIFT have been shipped to India, Thailand and Vietnam. Further international distribution may occur through private enterprise.

Funding and economic change may make either government or private enterprise more effective at different locations for disseminating germplasm to best benefit poor consumers. However, it is essential that government mechanisms are established to ensure that the poor fish farmer benefits as well.

4.0. PUBLICATIONS

As mentioned earlier, during the entire duration of the Project, the one major activity that suffered the most was the preparation of scientific publications. Although a number of publications have been prepared, about 10 scientific papers are in various stages of preparation. ICLARM and its partners are committed to complete these papers as they are crucial to understand the whole range of research activities conducted under the auspices of the GIFT Project. The status in respect of each of the major publication activity is described below.

4.1. Preparation of the book "Aquaculture Genetic Research Initiatives in Asia, Africa and the Pacific".

Status of Implementation:

The first draft of the book (Genetic Research in Tilapias and other Farmed Fish in Asia, Africa and the Pacific) was completed in April 1996. Technical editing has been completed for the majority of the papers in the book. Copy editing is in-progress. As of December 1997, about 14 papers have been copy edited and returned to technical editors for incorporation of inputs.

4.2. Preparation of GIFT Manual of Procedures

Status of Implementation:

Revisions from the Validation course conducted in July/August 1997 (see section 3.3.4. for details) have been incorporated in the entire standardized package of materials. The United Nations TRAIN-X methodology of preparing a standardized training package was followed in revision of the manual.

4.3. Scientific Papers/Presentations

Status of Implementation:

The Project has published a number of studies/papers in primary journal/books/proceedings and likewise presented papers in international conferences. As of July 1998 several papers from major and complementary experiments are in various stages of completion for publication. A detailed list of published papers and the status of pending papers are in Annex 4 of this report.

5.0. CONCLUSIONS AND RECOMMENDATIONS

The selective breeding program for Nile tilapia developed and implemented by ICLARM and its partners is the first systematic attempt to apply the principles of animal breeding theory to tropical aquaculture in a developing country setting with the primary objective of demonstrating: (a) the importance of genetics in improving production efficiency in aquaculture, (b) methods and mechanisms for dissemination of results and products to reach targeted beneficiaries effectively, and (c) methods to estimate *ex ante* the potential benefits to a wide spectrum of aquaculture enterprises. Successful demonstration could mean not only better tilapia breeds for fish farmers but also new breeding procedures for use with other species. ICLARM and its partners have also proceeded a step further in developing the non-stock non-profit GIFT Foundation International as a mechanism for continuation of the GIFT Program for the benefit of the tilapia industry in the Philippines and elsewhere.

The program has demonstrated the enormous gains in economic performance of farmed tilapia that are possible from a systematic selection program. The GIFT program has stimulated planning of and implementation of national fish breeding programs in Bangladesh, India, Indonesia and Vietnam. An International Network on Genetics in Aquaculture (INGA) was established in 1993. Bilateral and multilateral research initiatives under the INGA have begun to include other fish species, especially the carps which account for most of the finfish production in Asia.

ICLARM is now well poised to establishing itself as one of world leaders in Fish Germplasm Enhancement and Breeding. The decade long experience and the significant lessons learnt in implementing the GIFT and the DEGITA projects, the research forum offered by the GIFT Foundation, the INGA and the GICA initiatives, mark the beginnings of exciting research and development opportunities for ICLARM. ICLARM's unique position in the Consultative Group on International Agricultural Research as the only center devoted to fisheries research also brings with it new challenges. In executing its new responsibilities, ICLARM will build on and forge new and innovative working relationships with national fisheries centers.

The selective breeding technology generated through the GIFT Project is not an end in itself. It is in many ways the beginning of new research challenges and directions for ICLARM and its partners. Some of the research challenges emanating from the GIFT Project are as follows:

The need for systematic documentation and conservation of tilapia genetics resources was felt very strongly throughout the course of GIFT project implementation. It was also recognized that there exists a vast but scattered body of local knowledge about fish genetic resources and habitats which also are in urgent need of conservation as the germplasm itself. ICLARM's FISHBASE program is a step in the right direction. This activity, however, should be strengthened and could benefit through appropriate partnerships with projects and agencies which are currently engaged in such activities.

The main bottlenecks experienced in the GIFT breeding program were: (a) the amount of time required in producing the required numbers of families; (b) the time required to rear these families to a size when they can be tagged; and (c) the number of families that can be effectively tagged; and (d) the tag-losses experienced during rearing. Although corrections for the effects of various factors affecting (initial size, initial age, etc.) the early life history through novel statistical procedures and manipulations were possible, it is important that procedures for synchronous maturation of breeders and more efficient tagging techniques be developed. DNA fingerprinting is one possible technique of enhancing the efficiency of the selection program.

The focus of the GIFT program has been on a relatively simple but extremely important breeding objective: growth rate, with recording on other traits such as survival, occurrence of diseases, and maturation rate. The purpose of this exercise was to demonstrate the effectiveness of a properly planned selection program. In future experiments, it is important that the breeding objectives (in relation to the targeted species) are carefully examined. This should necessarily include a formal study on the benefits to the fish farmers arising from improvements in each trait.

The rigorous genotype by environment interaction experiments conducted under the GIFT Program did not suggest the need for several different strains for different production systems. However, it should be recognized that the rice-fish farming system has received very little attention in the GIFT project. It is possible that the importance of genotype x environment may emerge later, particularly when the GIFT strain is used in other countries.

The GIFT breeding nucleus should remain open in the medium term. There should be continuous comparative evaluations to screen potential genes for possible inclusion in the on-going selection program, and if the potential benefits from such genetic mixing will outweigh setbacks. Sex manipulations procedures (hormonal sex reversal, ploidy induction, or YY-male technologies) are important in the distribution process and should not be done in the breeding nucleus.

Finally, it appears that fish breeding research has come of age when overall funding from public multilateral development assistance sources to agricultural research has been declining. As economies are transformed and agriculture in general becomes more commercialized, private sector is also expected to play an increasingly important role in technology generation, in food production, and in ensuring food security. For ICLARM to advance its strategic research agenda of improving productivity in aquaculture to close the looming gap in demand for and supply of food fish estimated at about 100 million mt by the year 2025 and also to establish a strong research base for the evolving tropical aquaculture industry, it is important that ICLARM reaches out to building strategic alliances with the private sector. To this end, ICLARM has developed a pro-active policy

document describing the specific guidelines for mutually beneficial relationships with the private sector.

In summary,

1. Strain variation exists for growth rate of Oreochromis niloticus.
2. Immediate impact can be made by using the best strains.
3. In general, wild African strains grow better than domestic Asian strains.
4. Genotype-environment interactions occur for growth rate in all tilapia, and need to be considered.
5. Intraspecific crossbreeding improves growth rate up to 12% in Nile tilapia.
6. Seven generations of selection can improve growth rate by 77% or more (up to 123%) in Nile tilapia.
7. GIFT line grows faster than most genotypes of Nile tilapia in Asia and in most environments.
8. Fast growth and late sexual maturity are negatively correlated in Nile tilapia.
9. GIFT tilapia are positively impacting food production in Asia.

The successful GIFT program leads to the need to address the following research issues.

1. Selection to improve growth rate should be continued for GIFT Nile tilapia as selection plateaus have not been reached.
2. Crossbreeding and selection should be combined and evaluated to maximize genetic gain for growth in tilapia.
3. YY male technology and sex reversal should be combined with GIFT selection procedures and evaluated.
4. Correlated responses to selection for other important economic traits should be evaluated.
5. Additional heritabilities and genetic correlations should be calculated for the design of multiple trait selection programs.
6. Enough time has elapsed since the original distribution that post impact assessment of GIFT Nile tilapia should be conducted. In fact, since GIFT fish have been distributed by both government and private enterprise, an opportunity to assess the relative merits of different dissemination exists.
7. Research techniques should be evaluated to allow transfer of the successful GIFT technology to Africa. The UNDP/TCDC program is a possible mechanism to accomplish this.

ICLARM is seeking to address these research needs with the appropriate partner institution and follow-up funding is needed to maximize the benefit from the groundwork laid by the GIFT project.

6.0. EXPERIENCES AND LESSONS LEARNED

The GIFT Project has been a major multidisciplinary and multi-institutional initiative encompassing a range of complex research, training and human capacity strengthening activities. It was one of the longest running program implemented by ICLARM and its partners. The GIFT project as a whole, like any pioneering projects, is a story of research successes and also setbacks. The project evolved continuously through rigorous peer review and by taking stock of experiences gained at the end of each year of implementation. Almost every step that the project took along its long journey of ten years has been a learning experience. The team has braved many challenges, including some natural calamities (earthquakes, typhoons, and the Mount Pinatubo eruption).

The First Phase of the GIFT Project was almost exclusively devoted to development of research methods, development of strategic research directions, and development of equitable collaboration mechanisms among partners. This period was also punctuated with some serious discussions and debate on the GIFT Project's chosen research direction and strategy.

Although tilapias in general have the rare distinction of being the subject of more research than perhaps any other tropical farmed fish, it was recognized by the team that the literature on tilapia biology of some 2000 published papers lacked an interdisciplinary approach to tilapia farming. There was very little information that the project team could gain for direct implementation in its activities. The team had to invent and adapt methods for controlled mating, mass marking of individual fish, sampling schemes, anesthesia, recording of traits and ancestry. This had to be accomplished within a short span of about 8 months before major project experiments could begin. In preparation for the collection and transport of new Nile tilapia germplasm from Africa, the team also had to develop very quickly quarantine procedures in consultation with fish health experts.

For such a pioneering work, it is not surprising that the debate on GIFT Project's research direction was fierce at times. Some were concerned that that the GIFT team would bring in new fish diseases and parasites along with the new germplasm from Africa. This spurred the team on to develop the best possible quarantine safeguards for its transfers of fish – an imperative to this day. It was also alleged that the Project's stated objective of developing improved breeds of Nile tilapia could destroy the existing tilapia biodiversity in Asia. In fact, the whole experimental design was questioned. The GIFT team considered all criticisms carefully and prepared objective documents and scientific reasoning against these criticisms. The senior advisors to the UNDP were also involved in executive level discussions on these issues. Independent expert panels were commissioned by UNDP to unequivocally address all challenges and criticisms leveled against the GIFT Project Research Plan. Eventually, the GIFT approach was "commended" by senior scientist peers from the world of animal and plant breeding.

The dedicated collaboration spirit exhibited among partners has always been commended in all international fora. Although the team represented a diverse group of individuals with different educational and cultural background, there was a deep sense of togetherness during the execution of the project. Planning and implementation of project activities have always been on a consultative basis. The dedication of team members to

the goals of the Project, as observed by the External Advisory Panel can be seen in the Message from the Chair of the EAP.

As the project evolved into its second Phase, the complexity of activities increased. In preparation for implementation of the GIFT Phase II, ICLARM and its partners developed Memoranda of Understandings defining in most exclusive terms the roles and responsibilities of each institution. The planning of project activities was through a Project Management Committee, which was tasked to prepare detailed annual plans and budgets for review by UNDP and the External Advisory Panel. This mechanism was highly effective.

The GIFT Phase II also saw the expansion of project activities (and experiences) to other countries participating in DEGITA and INGA projects. This, admittedly, led to dilution of research thrust in the GIFT Project and also stretching of resources. As a result, some major activities of the GIFT Project, especially, the preparation of scientific publications and thorough data analysis of experiments conducted, were severely compromised. The GIFT Project would have greatly benefited by inclusion of at least one additional scientist at ICLARM. Unfortunately, the Phase II of the Project also coincided with several new parallel initiatives in ICLARM and core funds were not readily available for this purpose.

Substantial resources were devoted towards the evolution of the National Breeding Program and later the GIFT Foundation in the Philippines during the GIFT Phase II. For nearly four years, from 1993-1996, ICLARM in cooperation with the Philippine national institutions vigorously pursued a public sector approach (involving BFAR and CLSU) to continue the GIFT legacy. Thereafter, the partners realized that the complex breeding activities of the project and the maintenance of the core of highly trained staff would require a more commercial-oriented approach. A non-stock, non-profit Foundation was therefore established by the partners. The experience of AKVAFORSK in setting up the pioneering salmon breeding program in Norway was extremely valuable in this context.

At the time of writing this report the GIFT Foundation was already in its 8th month of successful operation as an independent organization. All the 18 GIFT Project core-staff members are employed in the Foundation and continue to contribute to the mission of the Foundation. The Foundation has developed separate memoranda of agreements with the original GIFT Project institutions to continue strategic research and development activities.

One of the very first lessons the GIFT Foundation has learned is that the continued growth and success of the tilapia industry in the Philippines, and possibly elsewhere, cannot be solely fueled by a selective breeding program, no matter how successful it is in incorporating various economically important traits in its selection program. A selective breeding program without feedback from a vigorous tilapia production systems research program may fail to meet farmers' real needs and thus may fail to contribute significantly to industry growth and success. In order to strategically position itself, the Foundation is already developing strategies to develop production systems technology, product quality, and effective customer orientation.



FIGURES



Figure 1.

**Schematic diagram of the
Generation 1 experiment to
estimate the magnitude of genotype x
environment interaction**

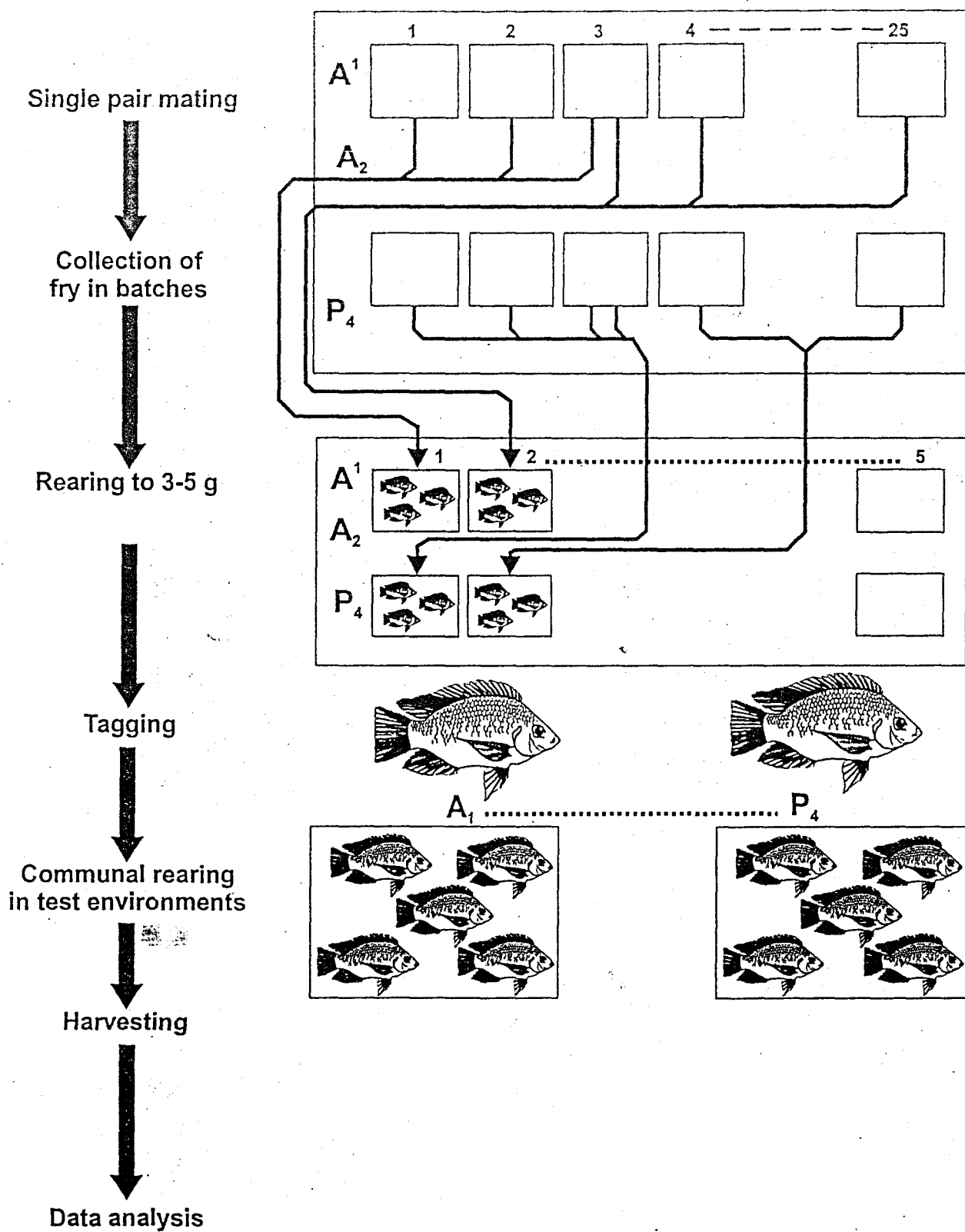


Fig. 1. Schematic diagram of the Generation 1 Experiment to estimate the magnitude of Genotype x environment interaction. A₁....P₄ refer to different Nile tilapia strains used.



Figure 2.

**Schematic Diagram of the Generation 2
experiment (8 x 8 diallel crossing) to
estimate the magnitude of additive- and
non-additive genetic effects**

Complete diallel crossing
(64-strain combinations)
eg: A₁A₁
Total hapas 384++



Rearing of batches
of fry to 3-5 g



Tagging



Communal rearing
in test environment



Harvesting



Data analysis

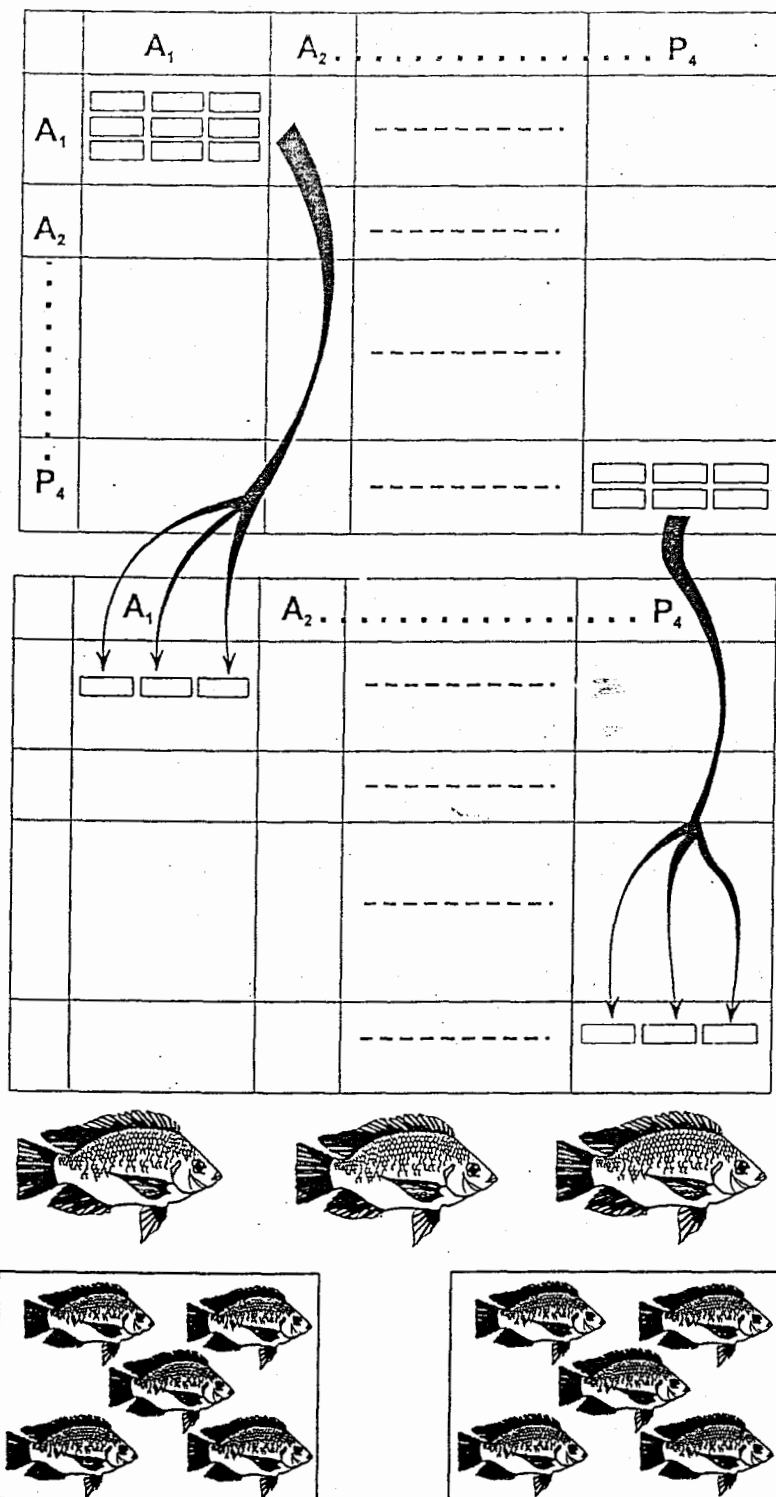


Fig. 2. Schematic diagram of the Generation 2 Experiment (8x8 Diallel Crossing) to estimate the magnitude of additive- and non-additive genetic effects.



Figure 3.

Schematic diagram of the

Selective Breeding experiments

(Generation 3 to 9) to estimate genetic

gains and response to selection

Nested mating
1 size: 2 different dams

Rearing of full sibs
to 3-5 g

Tagging

Communal rearing

Harvesting

Data analysis

- Estimation of breeding values
- Ranking of Potential Breeders
- Preparation of Mating Plan
- Selection of breeders

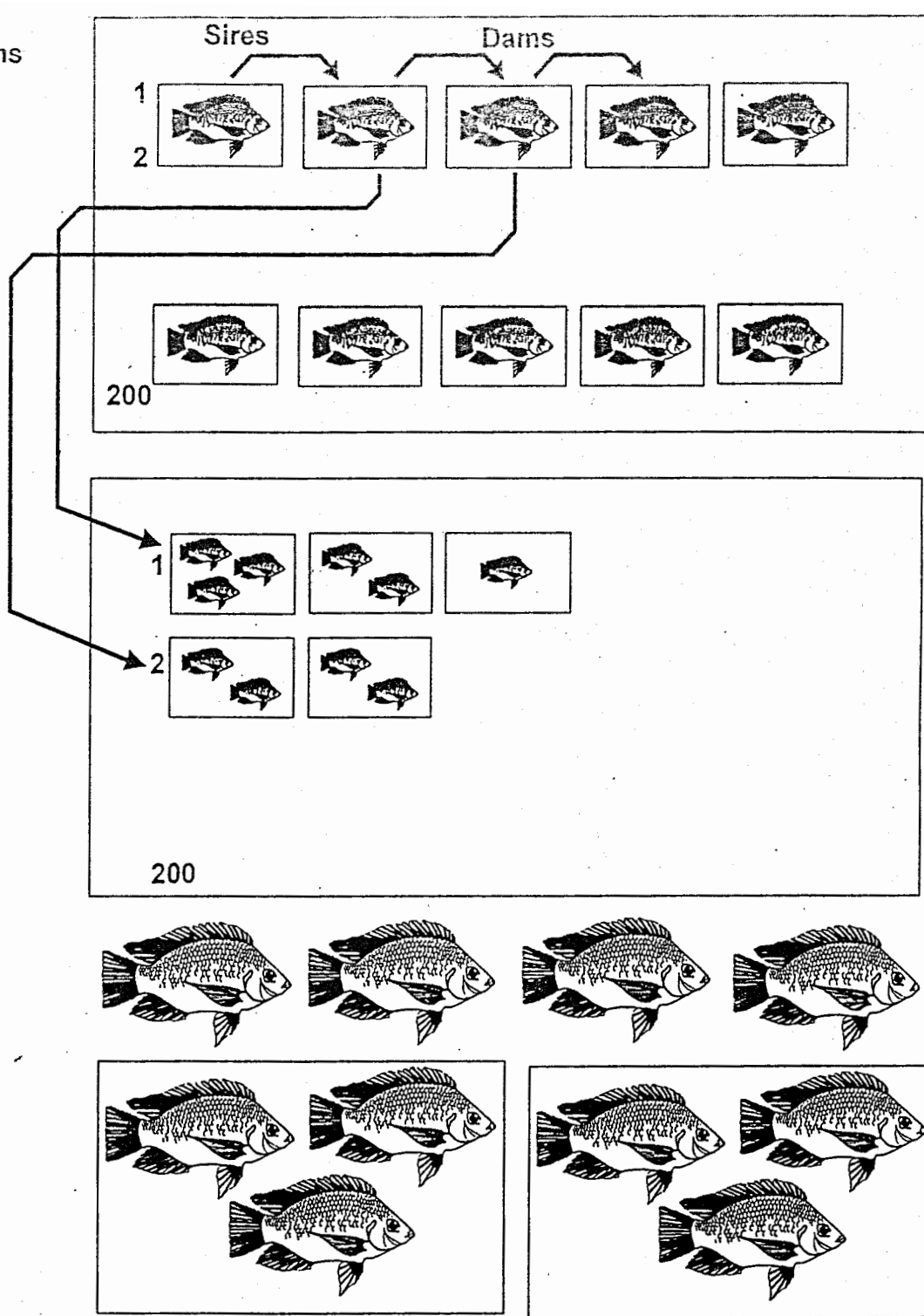


Fig. 3. Schematic diagram of Selective Breeding Experiments (Generation 3 through 9) to estimate genetic gains and response to selection.



Figure 4.

Observe response to selection across five generations of selection for growth performance. The cumulative response to selection in relation to the base population has been 85%

Accumulated selection response

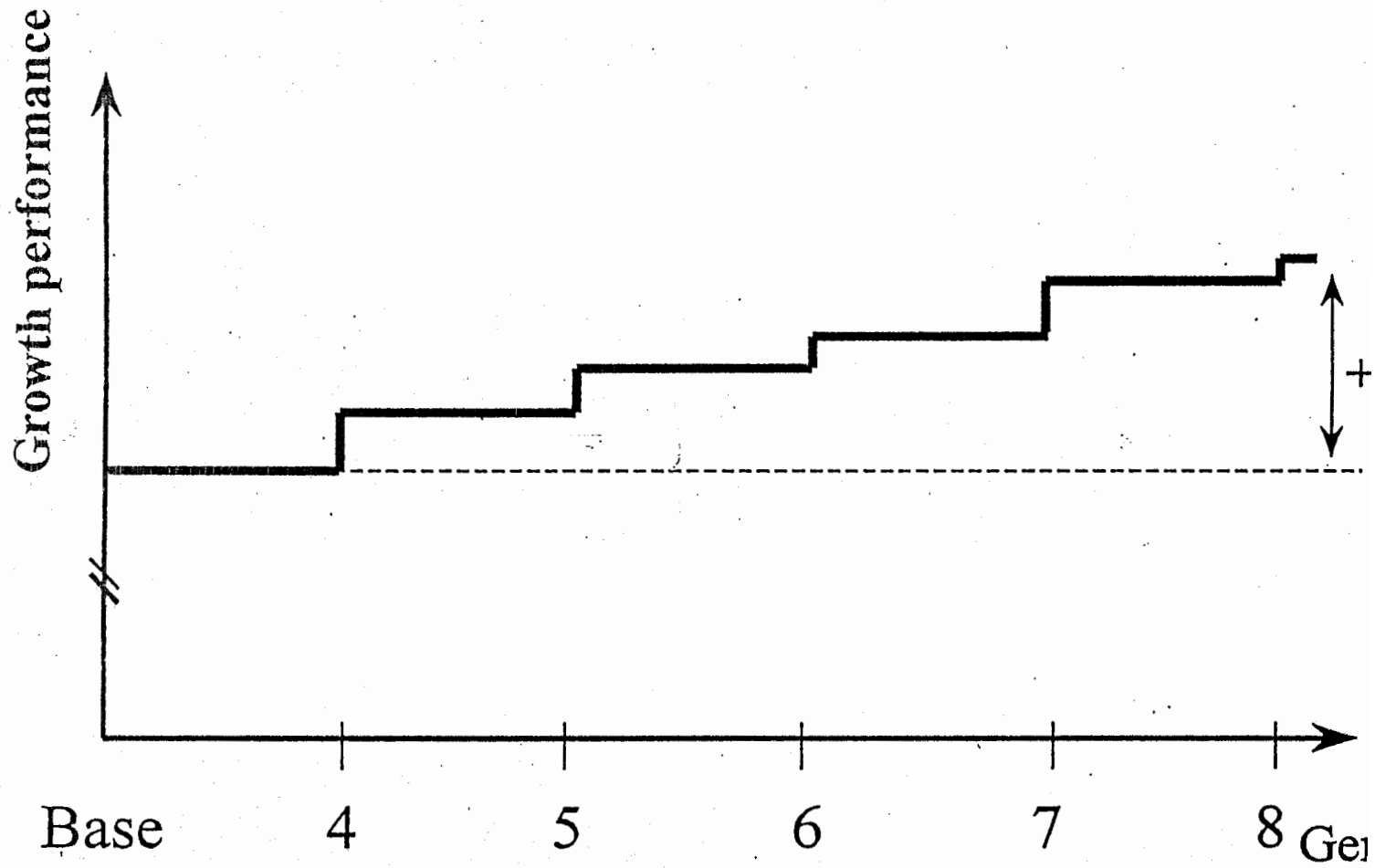


Fig. 4. Observed response to selection across five generations of selection for growth performance.

Selection Response

Generation	h^2 (se)	Selected, Mean, g	Control,* Mean, g	Response
3	.18 (.08)	-	-	-
4	.25 (.09)	48	38	+ 19.1
5	.18 (.08)	97	84	+ 13.5
6	.07 (.06)	146	133	+ 9.2
7	.13 (.07)	120	98	+ 17.8
8	.09 (.05)	157	147	+ 6, 2

* Average breeders in previous generation



TABLES



Table 1. Logframe:
Assessment of GIFT Project activities in relation
to its stated objectives

Table 1. Assessment of targeted activities, status of their implementation, and major outputs of the stated immediate objectives of the GIFT Project (March 1988 to December 1997)

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPLEMENTATION (as of December 1997)	OUTPUTS
3.1	CORE RESEARCH ACTIVITIES			
	<i>Immediate Objective 1: To develop improved breeds of tilapia and provide those breeds to national testing to the fish farmers</i>			
3.1.1	<i>Project Objective 1a: Documentation, collection, quarantine and genetic characterization of Nile tilapia</i>			
3.1.1.1	Documentation, Collection and Quarantine of Nile Tilapia Germplasm	Broaden the genetic base by bringing in pure Nile tilapia stocks from Africa; follow protocols of International Codes of Practice in importation of tilapia	Accomplished during Project Phase 1; initiated in 1988 and completed successfully in 1992; activity was interfaced with INGA activity during Project Phase II for further collections and transfers.	- Established diverse Nile tilapia germplasm (pure wild and improved) useful for further research to develop improved breeding programs - Developed quarantine and quarantine experience - Quarantine documented Procedures
3.1.1.2.	Biochemical and morphometric characterization of Nile tilapia germplasm held by the Project	Conduct biochemical and morphometric characterization of 8 founder stocks used as base population for GIFT strain	Done; technical details published in proceeding/ scientific journal	- Thorough biochemical and morphometric procedures 6 scientific publications
3.1.2	Maintenance of live and cryopreserved tilapia genebank	Establish protocols for maintenance of live cryopreserved genebank	Activity being continued. Note: on termination of the Project in Dec. 1997, the gene bank is being maintained by BFAR/ NFFTRC & GFII	Establishment of cryopreservation collections for research and national breeding

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPMENTATION (as of December 1997)	OUTCOMES
				• Protocols for live genebank • Protocols for developed Institute of University • Establishment cryopreservation • Duplication collections • Establishment Reference
3.1.3	Generation 1 (Genotype x Environment interaction experiment)	Estimate the magnitude of GxE interaction; decide whether or not specialized strain is necessary for each farm environment and its implication in breeding program	Successfully carried out from March to November 1989; technical details published in scientific journal	• Strategic development strain and programs to investigate system not performing mixed together population • 4 scientific
3.1.4	Diallel Cross Experiment (Generation 2)	Determine the breeding strategy: purebreeding or crossbreeding	Successfully carried out from January to October 1990; Technical details published in a scientific journal	• strategic development a) simple commercial

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPMENTATION (as of December 1997)	OU
				base pop genetic approach improve b) Based on perform 25 best and cross the G4 e chosen based p Three scier published
	Project Objective 1d: Building the base population and starting the selection program (Project's Experiment)			
3.1.5	Generation 3 (Building the base population)	Estimate the genetic parameters (heritability, genetic variation and genetic correlation) of survival, growth and reproduction traits	The study was conducted between November 1990 to June 1991; estimation of genetic parameters for generation data analysis across generation in-progress 3 datasets completed;	- Substantial growth and population - improved t performs t Philippine strain - two major in prepara
3.1.6	Project Objective 1e: Selection and estimation of genetic gains			
3.1.6.1	(i) Selection for growth and (ii) selection for two traits (spawning and frequency of spawning females)	Estimate the response to selection (magnitude of genetic gain)	selection for growth completed; analysis of datasets in progress; manuscripts based on reanalyzed datasets in-preparation	Genetically breed and gains.

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPMENTATION (as of December 1997)	OU
3.1.6.2	Estimation of the magnitude of genotype (families) x season interaction	determine if there is a need to establish two separate lines depending on performance during major growing seasons	selection for two traits (growth and frequency of spawning females) is being continued under GIFT Foundation two separate experiments conducted in 1993 and 1995; preliminary analysis done; pre-corrections of data made and model for data analysis	- Five genera completed. 12-17% per five genera was observed increase in relative to t - Initiation of developing tilapia and p multi-trait s growth and - Standardiz controlled mass indiv recording o data, growt - Improved r methodology software for genetic par response to applicable country aq and develo - Major scien complemen preparation - No Signific Strategic d separate lin considered

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPLEMENTATION (as of December 1997)	OUTCOMES
3.1.6.3	Estimation of genetic parameters for maturation	investigate genetics of frequency of spawning females per family	developed. Manuscripts based on re-analyzed datasets to be prepared. experiment completed. Re-analysis of data initiated; Manuscript to be prepared as soon as data analysis is completed	- Major paper preparation - Significant between groups - Strategic decisions include major economic trends - Major paper preparation
3.1.6.4	Response to selection for late and early spawning females		experiment completed; presented at 6th ISGA meeting	One paper in Aquaculture
3.1.6.5	Determination of indirect markers to improve survival	a) Evaluate lysozyme activity (LA) and spontaneous haemolytic (SH) activity b) Determine genetic variation in LA and SH	Experiment completed as part of Ph.D thesis by a student in AKVAFORSK	Two scientific review
3.1.6.6	<i>Specific genetic studies to design efficient selection program</i>			
	Replicate effects of separate rearing of families		Experiment completed; draft manuscript being reviewed	paper for preparation
	Sex reversal and evaluation of growth performance of monosex against mixed sex culture	Asses how the genetically improved breed of tilapia perform under the mixed or monosex culture technique	Experiment completed; Study presented at Genetics Workshop in Thailand in 1994; Paper reviewed and data were re-analyzed;	paper for preparation
	Effects of communal and separate rearing of test groups	Determine whether communal stocking of control groups has any effect on growth performance & thereby estimated response to selection	Experiment completed; manuscript under review. Revision of paper will be made based on revised analysis of data	paper for preparation

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPLEMENTATION (as of December 1997)	OUTPUTS
3.1.6.7	Growth performance in different farming systems	Determine response of the GIFT fish under a broad range of management schemes	Seven complementary studies completed	• Information performance breed under production • Two papers by FAC/CLS • Major paper
3.1.7	Commissioned Reviews to investigate the genetics of resistance to biotic and abiotic stress			
	Genetics of disease resistance	Commission on systematic study and review on genetics of parasites and disease resistance	Revised version of Review on genetics of parasites and diseases by AKVAFORSK (T. Gjedrem) submitted to ICLARM	Completion of r of parasites and of Nile tilapia" <i>Note: To be publ Research i Farmed Fi the Pacific</i>
	Genetics of salt tolerance	Commission systematic studies and reviews on salt tolerance	Not accomplished as a commissioned review due to lack of resources Special session on "Development of salt tolerant tilapia" held in conjunction with 3rd INGA Steering Committee in Cairo, Egypt, 8-12 July 1996	• Developme conducting development Nile tilapia of brackish
3.1.8	Project Objective 1e, Activity 3: Database Management			
3.1.8.1.	Standardization of datasets and preparation of data dictionary	Complete standardization of datasets generated since 1988 to enable efficient data analysis; prepare data dictionary	Accomplished as planned	• Standardize Generation • Developme and docum generations

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPMENTATION (as of December 1997)	OU
3.1.8.2	User - friendly software (within SAS) for routine analysis of genetics data	Produce stand-alone software for analysis of genetics data	Done	• Application GIFT Stat a manual
<i>Regional on-farm trials and impact assessment</i>				
3.1.9	On-farm trial and assessment of socioeconomic benefits	Evaluate performance of GIFT under on-farm conditions and asses its genetic, socioeconomic, and environmental impacts	GIFT Project conducted on-farm experiments from 1991-1993; the study was interfaced with ADB-DEGITA Project in 1994 and was completed in June 1997	• Developme methodolo station and performanc non-Gift st • GIFT strain DEGITA m • Comprehe GIFT strain conducted • Baseline d: Philippine generated • One major the Interna published • Another or preparation
3.2	TRAINING AND INFORMATION DISSEMINATION			
<i>Immediate Objective 2: To strengthen national institutions in aquaculture genetics research</i>				
<i>Activity 1: Training and Assistance to Plan National Fish Breeding Program</i>				
3.2.1	Training of Project Staff	Enhance the professional capabilities of technical personnel in national partner institutions and GIFT Project	Done; training on quantitative genetics conducted for leaders of GIFT partner institutions in Norway in 1988; intensive training on "Application of Quantitative Genetics in Aquaculture"	• Trained GI

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPMENTATION (as of December 1997)	OUTCOMES
3.2.2.	Training and supervision of graduate students	-	given to GIFT core research staff in 1995; Six training workshops conducted for staff from 1989 to 1993 aside from hands-on training on computer software. 5 students directly involved in Project completed MSc; out of 5 students, 3 received outstanding mark for thesis; Six FSP supported students likewise completed MSc	• Advisory se MSc studen
3.2.3	Training in support of national fish breeding programs	Establish national breeding programs	Substantial progress made in establishment of GIFT Foundation and initiation of national breeding program in Indonesia and Vietnam during the GIFT Phase II	Nineteen se workshops Philippines September formulate d of GIFT fish on GIFT per farmers and Station and the GIFT Fc International: • About 688, ; disseminate Governmen • Developmer the establis Foundation for dissemin sustainabili • Establishm Foundation (GFII)

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPLEMENTATION (as of December 1997)	OUTCOMES
3.3	INTERNATIONAL COOPERATION			- Potential in Foundation - Planning a national fish in Bangladesh, Thailand and stimulated
<i>Immediate Objective 3: International Cooperation and Networking</i>				
3.3.1	Establishment of a formal forum for exchange of methods and genetic materials	Provide support as needed	Done, significant technical and logistical support provided during the INGA Steering Committee meetings held in 1994 (Thailand), 1995 (India) and 1996 (Egypt); Assistance provided in formal presentation of INGA during donor consultation meeting held in Paris in 1996	- INGA established - Continuous logistical support
3.3.2	Production and distribution of germplasm for evaluation		Accomplished through the INGA Project	A total of 45 distributed in Indonesia, through INC Projects.
3.3.3	Preparation of Standard Research Methodologies and Reporting Procedures for Evaluation of Fish Genetic Materials		Completed and adopted by INGA member countries	Standard Research Methodologies and Reporting Procedures for Evaluation of Fish Genetic Materials in coordination

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPMENTATION (as of December 1997)	OU
3.3.4	Preparation of GIFT Manual of Procedures		Volume 1 completed; validation process conducted for researchers in Philippine national institutions; revisions based on results of validation incorporated	• GIFT Manual (volume 1);
3.3.5.	Training of national program scientists	Enhance the professional capabilities of technical personnel in national institutions through training	Done, One intensive training program (four weeks) on quantitative genetics implemented for national participants from INGA member countries; Two separate short-term trainings/ study tours organized for 2 scientists from RIA, Vietnam and 3 scientists from CIFA, India	• Trained national scientists
3.4	DEFINITION OF METHODS AND APPROACHES FOR APPLICATION TO OTHER SPECIES			
3.4.1.	Strategic planning for research, training and networking	Organize consultative workshops to develop strategic plans for fish genetic research	Consultative workshop to develop "Strategies for Collaborative Research and Training in Application of Genetics to Increase Sustainable Aquaculture Production" organized in June 1995 in India; Paper on ICLARM's future directions for research in germplasm enhancement and breeding conceptualized by Project Leader Developed a proposal on "Genetic improvement of Carps in Asia" which led to 5-year ADB	• Technical Assistance to implement 1 Improve 5 countries • Provision of inputs during Workshop 1 Assistance Improve held in India

CODE	ACTIVITY	PLANNED TARGET	STATUS OF IMPLEMENTATION (as of December 1997)	OU
4.0	PUBLICATIONS			
4.1	Book (Genetic Research in Tilapias and other Farmed Fish Species in Asia, Africa and the Pacific")		technical editing of the initial draft completed; copy editing in-progress	
4.2.	GIFT Manual of Procedures		see section 3.3.4 above for details	see section 3.3
4.3.	Scientific Papers		completion of all pending papers from major breeding experiments and complementary experiments in-progress	Note: See details of the report



**Table 2. Details of Nile tilapia
(Oreochromis niloticus) germplasm
collected by the GIFT Project from Africa**

Table 2. Details of Nile tilapia (*Oreochromis niloticus*) germplasm collected by the GIFT Project from Africa.

Strain Code	Collection sites	Date	Numbers collected
E1	Egypt (First Collection)	May 1988	
	1) Lake Manzallah		30 breeders
	2) Creeks along Desert road to Port Said		25 breeders
	3) Lakes around Alexandria		70 fingerlings
E2	Egypt (Second Collection)	August 1989	
	1) Abassa		60 breeders
	2) Ismailia		90 breeders
E3	Egypt (Third Collection)	October 1992	
	1) Monsour		22 breeders
	2) Manzalla		41 breeders
	3) Timsah Lake		63 fingerlings
	4) Ismailia		7 fingerlings
	5) Abassa		153 fingerlings
	6) Mariut		266 fingerlings
	7) Suez Canal		4 breeders
	8) Idku		10 breeders
Gh	Ghana Volta River System	October 1988	200 fingerlings
Ke	Kenya First generation progeny from a founder stock collected in August 1988	August 1989	800 fingerlings
Se	Sénégal	October 1988	
	1) Dagana		120 breeders
	2) Dakar-Bangos		40 breeders
	3) Mbane		40 fingerlings



**Table 3. Origin of the four farmed
Nile tilapia (Oreochromis niloticus)
strains used by the GIFT Project
in the Philippines**

Table 3. Origin of the four farmed Nile tilapia (*Oreochromis niloticus*) strains used by the GIFT project in the Philippines.

Strain Code	Popular name	Origin
Is	Israel	Derived from founder stocks of Ghana origin kept in Israel. The original founder stock (1974) was 9 females and 2 males. Fry from 100-200 single pair mating shipped to the Philippines in 1979.
Si	Singapore	Descended from a founder stock of Ghana origin shipped from Israel to Singapore, then to the Philippines in 1979.
Th	Thailand	Egypt origin. Introduced in the Philippines from Thailand in 1987. The Thailand founder stock was introduced from Japan in 1965 (50 fish formed the founder stock, however, the number which actually survived to breed is not clear).
Tw	Taiwan	Descended from the founder stocks introduced to the Philippines from Taiwan in 1983-84; previous history not certain but most likely of Ghana origin.



**Table 4. List of Nile tilapia
(Oreochromis niloticus) germplasm
maintained by the GIFT Project at
NFFTRC/BFAR and GFII Facilities**

Table 4. List of Nile tilapia (*Oreochromis niloticus*) germplasm maintained by the GIFT Project at NFFTRC/BFAR and GFII¹ Facilities.

Origin	Date of Collection	Present No.	
		M	F
FOUNDER STOCKS ²			
EGYPT (First Collection)	May 1988	10	26
EGYPT (Second Collection)	August 1989	8	23
EGYPT (Third Collection)		82	127
1) Monsour	October 1992		
2) Manzalla			
3) Timsah Lake			
4) Ismailia			
5) Abassa			
6) Marriot			
7) Idku			
<i>Unreadable tags</i>			
GHANA			
1) Volta River System	October 1988	1	20
KENYA			
1) First generation progeny from a founder stock collected in August 1988	August 1989	32	18
SENEGAL			
1) Dagana	October 1988	30	6
2) Dakar-Bangos			
ISRAEL			
(Ghana)	1974	45	60
SINGAPORE			
(Ghana)	1979	46	51
TAIWAN			
(Ghana)	1983-84	37	26
THAILAND			
(Egypt)	1987	54	31

Table 4. Cont'd. / . . .

Origin	Date Produced	Present No.	
		M	F
REPLACEMENT STOCKS			
GHANA		31	32
KENYA		43	92
SENEGAL		8	6
ISRAEL		28	25
SINGAPORE		15	15
TAIWAN		12	11
THAILAND		36	9
EGYPT (Third Collection)		78	92
1) Monsour			
2) Manzalla			
3) Timsah Lake			
4) Abassa			
5) Marriot			
Unreadable tags			
SELECTED STOCKS ³			
G2 (Top ranking crosses)		14	66
G3 (Base population)		22	45
G4 (First selection)		28	37
G5 (Second selection)		46	61
G6 (Third selection)		46	79
G7 (Fourth selection)		43	127
G8 (Fourth selection - second best)		62	131
G9 (Fifth selection)		71	147

¹ Based on the latest actual inventory of stocks conducted in October 1997² Currently being maintained at the facility of NFFTRC/BFAR³ Currently being maintained at the GFII facility



**Table 5. Summary data on
cryopreserved sperm of Nile tilapia
(Oreochromis niloticus) stocks, held by
the GIFT Project at Nueva Ecija,
Philippines**

Table 5. Summary data on cryopreserved sperm of Nile tilapia (*Oreochromis niloticus*) stocks, held by the GIFT Project at Nueva Ecija, Philippines.

STRAIN	NO. OF FISH SAMPLES CRYOPRESERVED	STRAW NO. PER FISH
FOUNDER STOCKS		
ISRAEL	35	1-7
SINGAPORE	52	1-10
TAIWAN	49	1-7
THAILAND	28	1-7
EGYPT	22	1-7
GHANA	41*	1-7
KENYA	33	1-6
SENEGAL	21*	2-7
SELECTED STOCKS		
CROSSES	52	
BASE POPULATION	68	1-6
FIRST SELECTION	54	1-7
SECOND SELECTION	54	2-7
THIRD SELECTION	72	1-6
FOURTH SELECTION	53	1-7
FIFTH SELECTION	44	1-9
IDRC	83	2-6

* Includes Replacement Stocks



Table 6. Total number of individual records, number of test environments, number of sires and dams, used in dfREML analyses of growth performance

Table 6. Total number of individual records, number of test Environments, number of sires and dams, used in dfREML analyses of growth performance.

Generation	Number	Environment	Sires	Dams
3	13,544	9	34	85
4	15,994	8	111	193
5	9,112	2	149	234
6	7,323	2	174	244
7	7,128	3	107	171
8	8,017	4	127	180



Table 7. Estimated heritabilities (standard errors, se) for growth performance and observed response to selection, estimated as differences between progeny of selected individuals and progeny from breeders with average breeding values

Table 7. Estimated heritabilities (standard errors, se) for growth performance and observed response to selection, estimated as differences between progeny of selected individuals and progeny from breeders with average breeding values.

Generation	h ² (se)	Observed Response %
3	0.18 (0.08)	----
4	0.25 (0.09)	+19.1
5	0.18 (0.08)	+13.5
6	0.07 (0.06)	+9.2
7	0.13 (0.07)	+17.8
8	0.09 (0.05)	+6.2



Table 8. Estimates of survival in generation 3 through 8. Number of tagged individuals stocked (N-Stock), number of individuals recovered after rearing (N-Survival) and percentage survival (survival %) in each test environment. Explanation of environment codes and characteristics are in Attachment 6 of GIFT Final Report Part 2.

Table 8. Estimates of survival in generation 3 through 8. Number of tagged individuals stocked (n-stock), number of individuals recovered after rearing (n-survival) and percentage survival (survival %) in each test environment. Explanation of environment codes and characteristics are in Attachment 6 of GIFT Final Report Part 2.

Test Environment	N-Stock	N-Survival	Survival %
Generation 3			
BAIL	2,461	1,677	68
BFAR1	2,461	1,747	71
BFAR2	2,455	1,776	72
BFCG	1,209	1,016	84
FAC1	2,425	1,675	69
FAC2	2,431	1,554	64
FWAS	2,397	1,631	68
PNCG	1,167	768	66
SMAT	2,461	1,723	70
TOTAL	19,467	13,567	69
Generation 4			
BAIL	2,753	2,471	90
BFAR1	2,753	2,568	94
BFAR2	2,742	2,430	89
BFCG	2,737	2,615	96
FAC1	2,738	1,883	69
FAC2	2,738	2,251	82
FWAS	2,735	1,048	38
SMAT	2,747	1,758	64
TOTAL	21,943	17,024	77
Generation 5			
BFAR	5,630	4,834	86
FAC	4,541	3,827	84
TOTAL	10,171	8,661	85
Generation 6			
BFAR	5,274	3,732	71
FAC	4,300	3,076	72
TOTAL	9,574	6,808	71
Generation 7			
BFAR	2,168	1,133	52
FAC	4,454	3,450	77
PNCG	2,200	1,802	82
TOTAL	8,822	6,385	72
Generation 8			
BFAR	4,623	2,514	54
FAC	3,971	1,787	45
PNCG	1,927	1,049	54
RF	2,134	477	46
TOTAL	12,655	5,827	46



Table 9. Summary of training conducted for technical personnel in national institutions and GIFT Project

Table 9. Summary of trainings conducted for technical personnel in national institutions and GIFT Pro

Title of Training	Objectives	Dates Conducted	Venue	Participi
Research Planning Workshop	To familiarize GIFT Staff on quantitative genetics; Develop Project Research Plan	March to June 1988	AKVAFORSK, Ås, Norway	Leaders of all collaborating institutions
First training Workshop on Quantitative Genetics of Farmed Tilapias	Analyze growth data obtained from the project's first generation breeding experiment; Discuss workplans for the second generation experiment; Introduce the basics of quantitative genetics	29 October to 4 November 1989	FAC/CLSU, Muñoz, Nueva Ecija	GIFT core res representative SEAFDEC.
Introduction to SAS (Statistical Analysis System)	Orient Gift staff on the use of SAS software and apply in analysis of datasets	17 - 20 January 1990	FAC/CLSU Muñoz, Nueva Ecija	GIFT Core Re staff
Data analysis using SAS	Practice analysis of data using SAS	2 - 4 May 1990	FAC/CLSU Muñoz, Nueva Ecija	GIFT Core Re staff
Second training Workshop on Quantitative Genetics of Farmed tilapias	Review the basic concepts of quantitative genetics; analyze the growth data from the project's first and second generation experiments and define strategies for building the base population and planning of the selective breeding experiment.	22 - 27 August 1990	FAC/CLSU, Muñoz, Nueva Ecija	GIFT Core res
Third training workshop on Quantitative genetics of farmed tilapias	Dry-run presentations of the project paper for ISGA, Wuhan, China on 29 - April - 3 - May 1991	22 - 26 April 1991	FAC/CLSU, Muñoz, Nueva Ecija	GIFT Core res
Fourth Training Workshop on Quantitative Genetics of Farmed Tilapias	Analyze data from the Third generation breeding experiment; discuss workplans for the next breeding experiment	19 - 25 August 1991	FAC/CLSU Muñoz, Nueva Ecija	GIFT Core Re staff
Basic Statistical Method for Genetics Data Analysis	Practice application of statistical methods to fish genetics research	February / March 1992 (12 days)	FAC/CLSU Muñoz, Nueva Ecija	GIFT Core R staff

Title of Training	Objectives	Dates Conducted	Venue	Participants
Fifth Training Workshop on Quantitative Genetics of Farmed Tilapias	Formulate structure of Philippine National Tilapia Breeding Program: discuss results of previous breeding experiment	10 - 13 May 1993	FAC/CLSU Muñoz, Nueva Ecija	GIFT Core R staff; Private
Sixth Training Workshop on Quantitative Genetics of Farmed Tilapias	Analyze datasets from the Project's Fourth Generation Experiment; review on previous generation datasets and analysis.	18 - 23 May 1992	FAC/CLSU Muñoz, Nueva Ecija	GIFT Core R staff
Basic concepts of immunology in fish breeding	Discuss/apply specific and non-specific immunity, immunological methods and vaccination to fish breeding	March 10-11, 1994	NFFTRC/BFAR Muñoz, Nueva Ecija	GIFT Core R Staff, UCS and FAC/CLSU researchers
Application of Quantitative Genetics to Aquaculture^a	Develop a core group of scientists who would interact as equal partners, under the auspices of INGA in pursuing a focused research agenda in aquaculture genetics, with special emphasis on development of national fish breeding programs.	6 November-2 December 1995	IIRR, Silang, Cavite	GIFT Core R staff
Training of GIFT Methodology	Conduct the Initial Validation Course for the GIFT Manual of Procedures Volume I	6 July - 2 August 1997	NFFTRC/BFAR Muñoz, Nueva Ecija	Researchers, Instructors from Philippine National institutions
Training of Potential Instructors for Validation of GIFT Manual of Procedures	Orient the trainers who will be involved in delivery of GIFT Course and who will use the training material prepared according to UN TRAIN-X methodology	15-18 March 1998	NFFTRC/BFAR Muñoz, Nueva Ecija	GIFT Core R Staff

^a Similar training was participated in by 23 scientists from national institutions of INGA member countries.



**Table 10. Summary of training facilitated/
conducted by the GIFT Project in support of
the National Tilapia Breeding Program
in the Philippines**

Table 10. Summary of trainings facilitated/conducted by the GIFT Project in support of the national program in the Philippines.

Title	Date/Place Conducted	Participants	No. of persons Trained	Output
<i>Trainings to formulate distribution strategies of GIFT fish:</i>				
Consultative Planning Workshop	March 9-10, 1993 BFAR/NFFTRC Muñoz, Nueva Ecija	Regional Directors (Reg. I-V, DA), FAC/CLSU, ICLARM	14	- Classified DA-regional into Stations (ROS) and Provincial (PSS) - Proposed release of funds
Sixth Training workshop on Quantitative Genetics	May 10-13, 1993 BFAR/NFFTRC	GIFT Staff, ICLARM, AKVAFORSK, FAC/CLSU, BFAR/NFFTRC	18	Drafted PNTBP Structure Launch
Meeting to Discuss the Strategies for a Self-Sustainable PNTBP	Jan 20, 1994 Bureau of Soils, Quezon City	UNDP, ICLARM, BFAR, DA, BAR, DA-Regional Directors Farm Managers/Aquaculturist	24	Suggested strategies: - establish non-profit organization - non-profit fisheries organization - active participation of private
The PNTB: Strategies for Dissemination & the Role of Multiplier Stations	January 22, 1994 Bureau of Soils, Quezon City	DA-Regional Directors Farm Managers, Aquaculturist	15	- Defined functions and responsibilities - Outlined partnerships of research
	April 21-22, 1994 BFAR/NFFTRC	ROS (Reg. III, IV, VI, XII), NGO	11	-do-
<i>Trainings to gather feedback on GIFT fish performance from farmers and Research Outreach Stations</i>				
Genetic Research & the Tilapia Farming Industry	Feb. 24-25, 1994 BFAR/NFFTRC Muñoz, Nueva Ecija	Private hatchery operators, ROS (Reg. I-V), farmers, Technicians	66	Demonstrated GIFT methodology

Title	Date/Place Conducted	Participants	No. of persons Trained	Output
<i>Trainings to disseminate information on GIFT fish.</i>				
FSP-Hands-on Training on GIFT (First Batch)	July 4-9, 1994 BFAR/NFFTRC Muñoz, Nueva Ecija	DA Representatives - Regions I-XI, Agriculturists, Teachers	27	Lecture on GIFT methodology breeding
(Second Batch)	Oct. 21-25, 1994 BFAR/NFFTRC	Aquaculturists, Technicians, DA-Reg. I, III, IV, V, VII, VIII, IX, X, XII	25	
Training of FSP Extension workers	August 24, 1994 Sta. Barbara, Pangasinan	Fisheries Sector Program extension workers of Reg. I	30	Discussed the relevance of far orientation
Culture of GIFT Fish	August 28, 1995 Sta. Barbara, Pangasinan	LandBank Personnel	20	-do-
Economics & Culture Management	May 23, 1996	Agusan del Norte Farmers	58	Explained GIFT culture & mng
	May 25, 1996	Davao del Norte Farmers/ prawn operators North Cotabato Farmers	41	-do-
	May 26	Sultan Kudarat Farmers	34	Discussed Tilapia biology
	May 28	SOCSARGEN,	69	-do-
	May 29	Gen. Santos City	30	-do-
GIFT Orientation	September 26, 1997 Taal, Batangas	Cage operators	30	Discussed the cage culture of

Title	Date/Place Conducted	Participants	No. of persons Trained	Output
<i>Trainings which led to evolution/operationalization of GIFT Foundation International Inc.</i>				
Status of GIFT grow out in Demonstration and Private Farms	December 1, 1994 BFAR/NFFTRC Muñoz, Nueva Ecija	Private Hatchery operators	28	Feedback on GIFT performance discussed: • less fry production • late maturing • fast growing • white coloration after
	January 20, 1995	Private Hatchery operators	23	-do-
Status of ROS in the Multiplication & Dispersal of GIFT Fish	October 13-14, 1994 -do-	DA-ROS Region Representatives, Aquaculturists, LandBank of the Philippines, BFAR representatives	18	• Assessed multiplication & fish • Identified ROS problems
	May 29-31, 1995 -do-	-do-	15	• Presented regional profile
	May 13-17, 1996 -do-	DA-ROS (Reg. I, II, III, IV, V, VI, VII, VIII, IX, X, Aquaculturists, Agriculturists), BFAR, LBP	8	• Discussed status of ROS re defined specific roles when
Finalization of Legal Aspects & Distribution of GIFT Broodstocks to Private Licensed Hatcheries	June 15, 1995 BFAR/NFFTRC Muñoz, Nueva Ecija	Accredited Hatchery Operators, ICLARM, BFAR, FAC Representatives	7	• Signed letter of Commitment • Agreement to pay license &

Title	Date/Place Conducted	Participants	No. of persons Trained	Output
Special Meeting on Technical Issues of a Foundation	September 15, 1995 -do-	GIFT Staff, AVKAFORSK, BFAR Representatives	14	• Outlined the gene flow from farmers
Conceptual Framework for Philippine National Tilapia Breeding Program	February 26-28, 1996 Sulo Hotel, Quezon City	ICLARM, BFAR, FAC, AKVAFORSK, Hambrecht & Quist, Phils.	10	• Commercialization of research Business plan
Launching of Aquaculture Genetics Research	October 9, 1996 BFAR/NFFTRC Muñoz, Nueva Ecija	Philippine Govt. Officials, ROS, Farmers, Media Representatives	200	• Launched GIFT Foundation • Signed a 25-year collaboration genetics
First Strategic Planning: GIFT Foundation International Incorporated	December 16, 1996 -do-	Accredited Hatchery operators, BFAR, ICLARM Representatives	21	Drafted Vision, Mission and P
Guideline on GIFT Hatchery Management	June 25-26, 1997 -do-	Accredited Hatchery Operators	10 (representing 4 hatchery operators & hatchery technicians)	• Signed hatchery agreement



**Table 11. Dissemination of GIFT
broodstocks to Research Outreach
Station of the
Philippine Department of Agriculture**

Table 11. Dissemination of GIFT broodstocks to Research Outreach Station (ROS) of the Philippine Department of Agriculture*.

Region	Recipient	Number of fish distributed/year				
		1993	1994	1995	1996	1997
I (Ilocos Region)	ROS, Laoag, Ilocos	6, 600	-	67, 200	20, 000	20, 000
II (Cagayan Valley)	ROS, Isabela	35, 000	-	15, 000	-	40, 000
III (Central Luzon)	ROS, Zambales	-	2, 100	-	40, 000	-
IV (Southern Tagalog)	ROS, Laguna	-	24, 000	20, 000	-	-
V (Bicol Region)	ROS, Camarines Sur	-	14, 850	34, 000	30, 000	-
VI (Western Visayas)	ROS, Iloilo City	-	10, 000	-	30, 000	-
VII (Central Visayas)	ROS, Cebu City	-	4, 200	3, 000	10, 000	-
VIII (Eastern Visayas)	ROS, Leyte	-	22, 400	-	-	-
IX (Western Mindanao)	ROS, Zambales City Local Government Unit, Pagadian	-	6, 200 7, 000	- -	5, 000 -	- -
X (Northern Mindanao)	ROS, Agusan del Norte	-	12, 000	-	-	-
XI (Eastern Mindanao)	ROS, Davao	-	14, 000	10, 000	-	20, 000
XII (Central Mindanao)	ROS, Cotabato City	-	10, 000	-	8, 000	-
CAR	La Trinidad	-	12, 800	-	-	-
Total		41, 600	139, 550	149, 200	143, 000	80, 000
Grand Total		53, 350				

* Activity accomplished through NFFTRC



**Table 12. Dissemination of GIFT
broodstocksto private tilapia hatcheries
in 1995 accredited under the
National Tilapia Breeding Program
in the Philippines**

Table 12. Dissemination of GIFT broodstocks to private tilapia hatcheries in 1995 accredited under the national tilapia breeding program in the Philippines.

Name of Farmer/ Location of Farm	Area	Total no. of fish distributed
Agnes Tayag Arayat, Pampanga	1.87	26, 000
Michael Lim Looc, Calamba	0.70	12, 100
Jonathan Reyes Sta. Rosa, Nueva Ecija	1.0	12, 300
Jose Cervantes Bacolor, Pampanga	1.25	24, 000
Joel Garcia Capas, Tarlac	1.20	12, 000
Vic Zafra Sto. Tomas, Nueva Ecija	1.0	6, 500
Reynaldo Mencias Alcala, Pangasinan	1.0	6, 000
TOTAL.....		<u>98, 900</u>



**Table 13. Dissemination of GIFT brood
stocks to private tilapia hatcheries
in 1997, accredited under the GIFT
Foundation International, Inc.**

Table 13. Dissemination of GIFT broodstocks to private tilapia hatcheries in 1997, accredited under the GIFT Foundation International Inc.

Name of Farmer/ Location of Farm	Area	Total no. of fish distributed
Pedro Manese Tibagan, Tarlac	2.0	12, 000
Agnes Tayag Arayat, Pampanga	2.0	12, 000
Michael Lim Cabuyao, Laguna	1.0	6, 000
Jonathan, Reyes Sta. Rosa, Nueva Ecija	1.0	6, 000
TOTAL.....		<u>36, 000</u>



**Table 14. Record of Oreochromis niloticus
(GIFT strain) disseminated to
INGA member countries**

Table 14. Record of *Oreochromis niloticus* (GIFT strain) disseminated to INGA member countries.

Date Sent	Recipient		Details of Germplasm**
	Country	Institution	
July 1994	Bangladesh	Fisheries Research Institute (FRI), Mymensingh	GIFT strain-III generation selection
21 August 1996		Fisheries Research Institute (FRI), Mymensingh	GIFT strain-V generation selection (A wt. 30-40 g)
August 1996	Bangladesh	FRI, Mymensingh	GIFT strain-VII generation selection
June 1994	Peoples Republic of China	Shanghai Fisheries University, Shanghai	GIFT strain-III generation selection
			(a) GIFT strain-III generation selection (b) Egyptian strain
11 August 1997	Fiji	Ministry of Agriculture Fisheries and Forestry, Suva	GIFT strain-IX generation selection
June 1994	Indonesia	Research Institute for Freshwater Fisheries, Bogor	GIFT strain-III generation selection
11 August 1997	Indonesia	RIFF, Sukamandi	GIFT strain-IX generation selection
October 1994	Thailand	National Aquaculture Genetics Research Institute (NAGRI), Bangkok	GIFT strain-III generation selection
January 1995			GIFT strain-III generation selection
October 1994		Asian Institute of Technology*	GIFT strain-III generation selection
January 1995			GIFT strain-III generation selection
February 1996		NAGRI, Bangkok	GIFT strain-IV generation selection

* to test the feed conversion efficiency under AIT/ICLARM collaborative study

** the actual number of generations of selection equals x generation minus 2

Date Sent	Recipient		Details of Germplasm**
	Country	Institution	
May 1994	Vietnam	Research Institute for Aquaculture No. 1, Hanoi	a) GIFT strain-III generation selection b) Egypt strain
January 1996		Research Institute for Aquaculture No. 2, Ho Chi Min City	GIFT strain-IV generation selection
27 August 1996		Research Institute for Aquaculture No. 1, Hanoi	GIFT strain-V generation selection (A wt. 50-60g)
August 1996	Vietnam	RIA 1, Ha Bac	GIFT strain-VII generation selection



**Table 15. Training/Workshops organized/
facilitated by GIFT Project in support to
INGA and DEGITA Projects**

Table 15. Trainings/Workshops organized/facilitated by GIFT Project in support of INGA and DEGITA

Title of Training	Dates/Place Conducted	Resource Persons	Participants
<u>Formal</u>			
Application of Quantitative Genetics to Aquaculture	6 November - 2 December 1995 IIRR, Cavite	Senior Scientists from AKVAFORSK, ICLARM, University of Dublin, (Ireland)	Researchers from IN member countries (see attached list)
<u>Informal</u>			
Training/Study tour to familiarize with GIFT Project and Philippine aquaculture activities	April 1994 GIFT Project site, Muñoz, Nueva Ecija, Philippines	GIFT core staff/ICLARM scientists	Scientists of CIFA, Or India
Training to develop fish breeding plans for carp/tilapia in Vietnam	March 1996 RIA No. 1, Hanoi, Vietnam; RIA No. 2, Ho Chi Mihn, Vietnam	AKVAFORSK scientists	Scientists from RIA N and RIA No. 2

LIST OF PARTICIPANTS

Intensive Training on the Application of Quantitative Genetics to Aquaculture
IIRR, Silang, Cavite, Philippines
06 November - 02 December 1995

Bangladesh

Mr. Shyamal Chandra Mahata
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Fisheries Research Institute
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Fax no.: 880 91 55259

Mr. Ms. Shahidul Islam
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China

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Chinese Academy of Fisheries Sciences
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Ms. Olga S. Assemien
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Laboratoire de Physiologie de la Reproduction et de Genetique
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63-20-44

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Ghana

Mr. Felix Kleman Attipoe
Research Scientist
Water Research Institute (CSIR)
P.O. Box 38
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Fax no.: (233-21) 777 170/761 030
Tel no.: 233-21-775351/2

India

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Mr. Chindi Vasudevappa
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Fisheries Research Station, Hesaraghatta
University of Agricultural Sciences
Bangalore, Karnataka, India

Indonesia

Mrs. Lies Emmawati Hadie
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Bogor 16154 , Jawa Barat
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Malawi

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Fisheries Research Officer
Central and Northern Regions Aquaculture Centre
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Fax no.: 721 114

Philippines

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Ms. Ma Jodecel Danting
Researcher
NFFTRC/BFAR
CLSU compound, Munoz, Nueva Ecija
Tel no.: 3881455

Ms. Felicisima Longalong
Ms. Edna Dionisio
Research Assistant
Bureau of Fisheries and Aquatic Resources
National Freshwater Fisheries and Technology Research Center
Munoz, Nueva Ecija

Ms. Marietta P. Hechanova
Mr. Antonio Tadian
Freshwater Aquaculture Center
Central Luzon State University
Munoz, Nueva Ecija

Thailand

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Mr. Kongphop Ampolsak
National Aquaculture Genetics Research Institute
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Vietnam

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Tel. no.: 84-4-8780408

Dr. Nguyen Van Hao
Vice Director, RIA 2
116 Nguyen Dinh Chieu Street
1st District, Ho Chi Minh City
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**Table 16. Prior efforts of the GIFT Project
which led to the establishment of the
International Network on Genetics in
Aquaculture**

Table 16. Initiatives of GIFT Project which led to establishment of International Network on Genetics in Aquaculture.

Year	Activity	Output
1990	Asian Regional Workshop on Tilapia Genetics	Concluded that there was a strong need to establish international collaborative linkages that can help the evolution of national fish breeding programs in those countries
1992	First Meeting to Establish the International Collaborative Linkages in Fish Genetics Research	Defined the principles and mechanisms for establishing the linkage; led to the implementation of network formation functions carried out by the UNDP mission
1993	UNDP Mission Visit	Explored the possibility for aquaculture genetics in six countries in Asia (China, India, Indonesia, Philippines, Thailand and Vietnam) and four in Africa (Cote d'Ivoire, Egypt, Ghana and Malawi)
1993	INGA Network Planning Workshop	Recommended the establishment of a new genetics research, designated as the International Network on Genetics in Aquaculture; developed the guiding principles for establishing INGA.



ANNEXES



Annex 1.

**List of Personnel involved in the
GIFT Project (1988-1997)**

Genetic Improvement of Farmed Tilapia (GIFT) Project Staff List

NAME	POSITION	DATES OF SERVICE	
		STARTING DATE	CONCLUDING DATE
ICLARM			
Dr. Ambekar E. Eknath*	Project Leader	1988	December, 1997
Dr. Roger S. V. Pullin	Program Leader	1988	1996
Ms. Belen O. Acosta	Research Associate	1989	December, 1997
Ms. Josephine Capili	Sr. Research Assistant	May, 1988	November, 1991
Ms. Ravelina Velasco	Sr. Research Assistant	2 November 1988	December, 1997
Ms. Marietta P. de Vera	Research Associate	August, 1989	December, 1997
Ms. Madeline Dalusong	Research Associate		
Mr. Gaspar Bimbao	Research Associate	September, 1993	September, 1995
Ms. Carmela C. Janagap	Senior Research Assistant/Programmer	June, 1991	April, 1996
Mr. Divino Rosales	Programmer	1993	1994
Mr. Edwin de Guzman	Programmer	February, 1991	June, 1991
Ms. Ma. Josephine France Rius	Programmer	28 January 1992	December, 1997
Mr. Hernando Bolivar	Sr. Research Assistant	1989	December, 1997
Ms. Perla Virly	Secretary	July, 1991	December, 1997
Ms. Isa Manela	Secretary	July, 1988	June, 1991
Mr. Norberto Cabrera	Driver	September, 1992	December, 1997
Mr. Cirilo O. Federigan Jr.	Admin. Assistant	15 July, 1995	December, 1997
Mr. Michael Igharas	Clerk Typist	May, 1997	December, 1997
Ms. Florian Lopez	Clerk	March, 1996	30 May 1997
NFFTRC / BFAR			
Dr. Melchor Tayamen Jr.	Project Manager	1988	October, 1997
Mr. Ruben Reyes	Officer-In-Charge	1988	October, 1997
Ms. Jodecel Danting	Research Leader	1988	October, 1997
Ms. Edna Dionisio	Research Analyst	11 July 1988	December 1997
Ms. Felicisima Longalong	Research Analyst	3 January 1990	December 1997
Ms. Lilibeth Afan	Assistant Study Leader	July 1992	June 1997
Ms. Teresita Gonzales	Data encoder	16 January 1990	December 1997
Mr. Marlon Reyes	Foreman/Study Leader	January 1994	June 1997
Mr. Mario Danting	Accounting Clerk	January 1990	December 1997
Mr. Joseph Cruz	Field Assistant	August 1988	December 1997
Mr. Everlito dela Cruz	Field Assistant	August 1988	December 1997
Mr. Camilo Celestino	Field Assistant	August 1988	December 1997
Mr. Mario dela Cruz	Field Assistant	October 1991	December 1997

* Program Leader (Germplasm Enhancement and Breeding Program) from 1996 to October 1997.

NAME	POSITION	DATES OF SERVICE	
		STARTING DATE	CONCLUDING DATE
Mr. Baldwin Reyes	Field Assistant	1991	July, 1997
Mr. Danilo Beltran	Field Assistant	September 1991	December 1997
Mr. Rogelio Estrada	Driver/Field Assistant	September 1994	December 1997
Mr. Reginald Gonzales	Assistant Study Leader	1988	June 1997
Mr. Elmer Afan	Foreman/Study Leader	January 1994	June 1997
Ms. Grace Navidad			May 1995
Mr. Arnel dela Cruz	Field Assistant		
FAC / CLSU			
Dr. Ruben Sevilleja	Program Leader	1988	October 1997
Dr. Tereso Abella	Project Geneticist	1988	October 1997
Ms. Remedios Bolivar	Project Leader	1989	1994
Mr. Antonio Circa	Interim Project Leader	1988	October 1997
Mr. Saturnino Ladromo	Field Assistant	1989	1994
Mr. Rolando Villanueva	Field Assistant	01 July 1988	December 1997
Ms. Marietta Hechanova	Research Assistant	01 March 1993	December 1997
Mr. Leonardo Deguinat	Field Assistant	November 1993	December 1997
Mr. Antonio Taduan	Research Assistant	01 March 1993	December 1997
Dr. Marilou Abon	Consulting Statistician	01 June 1996	December 1997
Ms. Gina Encarnacion	Accounting Clerk	1988	October 1997
Ms. Angelee Romero	Messenger	01 August 1996	October 1997
Ms. Miguel May	General Clerk	1994	October 1997
AKVAFORSK (Norway)			
Dr. Erling Fimland	Director	1988	December, 1997
Dr. Trygve Gjedrem	Senior Scientist	1988	December, 1997
Dr. Hans Bentsen	Senior Scientist	1988	December, 1997
Dr. Bjarne Gjerde	Senior Scientist	1988	December, 1997
Dr. Terje Refstie	Senior Scientist	1988	December, 1997
Dr. Morten Rye	Senior Scientist	1988	December, 1997
UPMSI			
Dr. Julie Macaranas	Project Leader	April 1988	March 1991
Ms. Ma. Josefa Pante	Research Associate	April 1991	June 1992
Ms. Carmen Ablan R.	Research Associate	September 1988	June 1992
Ms. Liza Agustin	Research Associate	September 1988	August 1991
Ms. Ma. Theresa Rodriguez	Research Aide	November 1990	June 1992



Annex 2.

**List of GIFT Project Fixed Assets
and their Present Locations**

14th Floor, Multinational
Bancorporation Centre
6805 Ayala Avenue
Makati City, Philippines

Postal Address:
P.O. Box 2288
Manila, Philippines

Telephone: 845-2723
Telex: (632) 845-2205
Annex 2

Price Waterhouse



C E R T I F I C A T I O N

INTERNATIONAL CENTER FOR LIVING AQUATIC
RESOURCES MANAGEMENT, INC. (ICLARM)

We have reviewed the information shown in the accompanying summary of fixed assets acquired by the Center for the project "Genetic Improvement of Farmed Tilapia" or GIFT from January 1, 1993 to December 31, 1997. Our review included inspection of selected fixed assets, test of the accounting records and such other procedures as we considered necessary in the circumstances. We have found the accompanying summary of fixed assets to be in accordance with the project's books of account and accounting records.

Price Waterhouse

Makati City, Philippines
February 19, 1998

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
INFRASTRUCTURES:				
STAFF HOUSE COST			35,819.02	
	Concrete, w/ 3 bedroom, living/dining room and kitchen,			
	Located in a 8 hectare land owned by			
	Central Luzon State University, Muncz, Nueva Ecija			
RESEARCH AND BREEDING FACILITY			297,082.00	
	Located in a 8 hectare land owned by			
	Central Luzon State University, Muncz, Nueva Ecija			
Components:				
	73 Earthen ponds			
	12 Concrete tanks:			
	1 x 5 divisions			
	1 x 1 division			
	1 Earthen Water Reservoir, 1,500 sq. m.,			
	1 Overhead Concrete Water Tank (Tower)			
	Waterline system:			
	4" main 3" secondary waterlines			
	PVC water pressure-line type of pipes			
	water controlled by 4" and 3" Brass Gate valves			
	water is supplied/removed by use of waterpump			
	Perimeter Fence (made of concrete foundation, post and cyclone wire,			
	unfinished portion presently being continued by BFAR.)			
	1 Wet Laboratory (concrete, GI Roofing, w/ small room for bodega)			
	1 Sampling Hut (semi-concrete, nipa roofing)			
	2 Caretaker Houses (Concrete flooring, house made of nipa, wood & sawali)			
	1 Guard House (concrete)			
	Located at BFAR Facility, Muncz, Nueva Ecija			
	Concrete Tanks - "12 Circular and 38 Reference Tanks"			
	Renovation/improvement of cryo laboratory, Installation of Cryo units.			

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
Tag No.	DESCRIPTION	QTY.	AUDITED AMOUNT	LOCATION
	Installation of Hapa hangers, Nets			
GRAND TOTAL			332,901.02	
COMPUTERS				
3837	Acermate CPU 486DX4-100, 36MB			
3838	with colored monitor (Philips) ,hard disk, 850 MB			
3839 & 4175	Floppy disk drive, Mouse & keyboard			
4174	iPrinter, Citizen brand 200 GX	1set	4,148.10	3rd Floor - HQ
4225	AVR - Mitsui brand - 500 watts			
4080 - 82	Laptop computer (Toshiba) T4500C w/ complete RAM & Mouse	1set	3,640.90	3rd Floor - HQ
3864 -	Acer 486DX 16MB with 72-pin RAM, Hard disk Seagate			
3866	Floppy drive SVGA, Mouse, keyboard & color monitor			
3868	Printer, Citizen brand, Dot Matrix 80-col., 200 GX SN8505399	1set	1,620.59	BFAR
3869 -	Acer 486DX 16MB with 72-pin RAM, hard disk Seagate			
3871	Floppy drive, SVGA, Mouse, Keyboard & color monitor			
3873	Printer, Epson brand, Dot Matrix, 130., LQ1170,SN190038591	1set	1,509.47	BFAR
3960	Acer 486 DX 72-pin, 16MB with hard disk (Seagate)			
4191/92	Disk I/O controller IDE video card, SVGA colored			
& 4194	Monitor ISA floppy disk drive, mouse & keyboard	1set	3,819.17	BFAR
4202 &	"Thompson brand" with keyboard NTC,			
3961	Fujitsu M2613T hard disk and beltron mouse.			
3959	With AVR Powergrid	1set	3,819.17	FAC
3897	Pentium bridge 5/133 mhz processor, NTC			
4180	Floppy disk drive 3.5 1.44 MB, PC Lancard			
4179	3 COM etherlink and Sound blaster, Intel S/N			
4241	motherboard & mini tower case and mouse.	1set	837.29	3rd Floor - HQ
3813	Laptop, Acer brand Model U3865 SN#M017420			

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
AUDITED				
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
	with adaptor, bag, mouse and power cord/cable	1set	2,548.57	3rd Floor - HQ
4096 -	486-DX40 with 16 MB memory 4 x 4 - 30 pins Chicony Keypcard			
4099	Philips Monitor, conner CFS2101 hard disk & A4 tech Mouse.	1set	756.04	3rd Floor - HQ
5049	Notebook (Macintosh) brand SMB/530HD -			
	SN#SG5351EY4FV Powertock 5300 CS	1	5,357.06	3rd Floor - HQ
4253	Notebook (Pentium 150) digital Hi-note Ultra II LTS 5/150			
	colored with Floppy disk drive, spare battery & Mouse			
	(Updated: Sound card/Windows STAT AF FSP. ACCPCFF			
	& base) w/ serial no. 6R6440087 hard disk 1.3GB 32 MB			
	Includes license fee for windows			
3816	Printer, Epson LQ1170 -STYLUS 11			
	Digital 6x CD Mobile media	1set	8,540.96	FAC
4249	with AVR regulator - "Stancor brand"			
4168	386 with keyboard & Mouse (TVM) SN#30528716			
	with tower type CPU, 4MB SIMM RAM I/O Controller card			
4234	Printer, Citizen brand 200X			
4233	AVR - Strator - 1000 Watts	1set	2,127.93	3rd Floor - HQ
3925-3927	with VGA color 15" power grid Disk I/O controller, IDE			
4214	video card, accelerator, mini-tower w/ turbo case switch,			
4183	sound blaster, tape drive, (5100 business server 100 mhz			
4186	pentium 30 pin 16mb expandable to 128MB.			
4182	U P S	1set	1,955.52	3rd Floor - HQ
5040	Printer Stand			
3921	Acer brand with mouse SN54708455			
3923	and keyboard SNK6311644097			
3936	Printer, Dot matrix	1set	2,278.52	Cryo Lab/BFAR
3917-19	Thomson Monitor with keyboard & Mouse	1set	2,156.58	Cryo Lab/BFAR

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
3901	386 SX with Monochrome Monitor 30 - pin 2MB, Floppy			
3903	Floppy disk drive 5.25" 1.2 MB TEAC & 3.5" 1.44 MB TEAC			
3904	Disk I/O controller, IDE, video card			
3905	Printer, Epson Dot-matrix, 80-cpl, LX800SN18050936			
3907	with AVR - Stancor	1set	1,977.30	BFAR
3992	Computer, 486 DX 8 MB: 20-pins (Phillips) with 30-pin, SVGA			
	colored monitor, floppy disk loaded w/ video card, Power			
	to Power supply, Seagate hard disk, informat keyboard &			
3995	3 button mechanical mouse - 59558			
4020	AVR - Stac	1set	1,897.48	FAC
4074 to	New Computer w/ CD ROM and multi media			
4077	speaker, CPU, monitor and keyboard Pentium 166	1set	1,565.16	FAC
4165	Thompson PC+5/166 business comp 486mhz pentium with			
4167	I/O controller card, colored monitor color innovision, RAM 30-pin			
4166	16mb, Video /sound floppy disk drive 3.5" 1.44 MB, Mouse and			
4235	keyboard - 101 keys SNC950979153	1set	1,897.52	3rd Floor - HQ
5050	IOmega ZIP 100 Parallel Port Drive - SN#PRDH41A09X	1	0.03	3rd Floor - HQ
4087 & 4084	Computer with Tower CPU, DX - 40.1 unit 3COM and (1)			
4187	Tower, CPU etherlink III combo lincard (Beitron) AOC Monitor			
4088	AVR - Stavor brand			
4089	Printer, Epson FX1050	1set	2,635.43	3rd Floor - HQ
4243	Computer, 386 SX 30-PIN 2MB with Philips monitor Chicony			
4244	keyboard SN812207987 Floppy disk drive 5.25" 1.2MB TEAC.			
4255	TEAC, 3.5" 1.44 MB Disk I/O Controller, card and seagate hard disk			
4247	with UPS - PK2000 SN0020369	1set	539.57	FAC
4071	HP laserjet 6L printer with cable	1	534.09	FAC
5051	Printer, automatic feeder	1	501.99	Guesthouse

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
4178	HP Laserjet Printer 4P	1	1,280.01	3rd Floor
3958	Printer, Citizen	1	663.59	BFAR
4246	Printer, Epson, Dot-matrix 80-cd.LX800 SN2307255	1	663.59	FAC
4225	AVR - Mitsui brand - 500 watts	1	0.04	3rd Floor - HQ
4183	AVP - 500 Watts	1	0.04	3rd Floor - HQ
4264	UPS - APC Back-up 600, SN93070756500	1	0.04	BFAR
3999	UPS - PK2000 SNC020381with regulator	1	0.04	BFAR
4265	AVR Powergrid - 500 Watts	1	0.04	BFAR
4050	AVR - Hertz brand	1	0.04	BFAR
4072	UPS - APC 600 W	1	0.04	FAC
4250	AVR - Stavel 500 watts	1	0.04	FAC
SUB-TOTAL			59,372.04	
LAB/SCIENTIFIC EQUIPMENT				
3934	Cryo. Liquid Nitrogen tank	1	3,177.24	BFAR
3189	Cryogenic equipment Kryo 10 Series	1	805.59	BFAR
3007	Cryogenic Equipment, low level alarm	1	805.59	BFAR
4056	Timer - "Smiths"	1	0.04	BFAR
4266	Timer - "Smiths"	1	0.04	BFAR
4261	Grass Cutter - 2 HP	1	389.01	FAC/FIELD
4262	Grass Cutter - Kawasaki TD40 SN677434	1	320.97	FAC/FIELD

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
3982	Grass Cutter - Kawasaki	1	0.04	BFAR/FIELD
3983	Grass Cutter - Kawasaki	1	0.04	BFAR/FIELD
4062	Generator, diesel engine - Kubota RK125	1	1,526.24	BFAR/FIELD
5012	Generator, diesel engine - Kubota ER1400N-SN#19167-8862	1	3,024.50	BFAR/FIELD
5011	YSI dissolved oxygen meter model	1	2,121.56	FAC
4254	Pipe Threader set	1	567.72	BFAR
3962	GI Pipe wrench - short	1	0.04	BFAR
4107	GI Pipe wrench - long	1	0.04	BFAR
3964	GI Pipe wrench - field	1	0.04	BFAR
5013	GI Pipe wrench	1	0.04	BFAR
3896	Centrifugal, water pump - Sakai 4 x 4 (SET)	1	0.04	BFAR
5014	Pump, centrifugal - Taiwan 5 x 5	1	957.12	BFAR
3996	Water pump, centrifugal - Sakai	1	0.04	FAC
3997	Water pump, centrifugal - Kubota RK80	1	1,579.55	FAC
3967	Portable battery airator	1	0.04	Wet Lab
3965	Portable battery airator	1	0.04	Wet Lab
3966	Portable battery airator	1	0.04	Wet Lab
4112/3984	Marina submersible pump - Yamato	1	206.98	Cryo Lab
3989	Refrigerator, Westinghouse	1	305.13	Staffhouse
4061	Refrigerator	1	305.13	Guesthouse
4224	Salinity meter - hand refractometer - "Atago"	1	178.50	3rd Floor - HQ
4163	Salinity meter -	2	357.00	FAC
4052	Sewing Machine - SNF6262004 Sew Master	1	0.04	Wet lab

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
4268	Identity portable reader AVID - SN006147	1	840.00	Cryo Lab
4270	Identity portable reader AVID - SN004092	1	840.00	Cryo Lab
4269	Identity portable reader AVID - SN006354	1	840.00	Cryo Lab
3935	Tank, lab 30, liquid nitrogen	1	748.28	Cryo Lab
3937	Tank, lab 30 liquid nitrogen	1	748.28	Cryo Lab
4046	Tank, lab 30 liquid nitrogen	1	1,059.08	Cryo Lab
3921	Tank, lab 30 liquid nitrogen - red small tank	1	1,059.08	Cryo Lab
3924	Tank, Dewar SXR 35/10	1	1,059.08	Cryo Lab
3927	Tank, Dewar MVE XC 47/11-10	1	1,059.08	Cryo Lab
3928	Tank, Dewar MVE XC 47/11-10	1	1,059.08	Cryo Lab
3929	Tank, Dewar MVE XC 47/11-10	1	1,059.08	Cryo Lab
3925	Tank, Dewar MVE XC 47/11-10	1	1,254.83	Cryo Lab
3926	Tank, Dewar MVE XC 47/11-10	1	1,254.83	Cryo Lab
3937	Tank, cryogenic storage	1	2,490.42	Cryo Lab
3938	Tank, cryogenic storage	1	2,490.42	Cryo Lab
3922	Tank, cryogenic storage	1	2,975.10	Cryo Lab
3923	Tank, cryogenic storage	1	2,975.10	Cryo Lab
3881&5055	Wheel borrow	2	0.08	BFAR/Field

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
4223	Calipher - "Manostat"	1	0.04	3rd Floor - HQ
3917	Compressor Motor (Robin EY15DS/N)	1	0.04	Stockroom
3918	Compressor Motor (Robin EY15DS/N)	1	0.04	Stockroom
SUB-TOTAL			43,037.45	
TRANSPORTATION EQUIPMENT				
5034	Toyota Land Cruiser with Plate no. OEV-18384	1	22,058.73	Makati - HQ
5035	Mitsubishi, L200 (Blue) with plate no. OEV-18369	1	19,509.54	Munoz
TOTAL			41,568.27	
FURNITURE AND FIXTURES				
3844	Board, white	1	0.04	3rd Flr. - HQ
3810	Board, white	1	0.04	FAC
4151	Board, white with aluminum frame	1	0.04	BFAR
3998	Board, white	1	0.04	FAC
4110/3950	Board, white with stand	1	0.04	BFAR
4111/3951	Board, white wall type	1	0.04	BFAR
3952	Board, white, (Field)	1	0.04	BFAR
3953	Board, white	1	0.04	BFAR
3811	Book shelves (wooden)	1	133.15	FAC
3817	Book shelves (wooden)	1	133.15	FAC
3842	Book shelves	1	133.15	3rd Flr. - HQ
3843	Book shelves (wooden) - Lixas double walled 11 x 48 x 83.5	1	0.04	3rd Flr. - HQ
3900	Book shelves (wooden) - Narra	1	0.04	3rd Flr. - HQ
3901	Book shelves (wooden) - Lixas double walled 11 x 48 x 83.5	1	0.04	3rd Flr. - HQ
3841	Book shelves (wooden)	1	133.15	3rd Flr. - HQ
3923	Book shelves (wooden)	1	133.15	3rd Flr. - HQ
3924	Book shelves (wooden)	1	133.15	3rd Flr. - HQ
3854	Book shelves (wooden)	1	133.15	3rd Flr. - HQ
3855	Book shelves (wooden)	1	133.15	3rd Flr. - HQ
3955	Wooden shelves w/ hanging glass door (5x6)	1	133.15	BFAR
4105	Wooden shelves w/ hanging glass door	1	133.15	BFAR

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
4120	Table, Monobloc	1	0.04	BFAR
5036	Table, Monobloc	1	0.04	BFAR
4152	Two seater chair, monobloc	1	0.04	BFAR
5037	Two seater chair, monobloc	1	0.04	BFAR
4058-63	Chairs, Monobloc	12	0.48	BFAR
	Chairs, Monobloc	6	0.23	BFAR
4079	Chair, executive	1	0.04	3rd Flr. - HQ
3823	Chair, executive	1	0.04	FAC
3820	Chair, Visitor's	1	0.04	FAC
3821	Chair, Visitor's	1	0.04	FAC
3851	Chair, Visitor's	1	0.04	3rd Flr. - HQ
3990	Chair, computer (Ergo) 1	1	115.14	FAC
3990	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	GFII
4127	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	BFAR
4280	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	GF11/BFAR
4134	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	BFAR
5038	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	BFAR
4122	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	BFAR
4119	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	BFAR
3186	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	FAC
4263	Chair, computer (Ergo) - Low back w/ adjustment	1	0.04	FAC
3918	Chair, computer PC1513 A	1	115.14	FAC
3949	Chair, computer (Ergo)	1	115.14	3rd Flr. - HQ
3847	Chair, computer (Ergo)	1	115.14	3rd Flr. - HQ
3898	Chair, computer (Ergo)	1	115.14	3rd Flr. - HQ
3919	Chair, computer	1	115.14	3rd Flr. - HQ
3929	Chair, computer (Ergo)	1	115.14	3rd Flr. - HQ
4219	Chair, computer (Ergo)	1	115.14	3rd Flr. - HQ
3859	Chair, secretarial	1	115.14	3rd Flr. - HQ
3946	Electric Fan, desk	1	0.04	3rd Flr. - HQ
3991	Electric Fan, stand	1	0.04	FAC

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
3849	Electric Fan, desk	1	0.04	3rd Flr. - HQ
3849	Filing cabinet, steel with 2 drawers	1	0.04	3rd Flr. - HQ
3860	Filing cabinet	1	0.04	3rd Flr. - HQ
4216	Filing cabinet with 4 drawers	1	143.13	3rd Flr. - HQ
3944	Filing cabinet with 4 drawers gauge w/ built in handle	1	0.04	3rd Flr. - HQ
3930	Filing cabinet with 4 drawers	1	127.36	BFAR
3905	Filing cabinet, steel with 4 drawers	1	143.13	3rd Flr. - HQ
3904	Filing cabinet, steel with 4 drawers	1	143.13	3rd Flr. - HQ
4239	Filing cabinet, steel with 2 drawers	1	143.13	3rd Flr. - HQ
4103	Filing cabinet, steel with 4 drawers	1	143.13	BFAR
4104	Filing cabinet, steel with 4 drawers	1	143.13	BFAR
2985	Filing cabinet, steel with 4 drawers	1	143.13	BFAR
2986	Filing cabinet, steel with 4 drawers	1	143.13	BFAR
3908	Filing cabinet, wooden	1	143.13	3rd Flr. - HQ
4138	Wooden cabinet painted yellow - fabricated	1	0.04	BFAR
3824	Cabinet (no description if steel or wooden)	1	0.04	FAC
3848	Table, computer	1	140.62	3rd Flr. - HQ
4130	Table, computer (Sanwa)	1	140.62	BFAR
4129	Table computer (wooden, fabricated)	1	140.62	BFAR
3834	Table computer	1	140.62	3rd Flr. - HQ
5039	Table, computer	1	140.62	BFAR
3902	Table, computer	1	140.62	BFAR
3852	Table, Office	1	0.04	3rd Flr. - HQ
4217	Table, Office	1	0.04	3rd Flr. - HQ
3921	Table, Office (Small)	1	0.04	3rd Flr. - HQ
3922	Table, office	1	0.04	3rd Flr. - HQ
4217	Table, office	1	0.04	3rd Flr. - HQ
3930	Table, office	1	0.04	3rd Flr. - HQ
3943	Table, office	1	0.04	3rd Flr. - HQ
5001	Table, office (Single Pedestal w/ carvings - 24" x 29" x 48")	1	0.04	GF11
5002	Table, office (Single Pedestal w/ carvings - 24" x 29" x 48")	1	0.04	GF11
5003	Table, office (Single Pedestal w/ carvings - 24" x 29" x 48")	1	0.04	GF11

LIST OF ASSETS IN CUSTODY	
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AS OF DECEMBER 31, 1997

IN US DOLLAR

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GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
		AUDITED		
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
4237	Filer, tray ETP MT5	1	0.04	3rd Floor - HQ
4238	Filer, tray ETP MT5	1	0.04	3rd Floor - HQ
3833	Book shelves	1	133.15	3rd Flr. - HQ
3933	Book shelves	1	133.15	3rd Flr. - HQ
3830	Chair, Visitor's	1	0.04	3rd Flr. - HQ
3910	Chair, Visitor's	1	0.04	3rd Flr. - HQ
3825	Electric Fan, desk Rota - aire	1	0.04	FAC
3909	Filing cabinet, steel with 2 drawers	1	143.13	3rd Flr. - HQ
3899	Table, office	1	0.04	3rd Flr. - HQ
3950	Table, typing	1	0.04	3rd Flr. - HQ
3956	Clock, wall	1	0.04	BFAR
3902	Lamp, emergency	1	0.04	3rd Flr. - HQ
3906	Lamp, emergency	1	0.04	3rd Flr. - HQ
5042	Electric airpot - Kyowa	1	0.04	Guesthouse
4065	Electric airpot - "Zojirushi"	1	0.04	BFAR Staff house
4113	Rack with roller	1	0.04	BFAR
4240	Boston Puncher	1	0.04	3rd Floor - HQ
5045	Paper cutter trimmer	1	0.04	FAC
SUB-TOTAL			5,837.64	
OFFICE EQUIPMENT				
3818	Airconditioner, National 1.0 HP w/ timer CW-903FPH	1	544.14	FAC

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
3827	Airconditioner, National 1.0 HP w/ timer CW-903FPH	1	544.14	Guesthouse
3881	Airconditioner, National 1.0 HP w/ timer SN#602103938	1	543.23	Wet Lab.
3916	Airconditioner, National window type CW2402 - 2H	1	1,005.11	Cryo Lab.
4049	Calculator, "Sharp" EL-5090 SN 92040038	1	0.04	BFAR
3880	Calculator, "Sharp" EL-5090 SN 92040040	1	0.04	BFAR
3826	Vault, Safe	1	0.04	3rd Floor - HQ
3911	Typewriter, electric	1	0.04	3rd Floor - HQ
2991	Typewriter, Portable 707 DLX	1	0.04	BFAR
4229	Camera - "Minolta" X - 700	1	0.04	3rd Floor - HQ
4066	Camera - "Pentax brand"	1	0.04	BFAR
4139	Calculator - "Sharp"	1	0.04	BFAR
3998	Calculator - "Casio" ,FX-82C	1	0.04	BFAR
4064	Calculator - "Casio" FX730P	1	0.04	BFAR
4252	Calculator - "Casio" FX-3600 PV (3/18/97)	1	0.04	FAC
3944	Cutter, Board 11 x 15	1	0.04	BFAR
5046	Calculator - "Aurora" HC107-T SN 101707	1	0.04	BFAR
SUB-TOTAL			2,637.14	
COMMUNICATION EQUIPMENT:				

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	
Tag No.	DESCRIPTION	QTY.	AMOUNT	LOCATION
4230	Modem Ext. 144 Accura (Acquired in 5/15/96)			
	Serial No. A0415300W867	1	0.04	3rd Floor -HQ
4063	Interface device (Fax/data) - Motorola brand	1	0.04	BFAR
5047	Carryphone & interface device with fax			
	modem external & power supply	1	0.04	BFAR
4220	Walkman - Aiwa brand JS345	1	189.00	3rd Floor -HQ
4153	Walkman - Sony brand	1	122.43	3rd Floor -HQ
3835	Walkman - Aiwa brand J0202MKIII with adaptor & recharger	1	189.00	3rd Floor -HQ
4226	Walkman - Aiwa brand JX629	1	0.04	3rd Floor -HQ
4227	Nokia Cellular Phone (Taken from Land Cruiser)	1	0.04	3rd Floor -HQ
3947	Nokia Cellular Phone (Taken from Munoz)	1	0.04	3rd Floor -HQ
4231	Power Supply for Nokia Mender TS-135	1	0.04	3rd Floor -HQ
3900	Power Supply for radio with circuit mender TG-135 SN00337SN9	1	0.04	BFAR
4279	Radio transmitter/receiver "Midland"			
	SN329058 Model #70-530B with handset	1	0.04	BFAR
3962	Radio power supply - PS 20 (Attached to			
	Radio transceiver/receiver)	1	0.04	BFAR
5048	Icom Radio transceiver (136-174 Mhz; 40 max chanel's)	1	0.04	BFAR
	DC Power of 6-16 volts with SN #22767			
4125	Icom Radio transceiver (136-174 Mhz; 40 max chanel's)	1	0.04	BFAR
	DC Power of 6-16 volts with SN #100376			

GENETIC IMPROVEMENT OF FARMED TILAPIA - GIFT PROJECT -				
LIST OF ASSETS IN CUSTODY				
PROJECT DURATION: JANUARY 1, 1993 - DECEMBER 31, 1997				
AS OF DECEMBER 31, 1997				
IN US DOLLAR				
			AUDITED	ASSIGNEE/
Tag No.	DESCRIPTION	QTY.	AMOUNT	STAFF CODE
4124	Icom Radio transceiver (136-174 Mhz 40 max chanel's)	1	0.04	Hernando Bolivar - 204
	DC Power of 6-16 volts with SN #100374			
	SUB-TOTAL		500.95	
	GRAND TOTAL		485,854.51	

GENETIC IMPROVEMENT OF FARMED TILAPIA

NOTES TO THE SUMMARY OF ASSETS IN CUSTODY

PROJECT DURATION: JANUARY 1, 1993 TO DECEMBER 31, 1997

Note 1 - Assets in Custody

International Center for Living Aquatic Resources Management, Inc. (ICLARM) became a member of the Consultative Group on International Agricultural Research (CGIAR) in 1993. Prior to its membership in CGIAR, project assets were recorded as capital expenditures (an expense account) at its full historical landed cost. As a result of its membership in CGIAR, the Center adopted CGIAR accounting policies in recording these fixed assets as assets in custody. The Center capitalizes assets costing at least US\$500 or its equivalent in pesos. Acquisitions costing below the said limit are assigned a minimal value of P1.00 for control purposes and the difference charged to the operating cost of the projects. Fixed assets in custody are considered "assets in trust" and are therefore not subject to depreciation.

Note 2 - Inventory of project asset

A physical count of the GIFT assets was conducted in August 1997 to ascertain existence of project assets and correctness of recorded balances.



Annex 3.

**Extended Executive Summary of
the DEGITA Project Final Report**

Dissemination and Evaluation of Genetically Improved Tilapia Species in Asia: Final Report

International Center for Living Aquatic Resources Management (ICLARM)
MC P.O. Box 2631, Makati City, Metro Manila
0718 Philippines

Executive Summary

The International Center for Living Aquatic Resources Management (ICLARM) and its research partners developed a Nile tilapia line through selective breeding that outperformed in growth and survival other widely farmed tilapia strains, in varied farming environments, in Asia. Following this successful development of a genetically improved Nile tilapia strain, popularly known as GIFT strain, it became important to spread this "new technology" as a means to increasing aquaculture productivity, improving rural nutrition and increasing farmers' income in Asia. This report highlights the results of the on-station and on-farm evaluation of the performance of the GIFT strain against the commercially available tilapia strains as well as its likely economic, social and environmental impact in Bangladesh, China, the Philippines, Thailand and Vietnam. The work was carried out under the Dissemination and Evaluation of Genetically Improved Tilapia Species in Asia (DEGITA) Project of ICLARM, funded by the Asian Development Bank (ADB) under regional Technical Assistance (RETA) No. 5558.

A unified methodology integrating the disciplines of genetics, economics, sociology and environmental science was developed to achieve the objectives of the DEGITA Project. The methodology was a combination of *ex-ante* and *ex-post* evaluation techniques, and consisted of the following activities: a) national review of the past performance and the current status of the tilapia industry; b) on-station evaluation of GIFT vs. non-GIFT ("control") strains; c) analysis of agroecological and socioeconomic environments of project sites; d) baseline survey of fish producers, consumers and traders; and e) on-farm trials of GIFT vs. non-GIFT strains. The impact evaluation framework used was holistic in nature. The impacts were measured at household (producers vs. consumers) and country levels using a combination of general equilibrium (GE) analysis, micro system analyses (village studies and farming system analysis), and ecological analysis.

In Asia tilapia farming is growing rapidly. It accounted for about 87 percent of globally farmed tilapia in 1994. Asia's tilapia production increased from 154,509 t in 1984 to 519,192 t in 1994, giving an average annual growth rate of 12%. In 1994, DEGITA-participating countries such as China (43 percent), the Philippines (19 percent),

and Thailand (11 percent), accounted for the majority of farmed tilapia production in Asia.

Depending on their natural endowments and investment capacity, fish farmers in these countries raise tilapia in ditches, ponds, cages/pens and concrete tanks. In Bangladesh, most fish farmers culture tilapia in relatively small ponds and ditches. In China and the Philippines, they raise tilapia in ponds and cages with a few in concrete tanks. In Thailand and Vietnam, most farmers grow tilapia in ponds. The majority of fish farmers in Bangladesh, China, Thailand and Vietnam raise tilapia in polyculture systems. Moreover, integrated fish farming (e.g. rice-fish farming, fish-duck farming, fish-pig farming) is becoming popular in all these countries.

The average size of fish farm varies considerably. Fish farmers in Bangladesh have the smallest average size at 500 m², while the farmers in the Philippines have the highest with fishponds at 15,000 m². Fish farmers in China, Thailand and Vietnam operate an average fishpond area of 13,700 m², 10,500 m² and 1,000 m², respectively. As in China and the Philippines, those operating cages are working on smaller area than those operating fishponds. Fish farmers generally have security of tenure as they own a very high proportion of their fish farm area (56-99 percent), except in China.

Freshwater aquaculture is fast becoming a highly market-oriented farming activity. About 94-97, 97, 98 and 88 percent of the fish farmers' production in China, the Philippines, Thailand and Vietnam, respectively, is being marketed. Only fish farmers in Bangladesh have a low marketable surplus (20 percent of their production).

Overall, China's pond (monoculture and polyculture systems) and cage productivity (monoculture) per production cycle was higher than in the other countries. In pond and cage monoculture, China (5,860 kg/ha; 5,613 kg/100 m²) has higher productivity than the Philippines (3,559 kg/ha; 853 kg/100 m²). Also, China's (6,593 kg/ha) pond productivity under polyculture was higher compared to Bangladesh (1,736 kg/ha), Thailand (6,290 kg/ha) and Vietnam (3,020 kg/ha). It must be noted, however, that those culturing tilapia in cages/pens and tanks have relatively higher production per unit area than those using ponds due to their higher stocking density and an input-intensive production system.

In pond monoculture, China (US \$ 3,104/ha) has a lower profitability in terms of net profit over variable cost than the Philippines (US \$ 3,421/ha). But the country's (US \$4,146/100 m²) profitability in cage monoculture is substantially higher than the Philippines (US \$524/100 m²). Thailand (US \$7,395/ha) and China (US\$ 3,997/ha) have a relatively higher profitability in pond polyculture than Bangladesh (US\$ 1,258/ha) and Vietnam (US\$ 1,321/ha). The cash cost of producing a kilogram of fish varies considerably (Bangladesh: pond polyculture- US\$ 0.16/kg; China: pond monoculture- US \$0.76/kg, pond polyculture- US \$0.66/kg, cage monoculture- US \$1.09/kg; Philippines: pond monoculture- US \$0.60/kg, cage monoculture- US \$0.61/kg; Thailand: pond polyculture- US \$0.45/kg; and Vietnam: pond polyculture- US \$0.27/kg). As much as 75

percent of the gross value of production in these countries can be considered a 'pure profit'.

Socioeconomic problems (cage/pond vandalism, poaching, high prices of fry/fingerlings, increasing cost of inputs, high capital requirement, high marketing cost, lack of credit, water supply shortage, proliferation of tilapia farms, farm fragmentation and limited market) affect tilapia farming productivity in ponds more seriously than in cages. On the other hand, cage operators are more dependent on natural endowments such as the existence of lakes and water reservoir, and are greatly affected by natural/environmental factors (flood, water quality/pollution, drought, typhoon, cold and sulphur upwelling) and technical/biological factors (high fish mortality, fish diseases, poor fry/fingerlings quality, slow growth, lack technical assistance/management expertise, small fish at harvest, harvesting difficulty and over growth of weeds).

Fish producers generally have a higher annual fish consumption per capita than non-producers, indicating the importance of fish farming for the supply of nutritional protein requirement in the rural areas. Results of DEGITA (1995-1996) show that consumers in Bangladesh (21.3 kg) have a higher annual per capita fish consumption than those in Thailand (18.6 kg) and Vietnam (13.9 kg). Per capita fish consumption per annum in China and the Philippines amounts to 35.8 kg and 44.6 kg, respectively. In all these countries, except the Philippines, carp is the freshwater species which is traditionally consumed.

Consumers in these countries show a strong preference for the following fish species: Bangladesh- ilish (54 percent), rohu (32 percent) and tilapia (7 percent); China-tilapia (29 percent), crucian carp (12 percent), grass carp (10 percent); the Philippines-tilapia (65 percent), milkfish (11 percent) and threadfin bream (5 percent); Thailand-snakehead (43 percent), tilapia (30 percent) and hybrid catfish (21 percent); and Vietnam-common carp (25 percent), mrigal (17 percent), rohu (16 percent) and tilapia (15 percent). This has significant implications for the development of the tilapia farming industry, not only in all these countries but also in Asia, as tilapia is now a widely accepted food fish. In the case of tilapia, most consumers show a strong preference for bigger fish with sizes ranging from 2 to 4 fish/kg. However, consumers in Bangladesh show a preference for fish of smaller size: 5 to 7 fish/kg of tilapia.

The fish marketing structure in Asia is evolving fast particularly in the DEGITA-participating countries. Fish marketing in China and Vietnam (especially in the North) used to be an activity controlled by the government. The government is now encouraging the participation of the private sector (i.e., traders, wholesalers and retailers) as is relevant in Bangladesh, the Philippines and Thailand. Mainly due to the large trading volume they are handling, fish trading is a very lucrative economic activity for traders in the Philippines and Thailand but not yet in Bangladesh. Fish traders are earning a monthly income of US \$600 in the Philippines and US \$641 in Thailand. If the fish traders in Bangladesh can increase the volume of the fish they trade, it can also be a lucrative

business in this country, as fish traders are earning a comparable margin per kg of fish traded (US \$0.19) to those in the Philippines (US \$0.27) and Thailand (US \$0.20).

Private tilapia hatchery operations are more important in the Philippines than in any other DEGITA participating country, where the industry is dominated by small scale family-based hatcheries. On the average, hatchery operators in the Philippines are operating an effective hatchery area of 1.07 ha, and more than 90 percent are owner operators. With an average hatchery operator incurring a total production cost of about US\$ 2,500 per ha. per production cycle, they are earning a net income of about US\$ 4,300 per ha. per production cycle, which is considered lucrative.

On-station performance of the GIFT fish against commercially available local strains was evaluated in diverse environments in: Bangladesh (warm humid tropics); China (warm cool humid subtropics, warm arid and semi-arid subtropics); Thailand (warm subhumid tropics); and Vietnam (warm humid tropics). The magnitude of impact of the GIFT strain over existing strains (Bangladesh: Thai strain; China: '1978' introduced, '1988' introduced, Egypt, Red and Tianjin strains; Thailand: Chitralada and Chitralada I strains; and Vietnam: Egypt, Thailand and Vietnam strains), varied among various countries according to the history of tilapia introduction, environments and farmers' experience. In Bangladesh, where Nile tilapia has a short history with very limited broodstock management, the GIFT strain appears to be 42 percent superior to the local strain in terms of growth. In China, the GIFT strain did not show superior performance relative to local strains under on-station conditions. This is plausible because China has good existing strains obtained directly from Africa (the origin of Nile tilapia) and there is a longer history of tilapia farming, greater climatic variation and, therefore, the possibility of both natural and artificial selection of existing strains to local environments. In Vietnam, the GIFT fish of the 4th generation of selection had 40 percent higher growth rate compared to existing strains, though the GIFT fish of the 2nd generation of selection did not show statistically significant superior performance in terms of growth. The on-station data did not show any superiority of the GIFT strain over existing strains in terms of survival.

Following farmers' practices in terms of levels of input use and growing periods, the on-farm performance of the GIFT strain against the 'best' existing local strains was assessed in all DEGITA-participating countries. The trial sites represent the major agroecological zones, farming systems (extensive to intensive) and types of culture (ponds and cages) of Asia. After accounting for the wide heterogeneity of input use level, production environments and other intrinsic and extrinsic factors, the percentage change in average weight at harvest due solely to the mean superiority of GIFT strain over the existing non-GIFT strains were highly significant across different countries. The magnitude of the effect of the GIFT strain on growth rate, however, varied among countries, ranging from 18 percent in China to 66 percent in Bangladesh. In terms of survival, the GIFT strain showed higher survival compared to the existing strains only in China (3 percent) and the Philippines (13 percent). Results show that the GIFT technology is neutral with respect to feed use, i.e., increase in output due to GIFT

technology per unit of feed used was constant across feed use levels in all countries, except Bangladesh

Cost of production per unit of fish produced was lower for the GIFT strain than for other Nile tilapia strains in current use: more than 30% lower in Bangladesh and the Philippines, and about 20% lower in China, Thailand and Vietnam.

The estimated yield potential of the GIFT strain was more than 50 percent higher than that of the best existing strain, ranging from about 54 percent higher in Vietnam to about 97 percent higher in Bangladesh. The estimated Yield Gap II, the difference between potential and actual yields on farmers' field, ranges from around 3,000 kg ha in China to around 850 kg ha in the Philippines (which has a shorter growing period and poorer production environment). These estimated gaps show the extent to which yield might be increased through the invention of appropriate technology to overcome production constraints such as disease, temperature extremes, water shortage, poor water quality and so forth.

As the price elasticity of demand for tilapia is elastic (close to -1 in absolute term), the rate of decrease in the price of tilapia would be lower compared to the rate of increase in productivity. In the Philippines, a 15 percent increase in average national productivity of tilapia (Scenario I, early stage of adoption, wherein the rate of adoption would be lower but productivity gain to adoptor would be higher) would result in about 11 percent decrease in tilapia price; while a 20 percent increase in national productivity of tilapia (Scenario II, later stage of adoption curve, where the rate of adoption would be higher but productivity gain to adoptor would be lower) would decrease tilapia price by about 14.5 percent. Similar trends is observed in all the participating countries under both technical change scenarios.

In terms of profitability, the level of increase for adoptors would be higher during the early stage of adoption (Scenario I). As more and more farmers adopt the technology, the supply would increase and the relative price of tilapia would decrease in the later stage of adoption (Scenario II). These effects, combined with the relatively lower increase in average productivity gain due to adoption of GIFT strain in poorer quality farms, would reduce the level of increase in profitability. In the Philippines, the 40 percent increase in productivity due to GIFT strain would result in about 84 percent increase in profitability (from a US \$4,052 per farm of 1.6 ha to US \$7,452) during the early stage of adoption and profitability would only increase by about 17 percent during the later stage of adoption. Similar trends are expected to be observed in other countries. But due to the lower market price of tilapia, profitability of tilapia farming for non-adopting farmers would be expected to decline. However, as the GIFT technology is scale neutral, non-adopting farmers are expected to follow the early adoptors. The government should remove supply side constraints, if any, so that every farmer has access to the technology and each of them face the same "effective" price of tilapia fingerlings.

Adoption of GIFT strain is expected to increase tilapia production in particular and overall fish production in general. In the Philippines, the 15 percent increase in average national tilapia productivity would increase national tilapia production by about 16 percent during the early stage of adoption (Scenario I). In Thailand, the 8 percent increase in average national tilapia productivity would result to a 9 percent increase in tilapia production in the country under Scenario I. At the household level, production of other fish species would decrease marginally for GIFT-adopting farmers, while for non-adopting farmers it would increase marginally for those culturing fish in polyculture systems in Bangladesh, China, Thailand and Vietnam. This, combined with the fact that tilapia has no major competitor in the resource use in the Philippines, implies that adoption of the GIFT strain would increase tilapia production in all countries without decreasing the production of other species.

The technical change such as that provided by the faster-growing GIFT fish is expected to increase tilapia consumption in all types of household (GIFT adopting fish farmers, non-GIFT adopting fish farmers and consumers), except for the non-adopting farmers of the Philippines, for whom the negative effect of lower income would probably be higher than the positive effect of the lower market price of tilapia. In China, where the national average tilapia consumption is about 1 kg/year/household, GIFT technology is expected to increase average tilapia consumption by about 7 percent. In the Philippines, average tilapia consumption per household is expected to increase from 10 kg/year/household to about 12 kg/year/household due to the GIFT technology. Overall fish consumption is also expected to increase for all non-producing consumers and GIFT adoptors, and even for most of the non-GIFT adopting tilapia producers, though the level of increase is expected to be very small, ranging from almost 0 percent in China to about 3 percent in Thailand.

At the household level, adoption of the GIFT strain would increase the overall welfare gain for all consumers and GIFT adopting fish producers. For subsistence farmers in Bangladesh with 500 m² pond area under polyculture systems, GIFT technology would increase the overall economic gain by about US \$5-6/household/year. In the Philippines, the overall economic gain of GIFT adoption, with 1.6 ha of pond area under tilapia monoculture, would be about US \$1,500 to US \$3,500/household/year, depending on the level of productivity gain. Although the overall gain to an average consumer in the DEGITA-participating countries would be very low (often less than US \$5/household/year per household, except in the Philippines) due to the relatively low level of tilapia consumption per household, the overall economic gain for the country is very high due to the very large number of tilapia consumers. More than 50 percent of the total economic gain due to the GIFT technology would go to consumers, except in Thailand, where the share is about 40 percent.

As tilapia production is dominated by small- and medium-scale farmers employing extensive or semi-intensive production systems, the major portion of the producers' benefit is expected to go to them. In Bangladesh, where tilapia is cultured mostly in seasonal ponds under extensive culture systems by poor farmers with an

average farm size of 0.6 ha (fish farm of 0.05 ha), most of the producers' gain would go to small farmers.

Tilapia is consumed mainly by relatively poor people because of its relatively low price, except in the Philippines. As the price elasticity of demand for fish is higher for poor people than for the rich, the major portion of the consumers' benefit will go to relatively poorer (not to the very poor who can not afford to buy any fish) consumers in all these countries, except the Philippines. As richer consumers consume more tilapia in the Philippines, both in terms of the absolute level of consumption and share of total fish consumption, they are expected to get the larger share of the total consumers' benefit.

Except for the Philippines, almost all the pre-harvest labor for tilapia culture comes from family sources. As adoption of the GIFT strain does not require additional care, it is expected that the same structure of labor use will prevail even after the introduction of the GIFT technology. This implies that demand for pre-harvest labor is not expected to increase due to adoption of GIFT technology in these countries, except the Philippines. However, with the increase in production due to the adoption of the GIFT strain, demand for labor in harvesting and post-harvesting activities will increase in countries like China, the Philippines and Thailand. As the GIFT strain is technology neutral with respect to feed use, demand for feed is not expected to increase under the present level of stocking density. Thus, the use of the GIFT strain will not directly affect the demand for feed. However, if farmers increase their stocking density and thereby increase feed requirement per unit area, and increase the area under tilapia operation, overall demand for feed would increase.

GIFT fish is not a genetically modified organism in the biotechnological sense, but only a strain of *O. Niloticus* produced by selective breeding. In Asia, *O. niloticus* is already widespread and there are no reports, even anecdotal ones, that it has been responsible for the decline in the indigenous species. As such, it is argued that the GIFT fish should not be harmful either. With proper management of culture practices, the GIFT fish is unlikely to harm biodiversity and/or environmental integrity. However, more detailed studies on the environmental impact of the GIFT fish are needed.

In order to improve and sustain the productivity of the improved strain, all the DEGITA participating countries are in the process of developing their own national tilapia breeding programs with GIFT fish as genetic material (with or without other genetic material).

As the final outcome of any technological improvement depends equally on the policy and institutional environments within which it is introduced, appropriate government policies should aim at ensuring an incentive structure that reflects the true social opportunity cost of resource use in tilapia culture. It is likely that the market would fail to allocate resources efficiently to the tilapia sector in the presence of negative externalities (e.g. pollution of the aquatic environment, displacement of existing fish species or loss of fish biodiversity, etc.) and absence of perfect market conditions. Under

such conditions, government interventions are necessary in providing public goods (research, extension and infrastructure) and access to information and inputs (e.g., credit, availability of quality fry/fingerlings during the early stage of adoption, etc.) to “get the price right”.



Annex 4.

**List of Scientific Papers Published
And Status of Pending Publications**

LIST OF PUBLICATIONS & SCIENTIFIC PRESENTATIONS FROM THE GIFT PROJECT

- Acosta, B. O., E. Dionisio and A. E. Eknath. 1996. Growth and food conversion of five strains of Nile tilapia (*Oreochromis niloticus*) fry, p.537. In: R. S. V. Pullin, J. Lazard, M. Legendre, J.B. Kothias and D. Pauly (eds.) The Third International Symposium on Tilapia in Aquaculture. ICLARM Conf. Proc. 41, 575 p.
- Agustin, L. R. Froese, A. Eknath and R. S. V. Pullin. 1993. Documentation of genetic resources for aquaculture - the role of FISHBASE, p. 63-68. In D. Penman, N. Roongratri and B. MacAndrew (eds). AADCP Workshop. International Proceedings: Genetics in Aquaculture and Fisheries Management, 31 August to 4 September 1992, University of Stirling, Stirling, Scotland, UK
- Bentsen, H.B., A. E. Eknath, M.P. De Vera, J.C. Danting, H.L. Bolivar, R.A. Reyes, E. E. Dionisio, F. M. Longalong, A.V. Circa, M.M. Tayamen and B. Gjerde. Genetic Improvement of Farmed Tilapias: Growth Performance in a Complete Diallel Cross Experiment with Eight Strains of *Oreochromis niloticus*. *Aquaculture* 160 (1998) 145-173.
- Bolivar, R.B., and A. E. Eknath, H.L. Bolivar and R.A. Abella. 1993. Growth and reproduction of individually tagged Nile tilapia (*Oreochromis niloticus*) of different strains. *Aquaculture*, 111: 159-169
- Bolivar, R.B., H.L. Bolivar and A. E. Eknath. 1994. Effect of sampling frequency on the growth and survival of Nile tilapia (*Oreochromis niloticus*) in hapas. *Asian Fisheries Science* 7(1994): 129-133.
- Bolivar, H. M. de Vera, R. Reyes, R.B. Bolivar, H.B. Bentsen and A. E. Eknath. 1995. Early growth and survival of eight strains of Nile tilapia (*Oreochromis niloticus*) and their crosses. In L.M. Chiu, A. E. Munro, T.J. Lam, T.W. Chen, L.K.K. Chong, J.K. Ding, K.K. Hooi, H.W. Khoo, V.P.E. Chong, K.F. Shin and C.H. Tan (eds.) The Third Asian Fisheries Forum Proceedings.. Asian Fisheries Society, Manila, Philippines. 1,135 p.
- Bolivar H.L., R.B. Bolivar, C.C. Janagap, R.A. Reyes and A. E. Eknath. Fry Production of Eight Strains of Nile Tilapia and their Crosses. Poster paper presented at the 5th International Symposium in Genetics in Aquaculture, June 19-25, 1994. Dalhousie University, Halifax, Canada.
- Chiayvareesajja, J., Roed, K. H., Danting, J. C. De Vera, M. P., and Bentsen, H. B., 1997. Genetic variation in lysozyme and spontaneous hemolytic activities of blood serum from Nile tilapia. Submitted to *Aquaculture*. (ICLARM Contribution No. 1362)

- Chiayvareesajja, J., Eknath A. E., Danting, J. C., De Vera, M. P., and Bentsen, H. B., 1997. Genetic association of serum lysozyme and serum spontaneous haemolytic activities with survival and body weight in Nile tilapia. Submitted to Aquaculture, special issue, Proc. 6th Symposium of Genetics in Aquaculture, Stirling, 1997. (ICLARM Contribution No. 1363)
- Circa, A.V., A. E. Eknath, H.H. Bolivar and T.A. Abella. Growth performance of seven strains of *Oreochromis niloticus* in ponds fertilized with on-farm crop residues. Paper presented at the Third International Symposium on Tilapia in Aquaculture, 11-16 November 1991, Abidjan, Cote d' Ivoire.
- Circa, A.V., A. E. Eknath and A.G. Tadian. Genetic Improvement of Farmed Tilapias: The Growth Performance of the Gift Strain of Nile Tilapia in Rice-Fish Environments. Poster paper presented at the 5th International Symposium in Genetics in Aquaculture, June 19-25, 1994. Dalhousie University, Halifax, Canada.
- Danting, J.C., A. E. Eknath and H.B. Bentsen. Evaluation of Growth Performance Testing Methods for Strain Comparisons of Nile Tilapia. Poster Paper presented at the 5th International Symposium in Genetics in Aquaculture, June 19-25, 1994. Dalhousie University, Halifax, Canada.
- De Vera, M.P. and A. E. Eknath. Growth Performance of Males and Females of Different Strains of Nile Tilapia in Different Culture Environments. Paper presented at the 5th International Symposium in Genetics in Aquaculture, June 19-25, 1994. Dalhousie University, Halifax, Canada.
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- Eknath, A. E. The Blue Revolution and Perspective of a Sustainable Marine Biotechnology. Paper presented at the International Marine Biotechnology Conference (IMBC '94), 8-12 August 1994 at University of Norway, Tromso, Norway.
- Eknath, A. E. Highlights of GIFT Project Experiences. Paper presented at the International Network on Genetics in Aquaculture - Network Planning Workshop, ICLARM HQ, Makati, Metro Manila, Philippines, 17-22 July 1993.

- Eknath, A. E. 1994. Genetic Improvement of Farmed Tilapia (GIFT) project: from modest beginnings to international network. Villacorta, L.G., Dureza, L.W. (eds.) Tilapia Farming: Genetic Improvement and Advances on Culture Technology. Third National Symposium and Workshop on Tilapia Farming, Iloilo City, Philippines, 25-27 November 1993. PCAMRD Book Series No. 18. 86 p.
- Eknath, A. E. and R. R. Velasco. 1993. Birth of the 'super tilapia'. Fish Farmer, p.29 and 33.
- Eknath, A. E., J.M. Macaranas, L.Q. Agustin, R. R. Velasco, M.C.A. Ablan, M.J.R. Pante and R. S. V. Pullin. 1991. Biochemical and morphometric approaches to characterize farmed tilapias. Naga. The ICLARM Quarterly 14(2):7-9.
- Eknath, A. E., H.B. Bentsen, B. Gjerde, M.M. Tayamen, T.A. Abella, T. Gjedrem and R. S.V. Pullin. 1991. Approaches to national fish breeding programs: pointers from a tilapia pilot study. Naga. The ICLARM Quarterly 14(2): 10-12.
- Eknath, A. E., J.B. Capili, J.C. Danting, E. E. Dionisio, R.A. Reyes, N. D. Gerundo, M.M. Tayamen and R S.V. Pullin. Experiences with the importation and quarantine of germplasm for developing a national tilapia breeding program in the Philippines. Paper presented during the Third Asian Fisheries Forum, World Trade Center, Singapore, 26-30 October 1992.
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MAJOR SCIENTIFIC PAPERS FOR COMPLETION AND THEIR CURRENT STATUS:

1. Genetic Improvement of Farmed Tilapias (GIFT) - Systematic Effects of Test Environments, Sex, and Age at Stocking on Growth in Nile Tilapia.

Status: Data analysis completed. Needs only some fine tuning of analysis to determine the relationship between mean weights at stocking, final weights at stocking and age groups. Outline of manuscript prepared.

2. GIFT - Composition and Genetic Parameters of a Heterogeneous Base Population for Selection.

Status: DFREML analysis for growth completed. GFCAT analysis for survival initiated.

3. GIFT - Heritability of Growth and Survival during Six Generations of Selection.

This paper also is now being considered under three separate communications: 3 (a) heritability of growth across generations; 3 (b) heritability of survival across generations; 3 (c) genetic correlations between growth and survival.

Status: DFREML analysis for growth in progress. GFCAT for survival to be initiated.

4. GIFT - Genotype x Environment Interaction for Growth of Replicate Families Tested in Different Tilapia Farm Environments.

Status: to be initiated.

5. GIFT - Genotype x Season Interaction for Growth of Replicate Families Tested during different Grow-out Seasons.

Status: Data analysis completed.

6. GIFT - Growth Performance of an Improved Breed of Nile Tilapia in On-Farm Trials in Asia.

Status: Data analysis complete. Writing of manuscript to be coordinated with DEGITA Project Leader, Madan Dey.

7. GIFT - Response to selection for growth.

Status: Estimates of observed response to selection is possible during the data analysis in respect of paper 3 described above. Model for estimating the predicted response to selection under review and or further refinement.

8. GIFT - Genetic Correlation between Growth and Frequency of Spawning Females in Nile tilapia.

Status: Only preliminary analysis done. (Focus is on generation 7 datasets).

9. Effect of common environment during early rearing on subsequent growth performance of Nile tilapia.

Status: Data analysis complete; outline of manuscript reviewed.

10. Combined Effects of Genetic Improvement through Selection and Sex Reversal on Survival and Growth of Nile Tilapia.

Status: Revised analysis complete; manuscript needs slight revision based on revised analysis.

11. GIFT - Estimation of Genetic Variation and Heritability for Age- and Size- at First Spawning.

Status: DFREML analysis completed. Genetic correlations to be carried out.

12. Response to Bi - Directional Selection for Frequency of Early Maturing Females in Nile Tilapia.

Status: Paper presented during ISGA 6; Manuscript submitted.

We expect completion of all manuscripts during 1999. Priority will be on the primary paper on selection response which will be completed by December 1998